

Logic Manager Service

LM13-500

LM Service
Logic Manager

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LM13-500
R500
9/95

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About This Publication

This publication provides instructions for use by the Logic Manager System maintenance personnel. It will help you to maintain, test, troubleshoot, and repair the equipment.

This manual supports TDC 3000^X software release 500.

Table of Contents

1 INTRODUCTION

- 1.1 Purpose of This Manual
- 1.2 Related Publications

2 EQUIPMENT DESCRIPTION

- 2.1 System Overview
 - 2.1.1 Logic Manager Processor
 - 2.1.2 Logic Manager I/O System
 - 2.1.3 Serial I/O System
 - 2.1.4 Parallel I/O System
 - 2.1.5 Redundant Control System
- 2.2 Equipment Overview
 - 2.2.1 Processor Card Files
 - 2.2.1.1 Nonredundant Processor Card File
 - 2.2.1.2 Redundant Processor Card File
 - 2.2.1.3 Processor Card File Option Modules
 - 2.2.2 I/O Card Files
 - 2.2.2.1 Nonredundant Serial I/O Card Files
 - 2.2.2.2 Redundant Serial I/O Card File
 - 2.2.2.3 Nonredundant Parallel I/O Card Files
 - 2.2.2.4 Redundant Parallel I/O Card File
- 2.3 Processor Card File Modules
 - 2.3.1 Logic Manager Module
 - 2.3.1.1 Front Panel
 - 2.3.1.2 Circuit Board Layout
 - 2.3.1.3 Configuration
 - 2.3.2 Redundancy Control Module
 - 2.3.2.1 Front Panel
 - 2.3.2.2 Configuration
 - 2.3.3 8 K Memory Module
 - 2.3.4 24 K Memory Module
 - 2.3.5 4 K x 4 K Register Module
 - 2.3.6 System Control Module
 - 2.3.7 Processor Module
 - 2.3.8 I/O Control Module
 - 2.3.9 Power Supply Modules
 - 2.3.9.1 115/230 Vac 15 A Power Supply Module
 - 2.3.9.2 24 Vdc 15 A Power Supply Module
 - 2.3.10 Serial Link Module
 - 2.3.10.1 Front Panel
 - 2.3.10.2 Configuration
 - 2.3.11 Parallel Link Driver Module
 - 2.3.11.1 Front Panel
 - 2.3.11.2 Configuration

Table of Contents

2.4	I/O Card File Modules
2.4.1	Enhanced Diagnostic Module
2.4.1.1	Front Panel
2.4.1.2	Configuration
2.4.2	Serial Link Selector Module
2.4.2.1	Front Panel
2.4.2.2	Configuration
2.4.3	Parallel Bus Selector Module
2.4.3.1	Front Panel
2.4.3.2	Configuration
2.4.4	Power Supply Modules
2.4.4.1	I/O 24 Vdc 8 A Power Supply Module
2.4.4.2	I/O 115/230 Vac 15 A Power Supply Module
2.4.4.3	I/O 115/230 Vac 8 A Power Supply Module
2.4.5	Serial I/O Modules
2.4.5.1	Front Panel
2.4.5.2	Configuration
2.4.6	Parallel I/O Module
2.4.6.1	Front Panel
2.4.6.2	Configuration
2.4.7	Process I/O Modules
2.4.7.1	Discrete Input Modules
2.4.7.2	Analog Input Modules
2.4.7.3	Discrete Output Modules
2.4.7.4	Analog Output Modules
2.4.7.5	Special Function Modules
3	FAULT ISOLATION
3.1	Failure Types
4	UCN EXERCISER
4.1	Overview
4.2	Applications
4.3	Request Data Reponse (RDR)Test
4.4	TriangulationTest
4.5	Test Operation
4.6	UCN Exerciser Test Procedure
4.7	Analyzing Results
5	UCN STATISTICS DISPLAYS
5.1	UCN Status Display
5.2	Logic Manager Detailed Status Display
5.3	Logic Manager Self Test Display
5.4	I/O System Status Display
5.5	Logic Manager Hardware Status Display
5.6	Revisions and Personality Display
5.7	Logic Manager Configuration Display

Table of Contents

5.8	UCN Stats Display
5.9	Maintenance Support Displays
5.9.1	Node Status Information Display
5.10	Soft Failure Display
6	DIAGNOSTICS
6.1	System Powerup Self-Test
6.2	Program Memory Check
6.3	On-Line Checks
6.4	I/O System Test
6.5	Monitoring Diagnostics
7	REMOVAL AND REPLACEMENT
7.1	Electrostatic Discharge Protection
7.2	Module Removal and Replacement
7.3	Battery Removal and Replacement
7.4	Card File Removal and Replacement
8	SPARE PARTS
8.1	Introduction
APPENDIX A	CALIBRATION PROCEDURES
A.1	Model 621-0010-A/V Analog Output Modules
A.1.1	Equipment Required
A.1.2	Calibration Adjustment Procedure

INTRODUCTION

Section 1

This section describes the purpose of this manual and lists all available manuals that may be required when servicing the Logic Manager System.

1.1 PURPOSE OF THIS MANUAL

This Logic Manager Service manual provides instructions for servicing the Logic Manager System. These instructions consist of fault isolation and removal/replacement procedures down to the Optimum Replaceable Unit (ORU) parts level.

1.2 RELATED PUBLICATIONS

The following Honeywell support documents are either required or will be of assistance in servicing the Logic Manager System.

Title	Binder
Logic Manager Site Planning	System Site Planning
Logic Manager Implementation Guidelines	Implementation/Logic Manager - 1
Logic Manager Installation	Implementation/Logic Manager - 1
Universal Control Network Site Planning	System Site Planning
LCN Site Planning	System Site Planning
LCN System Installation	LCN Installation
LCN System Checkout	LCN Installation

EQUIPMENT DESCRIPTION

Section 2

This section gives an overview of the Logic Manager System, the various configurations of the system, and a detailed description of the system components.

2.1 System Overview

The Logic Manager System (LM) performs digital and analog logic control functions as an integral component of the Honeywell TDC 3000^X Distributed Control System. The LM provides process automation for control, communication, monitoring of specific processes, and a peer-to-peer link with the Process Manager System (PM), or other LMs, on the Universal Control Network (UCN). Figure 2-1 is an overview of the network with Logic Manager and Process Manager Systems installed. The information necessary to install a Logic Manager System on a Universal Control Network can be found in the *Universal Control Network Site Planning* manual in the *System Site Planning* binder.

The Logic Manager System consists of a high-speed programmable controller processor, power supplies, option modules, serial and/or parallel I/O links to field devices, and a Logic Manager Module (LMM) that provides an interface between the processor and the UCN.

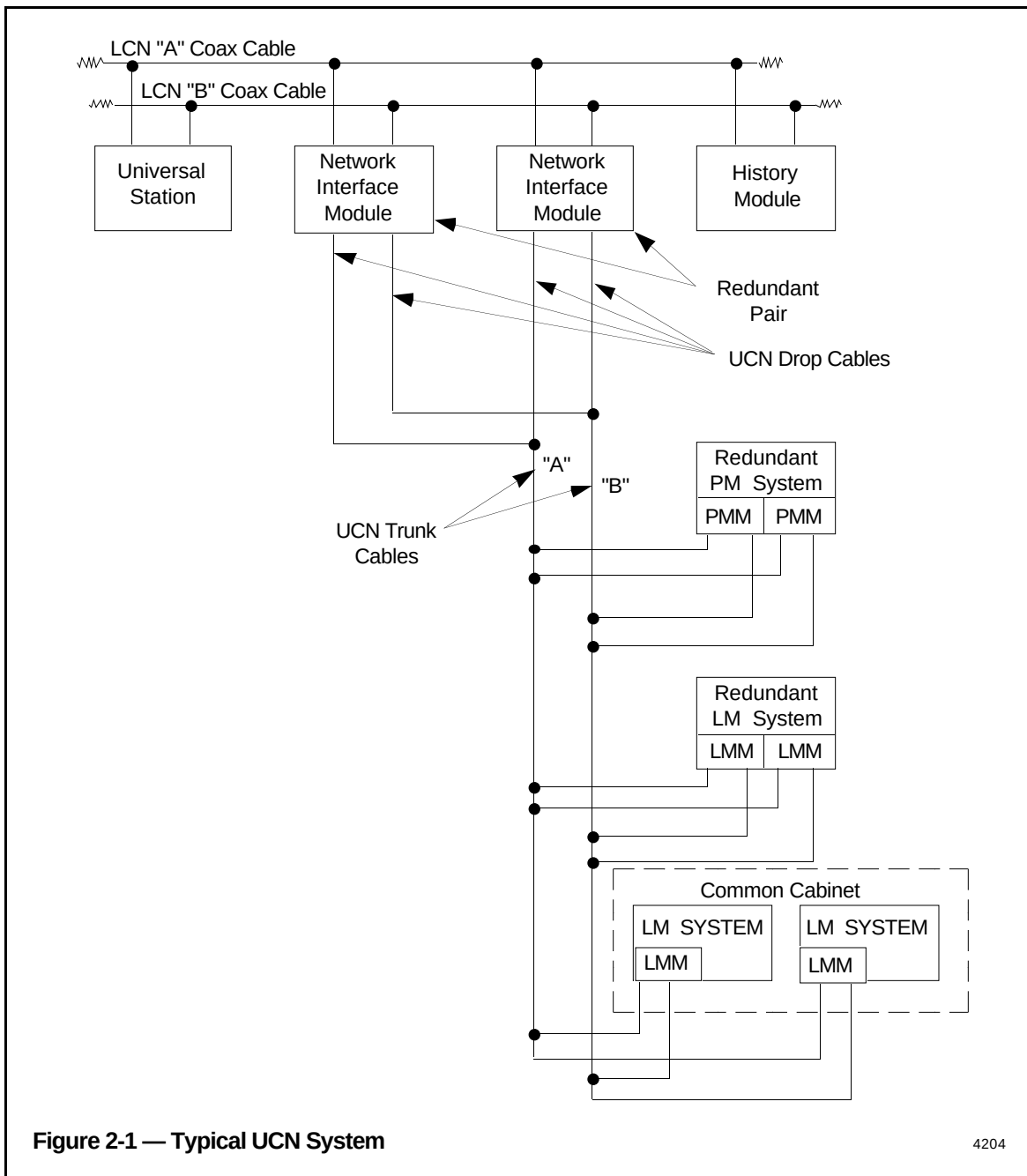
The LM is programmed with ladder logic using the model 623-60 MS-DOS Loader (refer to User Manual, number 623-8982 Rev. A, and 620-8983 Rev. C, 620 Redundant Control System User Manual). The Logic Manager Module in the Logic Manager Processor card file is configured via the Universal Station (US).

The Logic Manager System scans its parallel and/or serial I/O interface, executing the control program for I/O output devices, while providing input status collected from the I/O input devices. The Logic Manager System processes this information and converts it to UCN data format recognized by the TDC 3000^X interface.

The recommended Logic Manager System (LM) consists of a single or redundant Logic Manager Processor card file(s) (Honeywell 620-35 Processor hardware), and either, but not both, serial or parallel I/O card file(s) (Honeywell 621 I/O hardware). That is, serial and parallel I/O should not be mixed in the same “control group” of any Logic Manager System. However, although this serial/parallel mix is not recommended, it is technically permissible and can be accomplished with a “Special System.”

Failure in a parallel I/O system is potentially more hazardous than in a serial I/O system, in that there is a greater risk of causing total system failure in the parallel I/O architecture. For this reason, a redundant LM with parallel I/O is not recommended. Only serial I/O is supported with a redundant LM. A redundant LM with parallel I/O is a “Special System.”

Redundancy in the Logic Manager System involves a single I/O system controlled by dual (redundant) processors. The redundancy is managed by two Redundant Control Modules (RCM), one located in each Logic Manager Processor card file. Any optional modules that reside in one Logic Manager Processor card file must also be duplicated in the other Logic Manager Processor card file.



2.1.1 Logic Manager Processor

The Logic Manager Processor is composed of multi-modular units (cards) featuring remote serial and local parallel I/O capabilities. Its instruction set includes relay logic, timer, counter, and sequencer instructions in addition to four-function math, data handling capabilities, and comparisons. The memory capability is expandable to 32 K from the standard 24 K words of memory. With an I/O capacity of up-to-2048 points and an instruction set that includes matrices, subroutines, signed mathematical operations, and indirect addressing, the Logic Manager Processor is designed to service high speed and complex control applications.

Specifications for the Logic Manager Processor follow:

AC VOLTAGE	115/230 Vac $\pm$ 15% (selectable) 250 Vac max.
FREQUENCY	47 to 63 Hz
DC VOLTAGE	20 to 28 Vdc
POWER FAIL LEVEL	85 Vac for 115 Vac 190 Vac for 230 Vac 19 Vdc for 24 Vdc
POWER FAIL DELAY	11.5 ms (115 Vac)/7.0 ms (24 Vdc)
OPERATING TEMPERATURE	0° to 60°C ambient
STORAGE TEMPERATURE	-40° to +70°C
RELATIVE HUMIDITY	5% to 95% (noncondensing)
HEIGHT	27.2 cm (10.7 in.)
WIDTH	48.3 cm (19 in.)
DEPTH	19.1 cm (7.5 in.)
REAL I/O CAPACITY	2048
INTERNAL I/O CAPACITY (In Addition to Real I/O)	2048
DATA REGISTER	4096
REGISTER SIZE	17 bits
MEMORY SIZE (Words)	24 K standard, 8 K additional
MEMORY USAGE	1 word per ladder diagram element
MEMORY TYPE	CMOS RAM
BATTERY TYPE	Lithium – Size D in the Power Supply Module and size 2/3 A in the Logic Manager Module

BATTERY BACKUP (Minimum)	1.5 years
SCAN RATE (Relay Only)	2.5 ms/k
INDUSTRIAL RATINGS	NEMA-ICS2-203 IEC 2k V/5k V Isolation Test 255-4 ANSI C37.902a-1974 IEEE 518-1977 (Surge Withstand Test) UL 508 CSA C22.2-142-M1983 FM Class No. 3820

2.1.2 Logic Manager I/O System

The Logic Manager Input/Output (I/O) System offers several options for interfacing the Logic Manager Processor to the sensors and actuators of the plant floor. The sections that follow discuss the Parallel and Serial I/O Systems, which can provide three basic I/O configurations (local/parallel, remote/serial, or a combination of the two) for the Logic Manager Processor. The Logic Manager I/O System discussed is also an integral part of an additional Logic Manager System configuration, the Logic Manager Redundant Control System.

The Parallel and Serial I/O Systems include a wide variety of discrete, analog, special function I/O modules, and related I/O hardware (card files, power supplies, and I/O simulator assemblies).

Configurations and module types within the I/O system are determined by the specific application requirements and Logic Manager Processor capability.

The Logic Manager I/O System makes configuring a system for the Logic Manager both easy and cost effective. The I/O system offers a full range of 4-, 6-, 8-, 16-, or 32-point discrete, analog, and special function modules. The 19 inch I/O card file accommodates up to 12 I/O modules, the I/O card file Power Supply Module, and a serial or parallel interface module. A nonstandard half size (12 inch length) card file accommodates up to six I/O modules, the I/O card file Power Supply Module, and a serial or parallel interface module. Tables in subsection 2.4.7 of this manual offer a complete list of the currently available Process I/O modules.

The Input/Output System, an integral part of the Logic Manager System, provides the important physical link between the field devices and the control processor. This interface enables system inputs to be read and outputs to be controlled by the Logic Manager Processor.

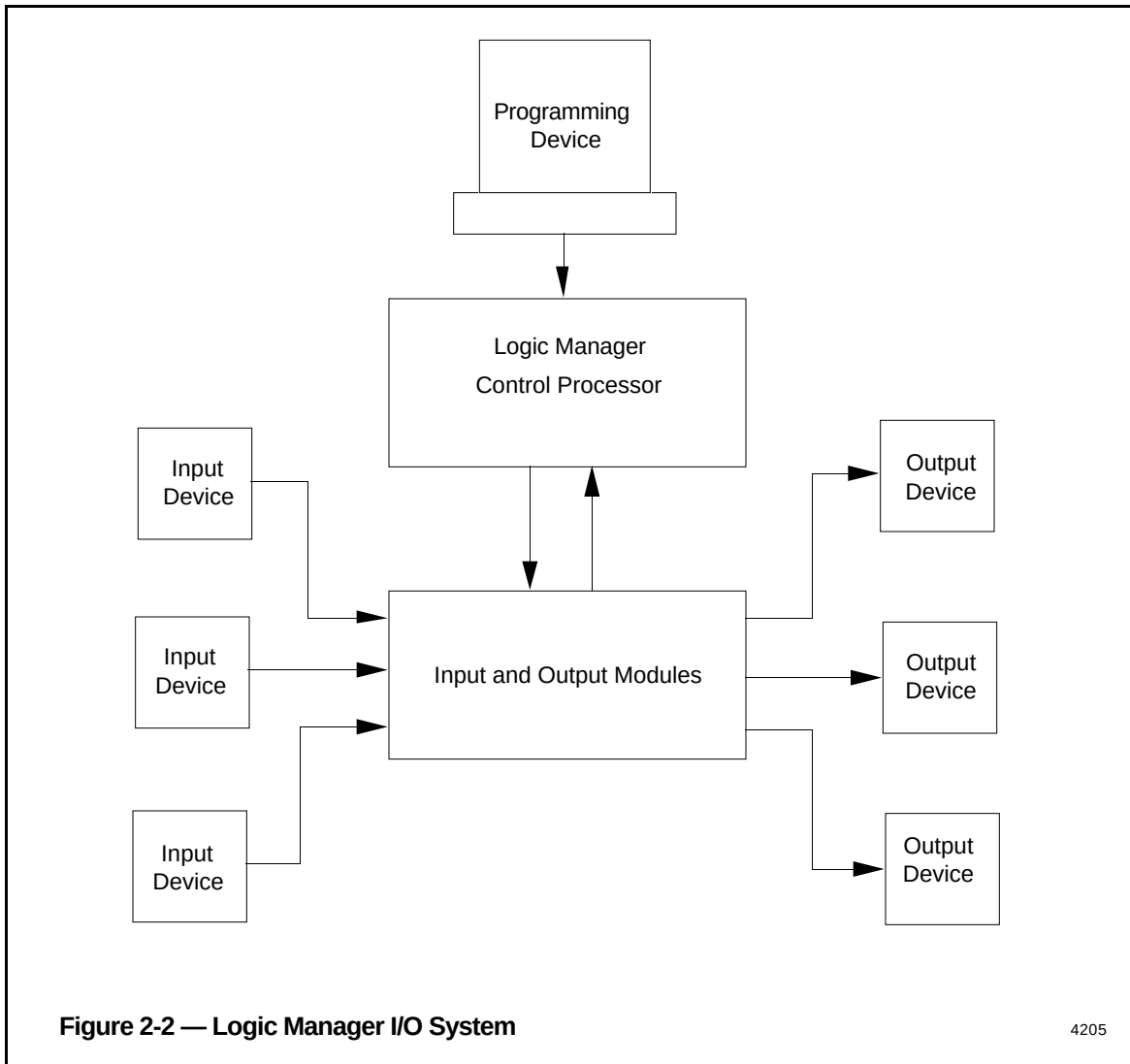
The I/O system consists of many different input and output modules with varying numbers of points and voltage types designed for compatibility with the many different kinds of field devices and power sources to which process I/O modules are connected. The Logic Manager Processor gathers and measures physical machine or process variables such as proximity (position of an object), pressure, temperature, fluid level, and voltage/current through the I/O interface. Based on these statuses or values, the Logic Manager Processor then issues commands to control the devices being monitored (e.g., alarms, control relays, lights, motors, pumps, and valves).

The discrete Input/Output interface is the most basic I/O interface and involves field devices (connected to I/O modules) that operate with distinct, single bit discrete signals (i.e., on/off, open/closed, extended/retracted). The state of these input signals is read by the I/O modules and is communicated to the Logic Manager Processor in a logic level format. Based on this input signal data, the processor then controls the system outputs as dictated by the user program being executed by the processor. Refer to Figure 2-2 for an illustration of the I/O interface.

Other more complex I/O interfaces include analog interfaces (the modules translate analog to digital or digital to analog signals) and multi-bit digital interfaces (multiple bits in parallel or serial formats, such as BCD inputs/outputs and pulse inputs/outputs). In most cases, these more complex I/O interfaces rely on PUSH/PULL communication between the I/O modules and the Logic Manager Processor. Some I/O systems may also contain I/O modules that perform specialized functions such as ASCII communication, system diagnostics, or high-speed control independent of the Logic Manager Processor.

The I/O system may be set up with the Logic Manager System in several configurations. A parallel configuration is used for local placement of the control processor and I/O. A serial configuration permits I/O to be located thousands of meters away from the processor. This configuration also minimizes field wiring. Logic Manager Processors can be configured as a “Special System” with a combination of parallel and serial I/O for maximum flexibility in a system layout, but this configuration is not recommended.

The I/O modules, with varying numbers of points (including the standard 8-, 16- and 32-point sizes) are designed for easy installation. The 8- and 16-point modules have terminal block sets that evenly split the I/O points between the top and bottom blocks of the module. These terminal block sets cut the size of each module's field wiring bundle in half. The blocks lock open without disturbing the wiring for easy module removal and replacement. The 32-point modules are equipped with easily removed connectors.



2.1.3 Serial I/O System

The remote serial I/O configuration allows the control system to operate over long distances. Serial I/O card files can be mounted close to the machine or process being controlled, eliminating long control wire runs. The serial I/O system may be set up in various multidrop configurations. See Figures 2-3 and 2-4 for two examples.

The key interface control modules in a serial configuration are the model 620-9939 Serial Link Module (SLM), and the model 621-9938R or 621-9940 Serial I/O Module (SIOM). One or two SLMs, which function as the masters for the serial links, may be placed in slots L and/or M of the Logic Manager Processor card file. The SIOM resides in the rightmost slot of each Serial I/O card file.

Each SLM sequentially communicates with every SIOM in its link. PUSH instructions and other output data are transmitted by the SLM to each SIOM, while PULL instructions and other input data are transmitted by each SIOM back to the SLM.

The serial I/O system requires two twisted pair cables, each pair individually shielded. The maximum cable length per channel is 3000 meters (10,000 feet).

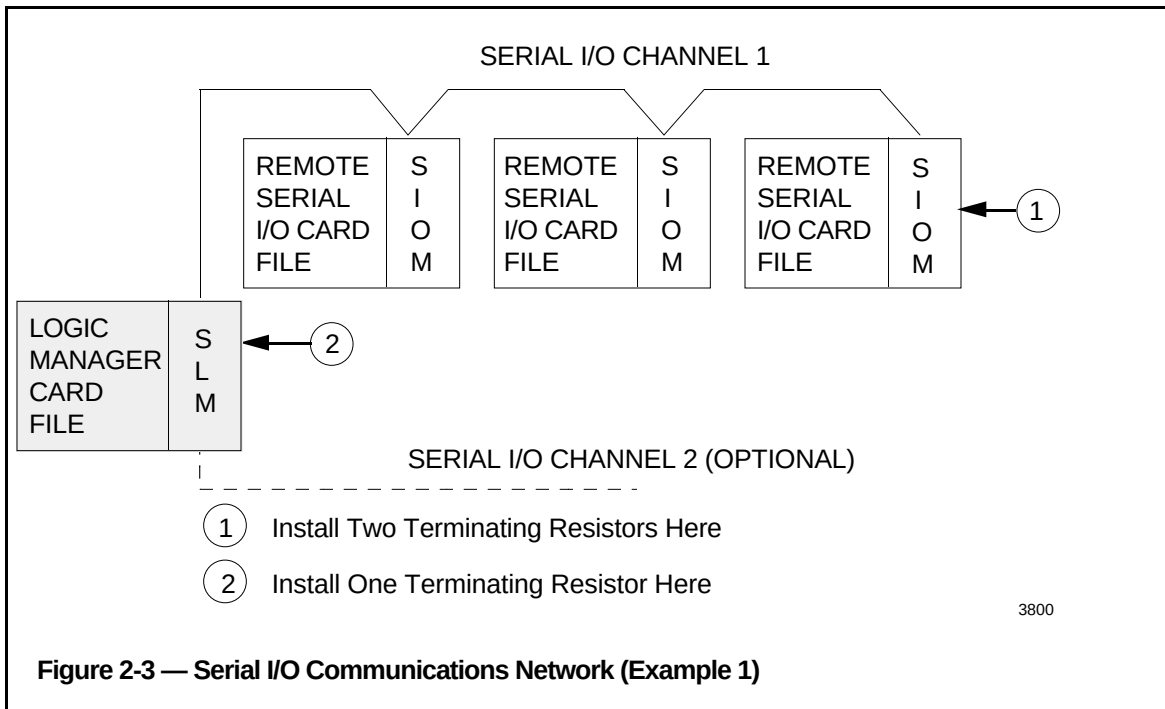
Specifications for the Logic Manager Serial I/O System follows:

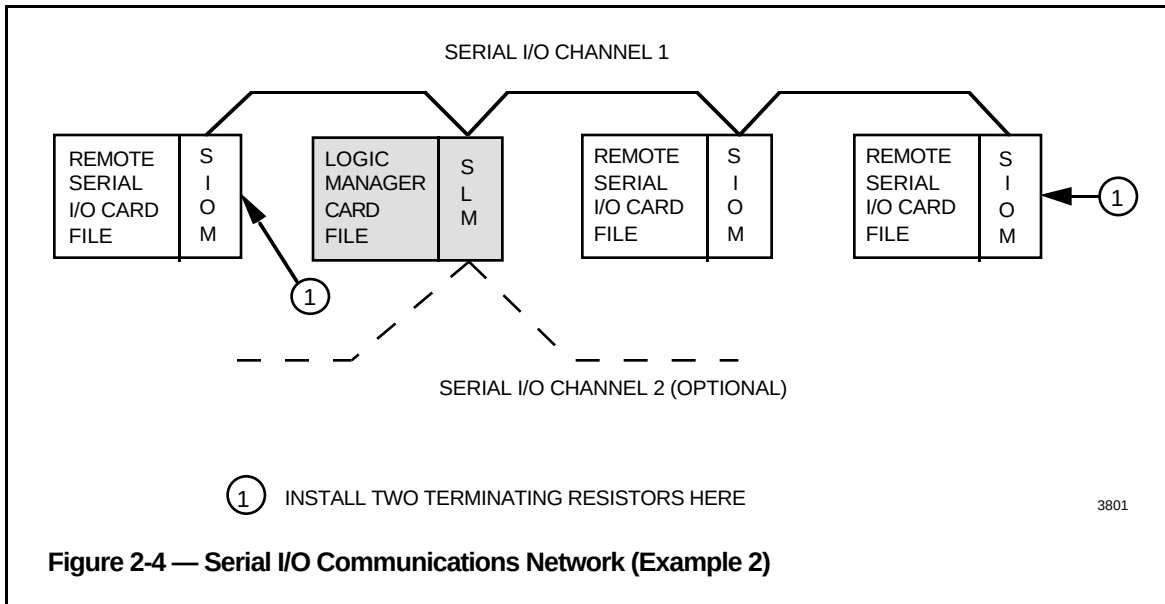
DATA TRANSFER MEDIUM	Serial 4-wire full duplex
DATA RATE	115.2 k baud
ELECTRICAL FORMAT	RS422
SERIAL CHANNEL SCAN RATE	Asynchronous to the processor scan. Dependent on the number of card files per link, number of card files containing PUSH/PULL modules, and the number of PUSH/PULL modules updated per scan (all or 1 per scan)
MAXIMUM DROPS PER CHANNEL	16
MAXIMUM I/O PER CHANNEL	2040
MAXIMUM CHANNEL LENGTH	1200 m /4000 ft. (Belden 9729, use 200 ohm terminating resistor *) or 3000 m /10,000 ft. (Belden 9182, use 300 ohm terminating resistor *)
ERROR CHECKING	Cyclical redundancy check plus data receive timeout
OPERATING CURRENT	621-9938R 2.06 A (+5 Vdc typical) 621-9939 2.60 A (+5 Vdc typical) 621-9940 2.03 A (+5 Vdc typical)

*Supplied with the 621-9939 module (SLM).

NOTE

Refer to Section 5 of the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder for cabling information and for rules regarding installation of terminating resistors.





2.1.4 Parallel I/O System

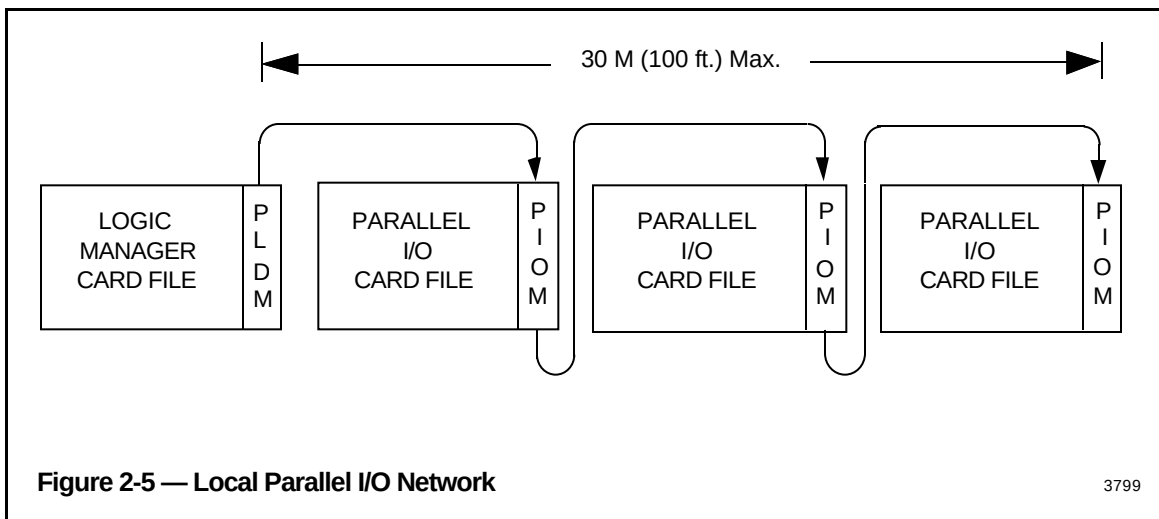
The local Parallel I/O System provides I/O update in real time. Up to 22 card files may be daisy chained in a Parallel I/O System layout. A parallel transmission length of up to 30 meters (100 feet) is allowed.

A typical parallel I/O layout is shown in Figure 2-5. The key interface modules in a parallel I/O configuration are the model 620-0088 Parallel Link Driver Module (PLDM) and the model 621-9937 Parallel I/O Module (PIOM). As shown in the figure, the first I/O card file in the local Parallel I/O System is connected to the D-type connector at the Parallel Link Driver Module. A second connector at the PIOM then connects to the next PIOM in the parallel link.

The Parallel I/O Module, installed in the rightmost slot of each I/O card file in the parallel I/O configuration, operates as an interface to the PLDM in the Logic Manager Processor card file and to the other PIOMs in other parallel I/O card files. The PIOM has two D-type connectors. The IN connector (male) is positioned at the upper portion of the module front plate, and the OUT connector (female) is positioned at the lower portion.

NOTE

Refer to Section 5 of the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder for cabling information, including model numbers for standard cables.



2.1.5 Redundant Control System

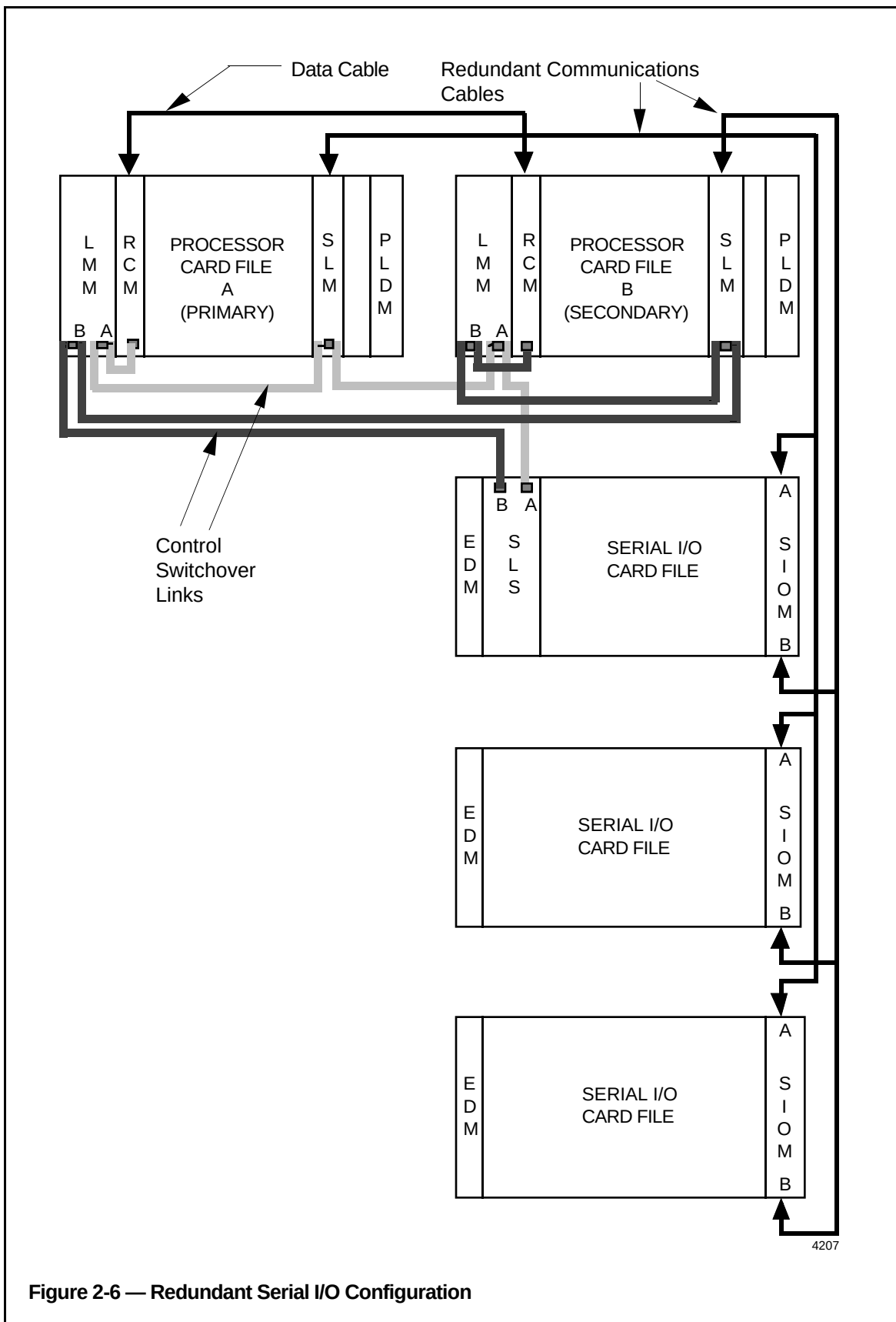
The Logic Manager Redundant Control System provides backup for the processors in the Logic Manager system. The redundancy system consists of two identically configured Logic Manager processors, each containing a model 620-0059 Redundancy Control Module (RCM). The RCMs are linked by a parallel data cable and initiate an orderly switchover of system I/O from the lead (primary) to the backup (secondary) processor if a critical processor or serial link fault occurs.

Both processors are connected to a common I/O system that may be parallel I/O, serial I/O, or a combination of the two. Serial I/O systems only are recommended. See Figure 2-6.

Serial I/O redundancy is accomplished through the model 621-9938R redundant Serial I/O Module (SIOM) and the model 621-9939 Serial Link Module (SLM). The optional model 621-9928R Serial Link Selector (SLS) is available to switch a communication link to the active processor.

CAUTION

Refer to the Honeywell 620 Redundant Control System User Manual, number 620-8983, for a proper understanding of a redundant system before maintenance is performed. Refer to the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder for detailed cabling information and for rules covering the installation of terminating resistors for the redundant system switchover communication links.



2.2 Equipment Overview

A Logic Manager System, which consists of Logic Manager Processor and I/O card files, can have multiple physical configurations. The system may reside in a single cabinet and the cabinet may be single access or dual access. The system may also reside in a complex of up to four dual access cabinets. However, part of the system may reside in the cabinet or cabinets, while the remainder of the system, the I/O portion, may be positioned externally, and/or remotely, in other cabinets, or panel mounted. In any case, redundant Logic Manager Processor card files must reside together.

2.2.1 Processor Card Files

The Logic Manager Processor card file is a 19 inch rack which may be frame or panel mounted. The card file includes the chassis, backplane, processor modules, I/O interface modules, a Power Supply Module, and front cover plates for the unused optional module slots. There are fourteen module slots, identified as A through N, with slot A on the left when facing the card file. The Logic Manager Processor modules are vertically positioned in the card file with the component side toward the left. The backplane connectors are offset to prevent inserting a module upside down.

The Logic Manager Processor card file has two configurations, nonredundant and redundant. Standard modules are required for both configurations. Other modules are optional and depend on the system requirement. In either case, slot positioning of the modules is defined for the configuration.

The following subsections define the two configurations of Logic Manager Processor card files and the additional option modules available for slots C and D.

2.2.1.1 Nonredundant Processor Card File

The basic nonredundant Logic Manager Processor card file contains the standard card modules listed below and is designated model ML-LMSA35 if it has an ac power supply module in slots J and K. If it has a dc power supply module in slots J and K, it is designated model ML-LMSD35. See Figure 2-7 for a front view of the card file and Table 2-1 for the card slot assignments. Option modules listed in subsection 2.2.1.3 can be added to the card file as required.

Table 2-1 — Nonredundant Processor Card File Slot Assignments

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A and B	Logic Manager Module (LMM)	620-0071	6, 8
C	Blank Metal Cover Plate	629-4004	1, 2, 4
D	Blank Metal Cover Plate	629-4004	1, 2, 4, 9
E	24 K Memory Module	620-0024	8
F	4 K x 4 K Register Module	620-0056	
G	System Control Module	620-0054	
H	Processor Module	620-0080	
I	2 K I/O Control Module	620-0085	
J and K	Power Supply Module (PSM) 115/230 Vac	620-0036	6
J and K	Power Supply Module (PSM) 24 Vdc	620-0047	6
L	Serial Link Module (SLM)	621-9939	8
M	Blank Plastic Cover Plate	621-9900	2, 3, 5
N	Parallel Link Driver Module (PLDM)	620-0088	7, 8

- Notes:
1. This slot is for option modules.
 2. Check DIP switch settings for option modules (see subsection 2.3).
 3. This slot is for the optional second model 621-9939 Serial Link Module.
 4. Install the model 629-4004 blank metal cover plate if this slot is empty.
 5. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 6. The module is double width.
 7. The Parallel Link Driver Module is required even when serial I/O is used exclusively because the PLDM performs some system control functions in addition to supporting parallel I/O.
 8. This module has its configuration DIP switches preset at the factory.
 9. The optional model 620-0027, 8 K Memory Module may be installed in this slot.

The Logic Manager Module has UCN drop cable connectors mounted on the front panel.

DIP switch settings on option modules added to the card file must be set at the time of installation. As two option slots are available, the option modules must be set to backplane address 2 or 3. If two option modules are implemented, one must be set to address 2 and the other set to address 3. The Logic Manager Module is factory preset to address 1.

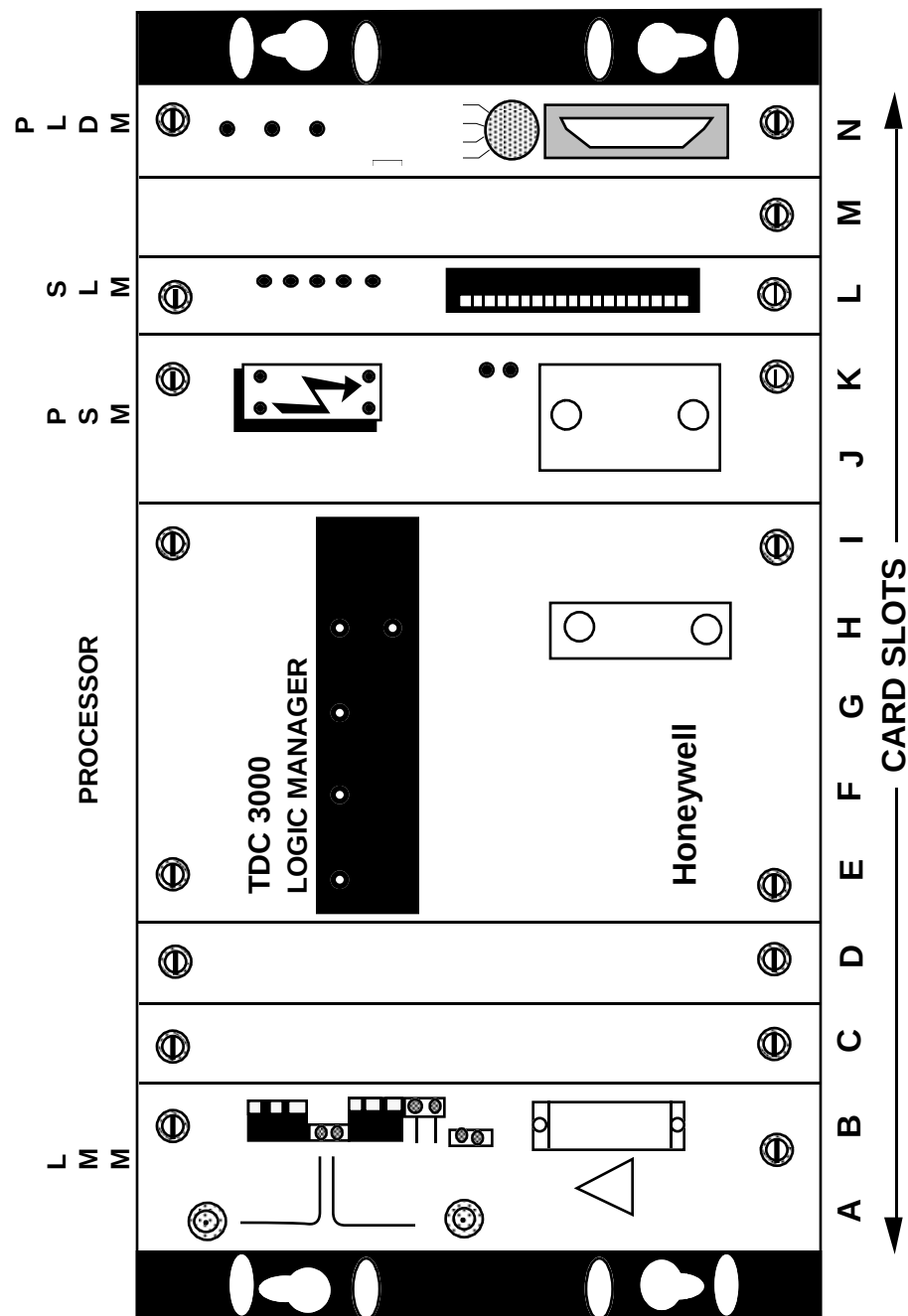


Figure 2-7 — Nonredundant Processor Card File Front View

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2.2.1.2 Redundant Processor Card File

A basic redundant Logic Manager System has two processor card files with standard card modules installed, as shown in Table 2-2 and Figure 2-8, and is designated a model ML-MLRA35 if it has an ac Power Supply Module in slots J and K. If a dc Power Supply Module is installed in slots J and K, the designation is model ML-LMRD35. A three meters (10 feet) long model 628-5000 Redundancy Data Control Cable is also included to connect the Redundancy Control Modules in each card file. Option modules listed in subsection 2.2.1.3 can be added to the card file as required.

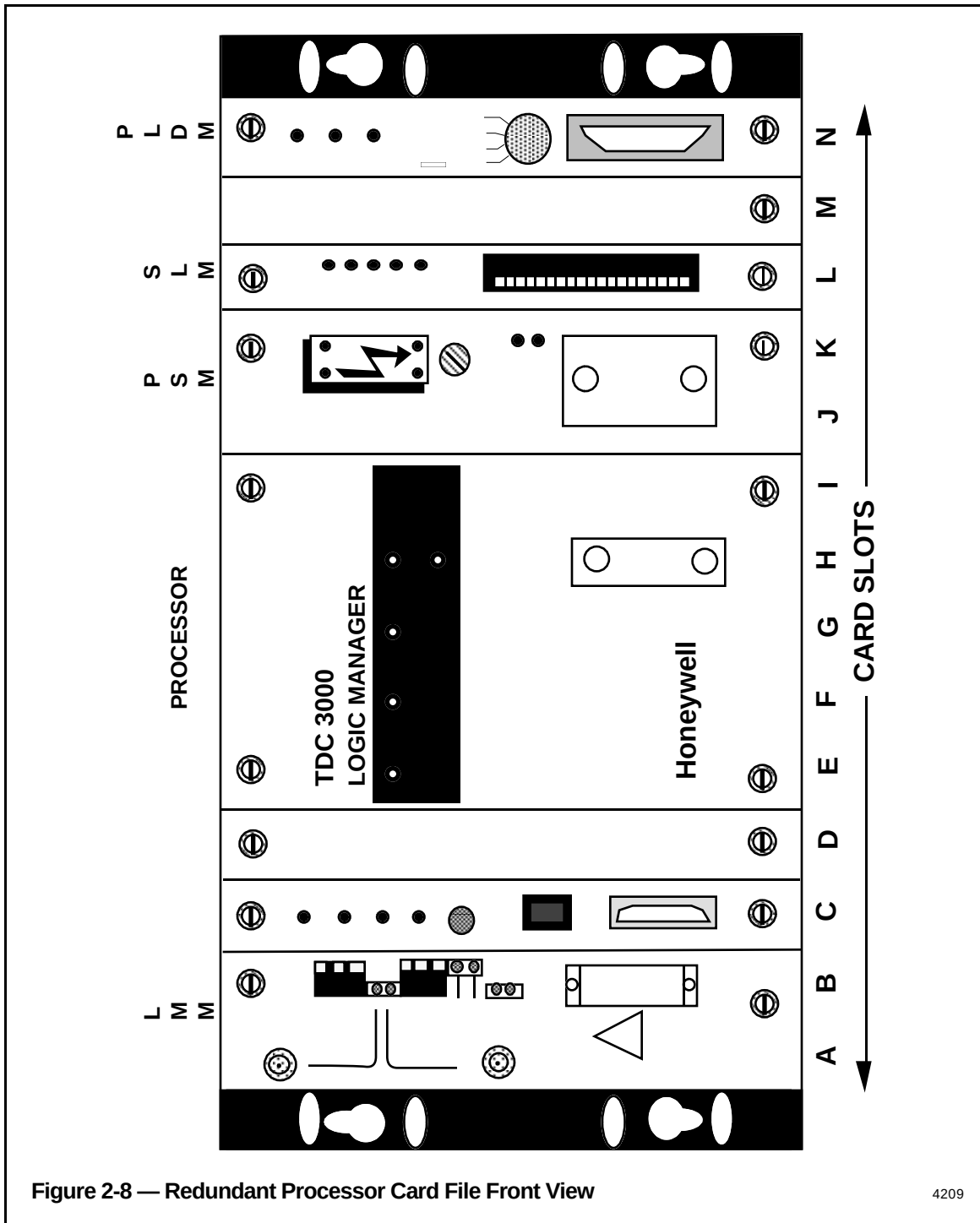
Table 2-2 — Redundant Processor Card File Slot Assignments

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A and B	Logic Manager Module (LMM)	620-0071	6, 8
C	Redundancy Control Module (RCM)	620-0059	8
D	Blank Metal Cover Plate	629-4004	1, 2, 4, 9
E	24 K Memory Module	620-0024	8
F	4 K x 4 K Register Module	620-0056	
G	System Control Module	620-0054	
H	Processor Module	620-0080	
I	2 K I/O Control Module	620-0085	
J and K	Power Supply Module (PSM) 115/230 Vac	620-0036	6
J and K	Power Supply Module (PSM) 24 Vdc	620-0047	6
L	Serial Link Module (SLM)	621-9939	8
M	Blank Plastic Cover Plate	621-9900	2, 3, 5
N	Parallel Link Driver Module (PLDM)	620-0088	7, 8

- Notes:
1. This slot is for option modules.
 2. Check DIP switch settings for option modules (see subsection 2.3).
 3. This slot is for the optional second model 621-9939 Serial Link Module.
 4. Install the model 629-4004 blank metal cover plate if this slot is empty.
 5. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 6. The module is double width.
 7. The model 620-0088 Parallel Link Driver Module is required even when serial I/O is used exclusively because the PLDM performs some system control functions in addition to supporting parallel I/O.
 8. This module has its configuration DIP switches preset at the factory.
 9. The model 620-0027 8 K Memory Module may be installed in this slot.

The Logic Manager Module has UCN drop cable connectors mounted on the front panel.

The Logic Manager Module is factory preset to address 1 at the factory. The Redundancy Control Module is preset to address 0 at the factory. DIP switch settings on option modules added to the card file must be set at the time of installation. As only one option slot is available, the option module must be set to backplane address 2 or 3.



2.2.1.3 Processor Card File Option Modules

The option modules listed in Table 2-3 may be used in the option slots C and D of the nonredundant processor card file or option slot D of the redundant processor card file as shown.

Table 2-3 — Processor Card File Option Modules

MODULE DESCRIPTION	MODEL NUMBER	REDUNDANT CARD SLOT	NONREDUNDANT CARD SLOT
8K Memory	620-0027	D	D
Control Network	620-0038	D	C or D
Modbus/RTU Computer Interface	620-0043	D	C or D
ABC Data Collection	620-0048	D	C or D
Dual Port ABC Data Collection	620-0052	D	C or D

If an optional communications module is used in a redundant serial I/O system, each of the redundant processor card files must contain an identical option module. Each option module port must be linked with an additional model 620-9928R Serial Link Selector Module in a serial I/O card file.

To utilize a model 620-0048 ABC Data Collection Module or model 620-0043 Modbus Computer Interface Module in a redundant Logic Manager System, a DCM or CIM must be installed in both processor card files and connected to an additional SLS in an I/O card file.

To utilize a model 620-0052 Dual Port ABC Data Collection Module, an identical DCM must be installed in both processor card files and connected to two additional SLS modules in an I/O card file.

2.2.2 I/O Card Files

The Logic Manager I/O card file has two sizes, the standard 19 inch and the 12 inch half-rack size. Both sizes may be panel mounted, but only the 19 inch size is cabinet mounted. The card file includes the chassis, backplane, I/O interface modules, process I/O modules, a Power Supply Module, and front cover plates for the unused module slots. The 19 inch I/O card file has fourteen module slots, identified as A through N, with slot A on the left side when facing the card file. The half size “Special” 12 inch card file has eight module slots, identified as A through H. A maximum of twelve and six slots, respectively, are available for process I/O modules. The I/O modules are vertically positioned in the card file with the component side toward the left. The backplane connectors are offset to prevent inserting a module upside down.

The Logic Manager I/O card file has two configuration types, serial and parallel. Standard modules are required for both configurations. Other modules are optional and depend on the system requirement. In either case, slot positioning of the modules is defined by the configuration.

The following sections define the different configuration types of Logic Manager I/O card files.

2.2.2.1 Nonredundant Serial I/O Card Files

The basic full size nonredundant Serial I/O Card File contains the standard card modules shown below and is designated model ML-LIOS01. Figure 2-9 is a view of the front of the card file and Table 2-4 defines the module slot assignments.

Table 2-4 — Full Size Nonredundant Serial I/O Card File Slot Assignment

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A	Enhanced Diagnostic Module (EDM)	621-0021	1, 3
B	Any Valid 621-xxxx I/O Module		2, 3, 4
C	Any Valid 621-xxxx I/O Module		2, 3, 4
D	Any Valid 621-xxxx I/O Module		2, 3, 4
E	Any Valid 621-xxxx I/O Module		2, 3, 4
F	Any Valid 621-xxxx I/O Module		2, 3, 4
G	Any Valid 621-xxxx I/O Module		2, 3, 4
H	Any Valid 621-xxxx I/O Module		2, 3, 4
I	Any Valid 621-xxxx I/O Module		2, 3, 4
J	Any Valid 621-xxxx I/O Module		2, 3, 4
K	Any Valid 621-xxxx I/O Module		2, 3, 4
L	Any Valid 621-xxxx I/O Module		2, 3, 4, 6
L and M	Power Supply Module (PSM) 115/230 Vac 15 A	621-9933	2, 5
M	Power Supply Module (PSM) 115/230 Vac 8 A	621-9934	2
M	Power Supply Module (PSM) 24 Vdc 8 A	621-9932	2
N	Serial I/O Module (SIOM)	621-9940	1, 3

Notes: 1. This is a standard card module slot.
 2. This is an option card module slot.
 3. Check DIP switch settings if applicable (see subsection 2.4).
 4. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 5. The module is double width.
 6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.

The basic half size nonredundant Serial I/O Card File contains the standard card modules shown in Table 2-5 and is designated model 621-9991. It is not a standard product, but a “Special,” and is described for reference only.

Table 2-5 — Half Size Nonredundant Serial I/O Card File Slot Assignments

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A	Enhanced Diagnostic Module (EDM)	621-0021	1, 3
B	Any Valid 621-xxxx I/O Module		2, 3, 4
C	Any Valid 621-xxxx I/O Module		2, 3, 4
D	Any Valid 621-xxxx I/O Module		2, 3, 4
E	Any Valid 621-xxxx I/O Module		2, 3, 4
F	Any Valid 621-xxxx I/O Module		2, 3, 4, 6
F and G	Power Supply Module (PSM) 115/230 Vac 15 A	621-9933	2, 5
G	Power Supply Module (PSM) 115/230 Vac 8 A	621-9934	2
G	Power Supply Module (PSM) 24 Vdc 8 A	621-9932	2
H	Serial I/O Module (SIOM)	621-9940	1, 3

- Notes:
1. This is a standard card module slot.
 2. This is an option card module slot.
 3. Check DIP switch settings if applicable (see subsection 2.4).
 4. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 5. The module is double width.
 6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.

All low level signal process I/O modules (analog, TTL, PIM, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal process I/O modules must be grouped together, starting at the rightmost slot and proceeding left.

All I/O slots must be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the slot and configured by the appropriate Serial I/O Module configuration DIP switches. Each empty slot must be assigned a quantity of 16 addresses to accommodate spares, allowing for future expansion. The only slots to be assigned an address of zero are those slots occupied by the left half of a double width module and both slots occupied by a double width module that requires no addresses.

I/O addresses 2040 through 2047 must not be assigned to any parallel or serial I/O slot of a system containing serial I/O.

All empty I/O slots must be covered with a model 621-9900 plastic cover plate.

The Power Supply Module selected for the card file must be compatible with the power input (115/230 Vac or 24 Vdc) and must supply enough current to the card file to satisfy the total load requirement.

All systems configured with a MiniCOP module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file option slots. The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP without a DCM.

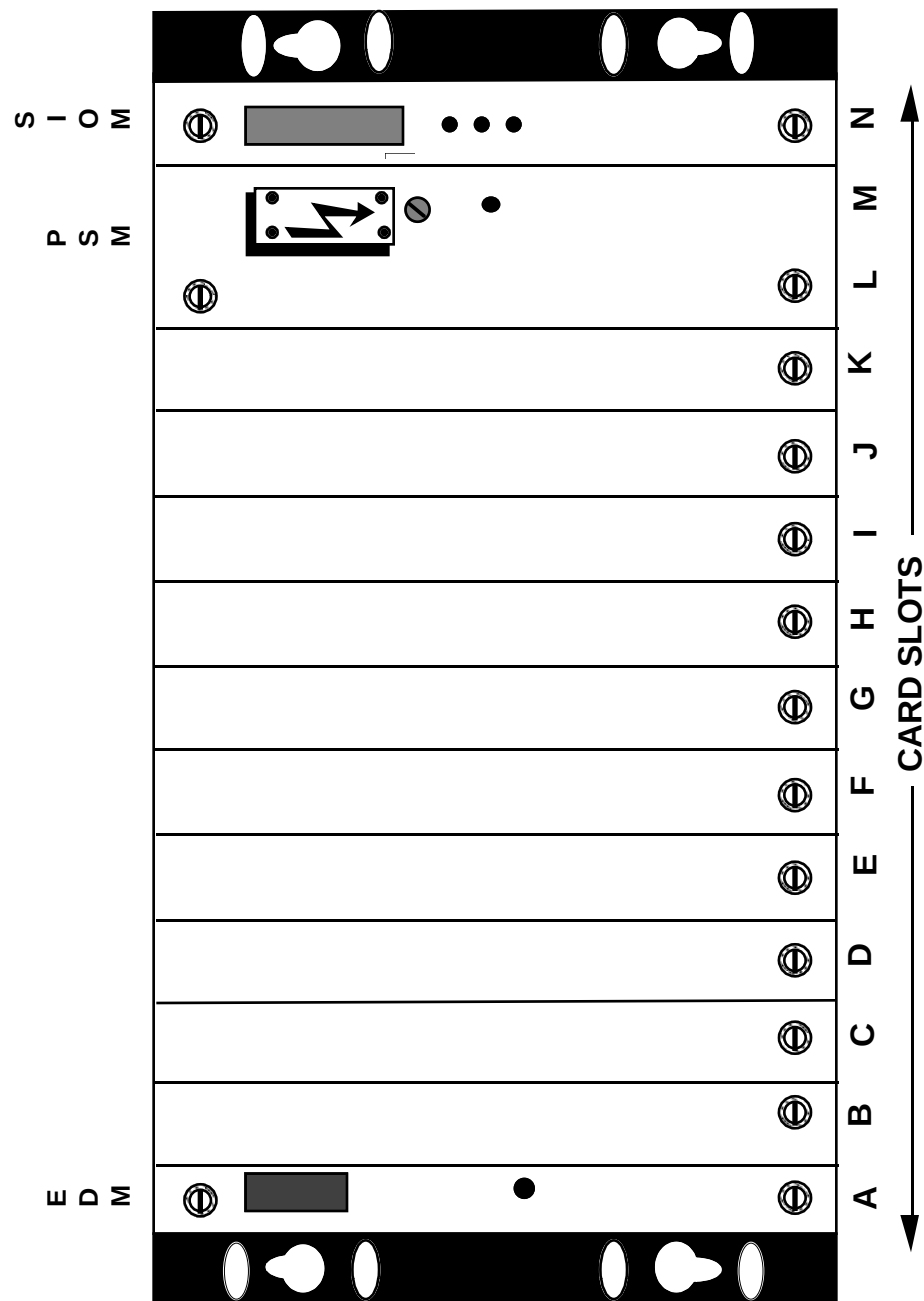


Figure 2-9 — Nonredundant Serial I/O Card File Front View

4210

2.2.2.2 Redundant Serial I/O Card File

The basic redundant Serial I/O Card File contains the standard card modules shown below. Figure 2-10 is a view of the front of the card file and Table 2-6 defines the module slot assignments.

Table 2-6 — Redundant Serial I/O Card File Slot Assignments

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A	Enhanced Diagnostic Module (EDM)	621-0021	1, 3
B and C	Serial Link Selector Module (SLS)	621-9928R	1, 3, 4
D	Any Valid 621-xxxx I/O Module		2, 3, 5
E	Any Valid 621-xxxx I/O Module		2, 3, 5
F	Any Valid 621-xxxx I/O Module		2, 3, 5
G	Any Valid 621-xxxx I/O Module		2, 3, 5
H	Any Valid 621-xxxx I/O Module		2, 3, 5
I	Any Valid 621-xxxx I/O Module		2, 3, 5
J	Any Valid 621-xxxx I/O Module		2, 3, 5
K	Any Valid 621-xxxx I/O Module		2, 3, 5
L	Any Valid 621-xxxx I/O Module		2, 3, 5, 7
L and M	Power Supply Module (PSM) 115/230 Vac 15 A	621-9933	2, 6
M	Power Supply Module (PSM) 115/230 Vac 8 A	621-9934	2
M	Power Supply Module (PSM) 24 Vdc 8 A	621-9932	2
N	Serial I/O Module (SIOM)	621-9938R	1, 3, 8

- Notes:
1. This is a standard card module slot.
 2. This is an option card module slot.
 3. Check the DIP switch settings if applicable (see subsection 2.4).
 4. At least one SLS is required for a redundant system. Additional I/O card files may contain a SLS. Otherwise, the slot can contain any valid 621-xxxx I/O module.
 5. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 6. The module is double width.
 7. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.
 8. Model 621-9938R has dual ports for processor redundancy.

All low level signal process I/O modules (analog, TTL, PIM, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal processor I/O modules must be grouped together, starting at the rightmost slot and proceeding left.

All I/O slots must be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the slot and configured by the appropriate Serial I/O Module configuration DIP switches. Each empty slot must be assigned a quantity of 16 addresses to accommodate spares, allowing for future expansion. The only slots to be assigned an address of zero are those slots occupied by the left half of a double width module and both slots occupied by a double width module that requires no addresses.

I/O addresses 2040 through 2047 must not be assigned to any parallel or serial slot of a system containing serial I/O.

All empty I/O slots must be covered with a model 621-9900 plastic cover plate.

The Power Supply Module selected for the card file must be compatible with the power input (115/230 Vac or 24 Vdc) and must supply enough current to the card file to satisfy the total load requirement.

All systems configured with a MiniCOP module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file option slots. The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP without a DCM.

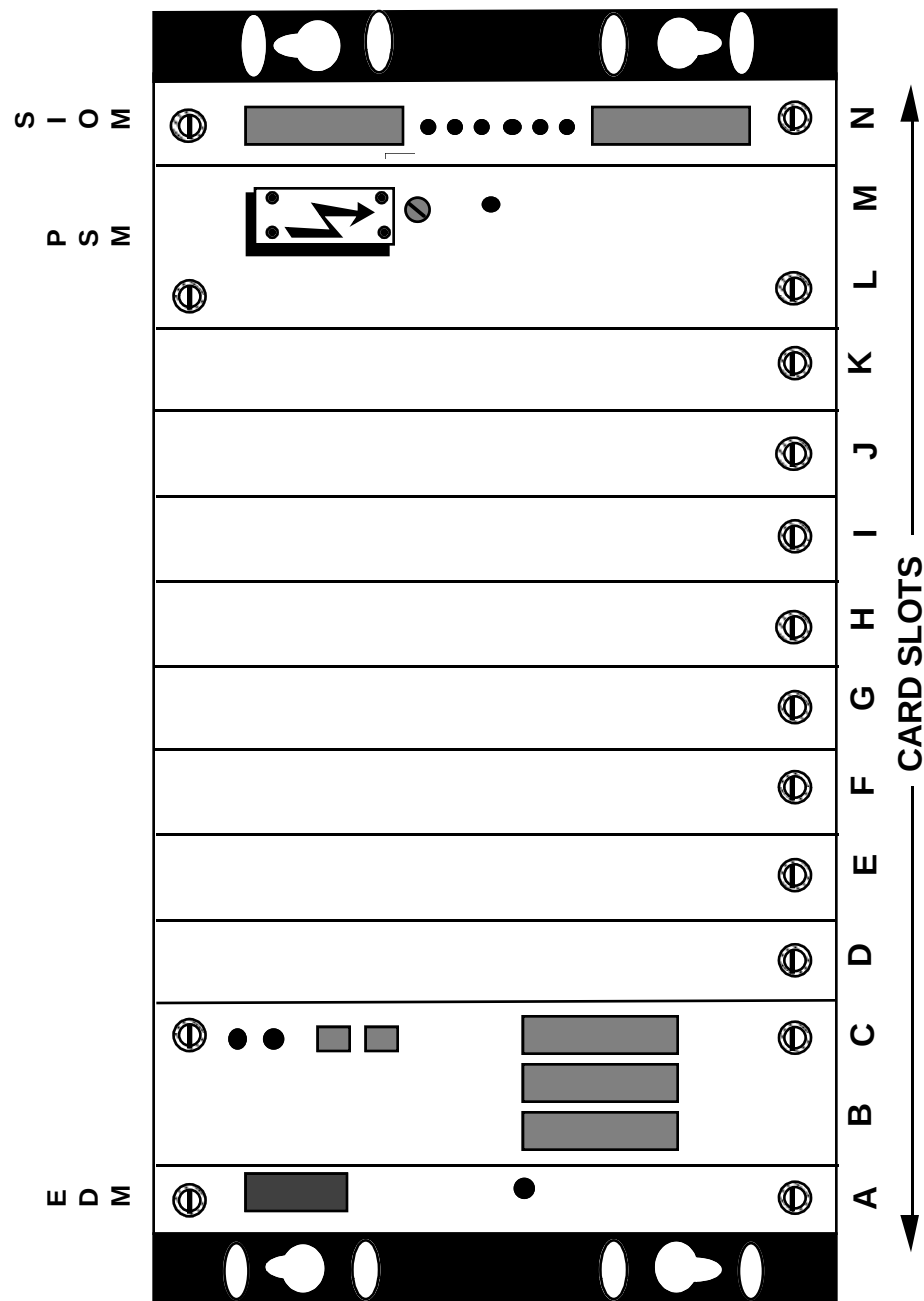


Figure 2-10 — Redundant Serial I/O Card File Front View

4211

2.2.2.3 Nonredundant Parallel I/O Card Files

The basic full size nonredundant Parallel I/O Card File contains the standard card modules as shown in Table 2-7 and is designated model ML-LIOP01. See Figure 2-11 for a front view of the card file.

Table 2-7 — Full Size Nonredundant Parallel I/O Card File Slot Assignments

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A	Enhanced Diagnostic Module (EDM)	621-0021	1, 3
B	Any Valid 621-xxxx I/O Module		2, 3, 4
C	Any Valid 621-xxxx I/O Module		2, 3, 4
D	Any Valid 621-xxxx I/O Module		2, 3, 4
E	Any Valid 621-xxxx I/O Module		2, 3, 4
F	Any Valid 621-xxxx I/O Module		2, 3, 4
G	Any Valid 621-xxxx I/O Module		2, 3, 4
H	Any Valid 621-xxxx I/O Module		2, 3, 4
I	Any Valid 621-xxxx I/O Module		2, 3, 4
J	Any Valid 621-xxxx I/O Module		2, 3, 4
K	Any Valid 621-xxxx I/O Module		2, 3, 4
L	Any Valid 621-xxxx I/O Module		2, 3, 4, 6
L and M	Power Supply Module (PSM) 115/230 Vac 15 A	621-9933	2, 5
M	Power Supply Module (PSM) 115/230 Vac 8 A	621-9934	2
M	Power Supply Module (PSM) 24 Vdc 8 A	621-9932	2
N	Parallel I/O Module (PIOM)	621-9937	1, 3

Notes: 1. This is a standard card module slot.
 2. This is an option card module slot.
 3. Check DIP switch settings if applicable (see subsection 2.4).
 4. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 5. The module is double width.
 6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.

The basic half size nonredundant Parallel I/O Card File contains the standard card modules as shown in Table 2-8 and is designated model 621-9991. It is not a standard product, but a “Special,” and is described for reference only.

Table 2-8 — Half Size Nonredundant Parallel I/O Card File Slot Assignments

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A	Enhanced Diagnostic Module (EDM)	621-0021	1, 3
B	Any Valid 621-xxxx I/O Module		2, 3, 4
C	Any Valid 621-xxxx I/O Module		2, 3, 4
D	Any Valid 621-xxxx I/O Module		2, 3, 4
E	Any Valid 621-xxxx I/O Module		2, 3, 4
F	Any Valid 621-xxxx I/O Module		2, 3, 4, 6
F and G	Power Supply Module (PSM) 115/230 Vac 15 A	621-9933	2, 5
G	Power Supply Module (PSM) 115/230 Vac 8 A	621-9934	2
G	Power Supply Module (PSM) 24 Vdc 8 A	621-9932	2
H	Parallel I/O Module (PIOM)	621-9937	1, 3

Notes: 1. This is a standard card module slot.
 2. This is an option card module slot.
 3. Check DIP switch settings if applicable (see subsection 2.4).
 4. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 5. The module is double width.
 6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.

All low level signal process I/O modules (analog, TTL, PIM, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal process I/O modules must be grouped together, starting at the rightmost slot and proceeding left.

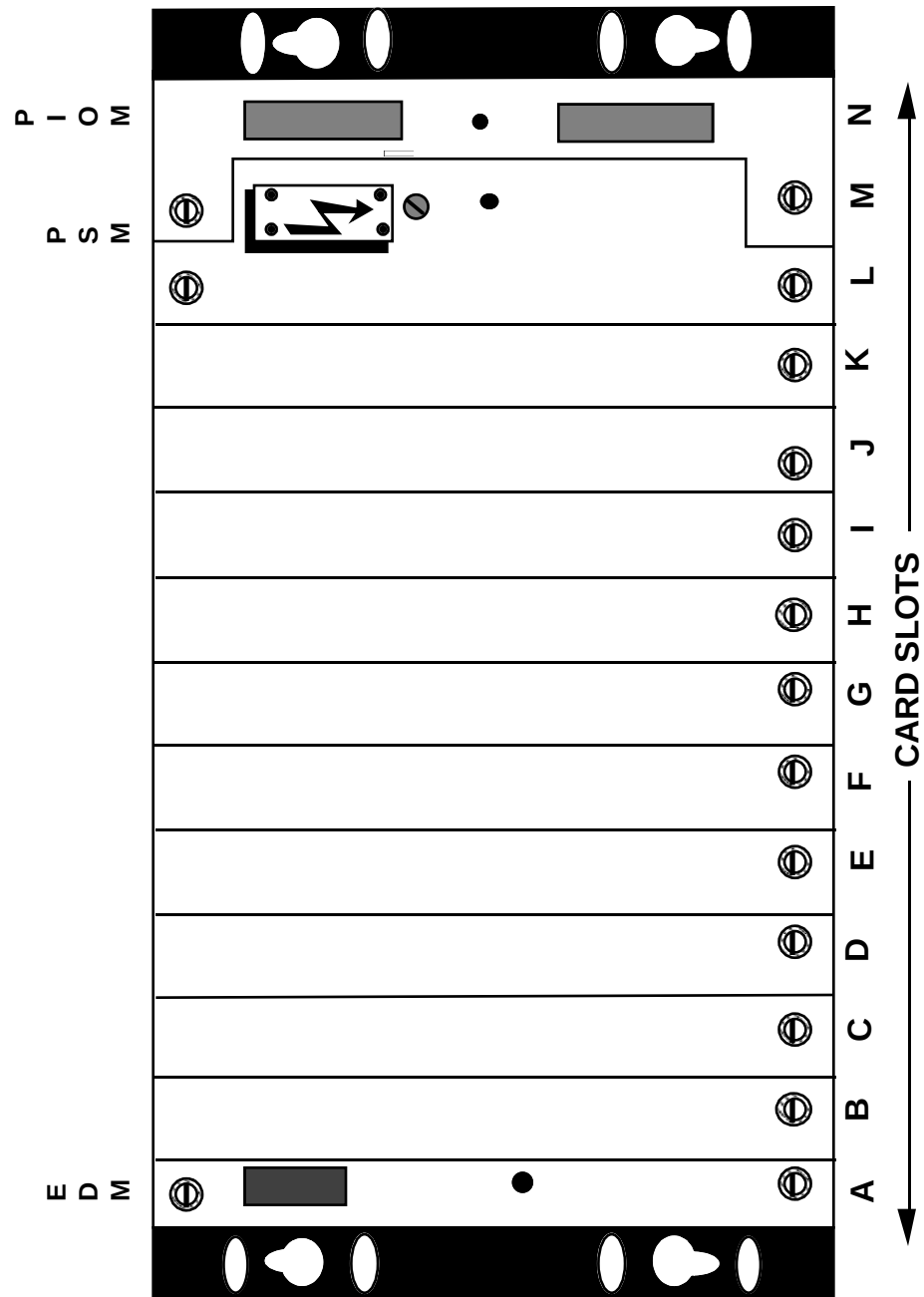
All I/O slots must be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the slot and configured by the appropriate Parallel I/O Module configuration DIP switches. Each empty slot must be assigned a quantity of 16 addresses to accommodate spares, allowing for future expansion. The only slots to be assigned an address of zero are those slots occupied by the left half of a double width module and both slots occupied by a double width module that requires no addresses.

I/O addresses 2040 through 2047 must not be assigned to any parallel or serial I/O slot of a system containing serial I/O.

All empty I/O slots must be covered with a model 621-9900 plastic cover plate.

The Power Supply Module selected for the card file must be compatible with the power input (115/230 Vac or 24 Vdc) and must supply enough current to the card file to satisfy the total load requirement.

All systems configured with a MiniCOP module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file option slots. The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP without a DCM.



4212

Figure 2-11 — Nonredundant Parallel I/O Card File Front View

2.2.2.4 Redundant Parallel I/O Card File

The basic redundant Parallel I/O Card File contains the standard card modules as shown in Table 2-9. See Figure 2-12 for a front view of the card file. It is not a standard product, but a “Special,” and is described for reference only.

Table 2-9 — Redundant Parallel I/O Card File Slot Assignments

CARD SLOT	MODULE DESCRIPTION	MODEL NUMBER	NOTES
A	Enhanced Diagnostic Module (EDM)	621-0021	1, 3
B and C	Serial Link Selector Module (SLS)	621-9928R	1, 3, 7, 8
D and E	Parallel Bus Selector Module (PBSM)	621-9929	1, 3, 7, 8
F	Any Valid 621-xxxx I/O Module		2, 3, 4
G	Any Valid 621-xxxx I/O Module		2, 3, 4
H	Any Valid 621-xxxx I/O Module		2, 3, 4
I	Any Valid 621-xxxx I/O Module		2, 3, 4
J	Any Valid 621-xxxx I/O Module		2, 3, 4
K	Any Valid 621-xxxx I/O Module		2, 3, 4
L	Any Valid 621-xxxx I/O Module		2, 3, 4, 6
L and M	Power Supply Module (PSM) 115/230 Vac 15 A	621-9933	2, 5
M	Power Supply Module (PSM) 115/230 Vac 8 A	621-9934	2
M	Power Supply Module (PSM) 24 Vdc 8 A	621-9932	2
N	Parallel I/O Module (PIOM)	621-9937	1, 3

- Notes:
1. This is a standard card module slot.
 2. This is an option card module slot.
 3. Check DIP switch settings if applicable (see subsection 2.4).
 4. Install the model 621-9900 blank plastic cover plate if this slot is empty.
 5. The module is double width.
 6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.
 7. This is a double width module.
 8. At least one SLS and one PBSM is required for a redundant system. Additional I/O card files may contain a SLS. Otherwise, the slots can contain any valid 621-xxxx I/O module.

All low level signal process I/O modules (analog, TTL, PIM, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal process I/O modules must be grouped together, starting at the rightmost slot and proceeding left.

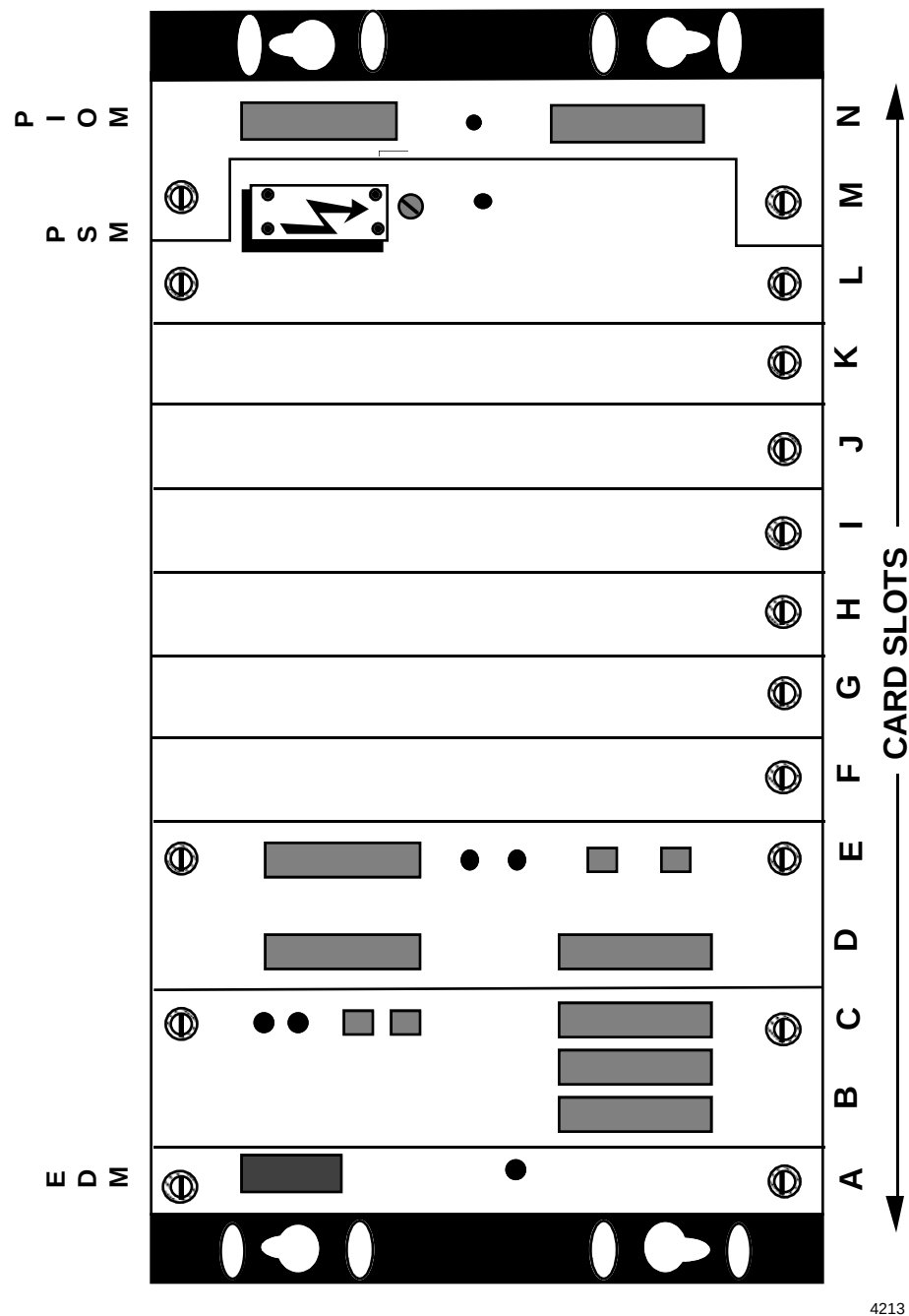
All I/O slots must be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the slot and configured by the appropriate Parallel I/O Module configuration DIP switches. Each empty slot must be assigned a quantity of 16 addresses to accommodate spares, allowing for future expansion. The only slots to be assigned an address of zero are those slots occupied by the left half of a double width module and both slots occupied by a double width module that requires no addresses, such as the Parallel Bus Selector Module.

I/O addresses 2040 through 2047 must not be assigned to any parallel or serial slot of a system containing serial I/O.

All empty I/O slots must be covered with a model 621-9900 plastic cover plate.

The Power Supply Module selected for the card file must be compatible with the power input (115/230 Vac or 24 Vdc) and must supply enough current to the card file to satisfy the total load requirement.

All systems configured with a MiniCOP module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file option slots. The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP without a DCM.



4213

Figure 2-12 — Redundant Parallel I/O Card File Front View

2.3 PROCESSOR CARD FILE MODULES

2.3.1 Logic Manager Module

The model 620-0071 Logic Manager Module (LMM) is a four board, double wide module that must be installed in slots A and B at the left front side of the Logic Manager Processor card file.

The LMM obtains power and Logic Manager System data signals via its backplane connectors. The LMM supports redundancy in the Logic Manager System by accepting Redundancy Control Module (RCM) inputs, which are used to activate one of two redundant LMMs connected to the Universal Control Network (UCN).

The LMM's database information is battery backed by its own internal lithium battery, which provides a minimum of six months data retention.

2.3.1.1 Front Panel

Figure 2-13 is a representation of the Logic Manager Module's front panel. The front panel has the following features:

- Two F-style coax connectors to link the module with the UCN A and UCN B trunk cables are located at the upper left of the front panel. These connectors provide the link between the LMM and the UCN trunk coaxial cables. Line protection is provided by a spark gap transient suppressor between the signal line and the shield for each connector.
- Two 3-contact detachable connectors, labeled **RCM A** and **RCM B**, are located at the upper right portion of the module front panel and provide connection points for the redundancy switchover control signals (RCM Selects). Pin 3, the top pin of each connector, is connected electrically to the front panel (chassis ground). The **RCM A** and **RCM B** connectors are linked to the redundancy connectors of the corresponding Redundancy Control Modules. The RCM input signals are used to select the lead (primary) and backup (secondary) processors.
- Six LED indicators which function as follows:
 - PASS** This green LED is illuminated following a successful startup and self-test. It indicates a normal operating state. The LED is extinguished when the module is initializing, is in the test mode, or if there is a hard fault shutdown.
 - BUS** This green LED is illuminated when communication on the Processor card file backplane is active. The LED is extinguished when the backplane is inactive.

NOTE

The **BUS** LED normally flickers at a frequency corresponding to the processor scan rate.

- BAT** This green LED remains illuminated as long as power is supplied to the Processor card file and the LMM's battery voltage level is sufficient to maintain the module memory data during a power failure. The LED is extinguished when the battery is low, or if there is no power supplied to the processor card file.
- UCN A** This green LED is illuminated when Channel A is selected for UCN signal reception. It is extinguished when Channel B is selected.
- UCN B** This green LED is illuminated when Channel B is selected for UCN signal reception. It is extinguished when Channel A is selected.
- XMIT** This green LED is illuminated when the module is transmitting data via the Carrier Band Modem to the UCN. The LED is extinguished when the transmitter is inactive.

- The LMM is shipped with a size 2/3 A lithium battery, model 629-3004, installed to provide backup power for the internal memory. The battery is accessed through a cover at the lower right of the module front panel. An insulating wafer prevents battery discharge during shipment/storage and must be removed before the system is operated. The LMM employs a pull strap for removal of the battery. Pull on the strap to remove the battery from its compartment and make sure it is properly replaced when the battery is reinstalled without the wafer. Observe the positive (+) and negative (-) markings in the battery compartment and be certain to orient the battery accordingly during installation.

The battery compartment is identified on the front panel of the LMM with an exclamation point symbol (!) and a reference to this manual. Refer servicing of the battery to qualified service personnel only.

WARNING

Lithium batteries are prohibited on passenger aircraft. The following conditions are potentially hazardous and must be avoided:

Recharge
Incineration
Short circuit
Forced discharge (reversal)
Use or storage outside of the specified temperature range
Puncturing, crushing, or disassembly



CAUTION

Retention of data in RAM is not guaranteed when the LMM is removed from the card file. Even if the battery remains installed, stray static charges may result in memory changes. A checksum fault would probably be detected during system startup.

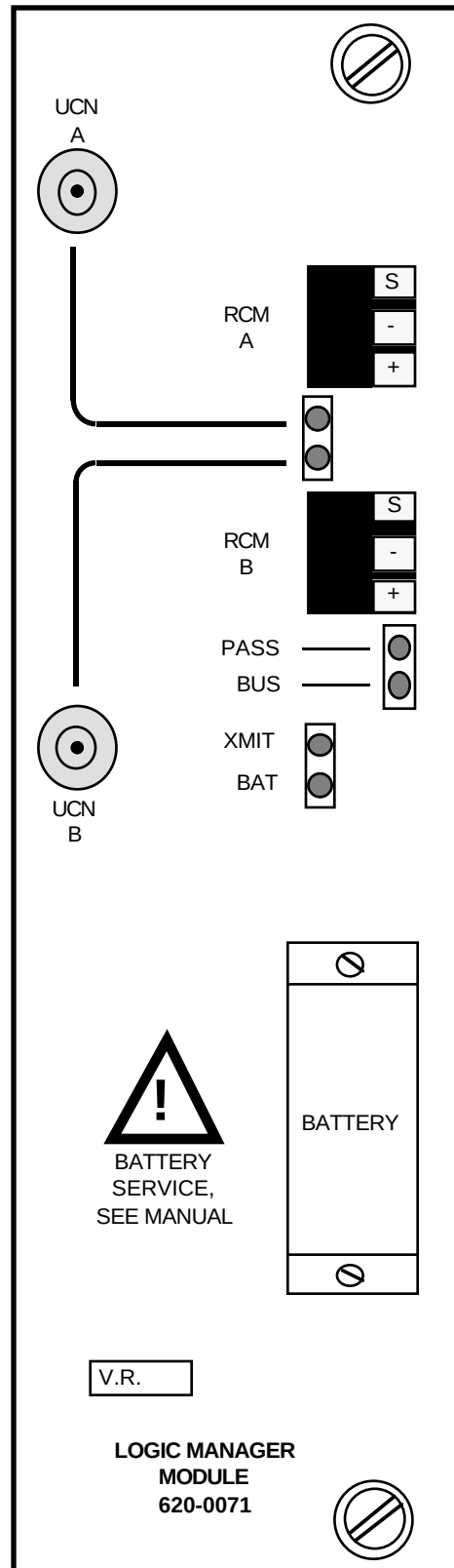


Figure 2-13— Logic Manager Module Front Panel

3903

2.3.1.2 Circuit Board Layout

The LMM consists of four printed circuit boards; a Processor Mother Board, a Processor Daughter Board, a Carrier Band Modem Board, and a Modem Power Board. See Figure 2-14. The Processor Mother and Daughter Boards contain configuration DIP switches that determine operating parameters. For configuration information concerning the LMM's two banks of DIP switches, see subsection 2.3.1.3.

Processor Mother Board

The Processor Mother Board contains the LMM's microprocessor unit and Logic Manager Module's processor interface. This primary board also contains 128 K words of RAM and 64 K words of ROM, that provides storage for the LMM's data. The board mates with the processor interface connectors at the backplane.

The board also contains DIP switch SW1, a 4-position switch that configures the module's backplane address. The switch should always be set to an assigned backplane address of one (1).

The **PASS** and **BUS** front panel LEDs are controlled by this board.

Processor Daughter Board

The Processor Daughter Board is linked to the Processor Mother Board with two ribbon cables. Data is passed from the Mother Board through the Daughter Board, which provides expansion RAM and the communication interface to the Carrier Band Modem Board.

The board contains DIP switch SW2, an 8-position switch that is used to select the LMM's node address on the UCN and to specify Lead/Backup (Primary/Secondary) status of the LMM in a redundant configuration.

The board also contains the module's system clock.

The Processor Daughter Board controls the **UCN A**, **UCN B**, **XMIT**, and **BAT** LEDs at the module front panel.

Carrier Band Modem Board

The Carrier Band Modem Board translates formatted digital data to and from the UCN. The board is directly linked to the **UCN A** and **UCN B** connectors at the module front panel.

The Carrier Band Modem Board transmits and receives data at a rate of 5 megabits per second in accordance with the IEEE 802.4 Communications Standard.

Modem Power Board

The Modem Power Board mates with the connector at the backplane, providing power for the Carrier Band Modem Board via an internal power cable.

2.3.1.3 Configuration

Four switch selectable functions must be configured on the LMM Processor Daughter Board. An 8-position DIP switch, SW2, provides the configuration capability. See Table 2-10. Position 1 selects the LMM's default to the primary (lead) or secondary (backup). Positions 2 through 6 determine the UCN node address. Refer to Table 2-12 for a detailed table of the UCN node address settings. Position 7 enables or disables the Network Test function. Position 8 is Factory Test Enable and should always be disabled.

Table 2-10 — LMM Daughter Board SW2 Settings

SW2 POSITION	STATE	FUNCTION
1	Open/Off *	Defaults to the primary LMM in a redundant configuration
	Closed/On	Defaults to secondary LMM in a redundant configuration
2	Open/Off *	Positions 2 through 6 are used to determine the LMM's node address on the UCN. See Table 2-12.
	Closed/On	
3	Open/Off *	
	Closed/On	
4	Open/Off *	
	Closed/On	
5	Open/Off *	
	Closed/On	
6	Open/Off *	
	Closed/On	
7	Open/Off *	Must be Off for normal operation
	Closed/On	Network Test Enable On = Enable
8	Open/Off *	Should always be Off
	Closed/On	Factory Test Enable On = Enable

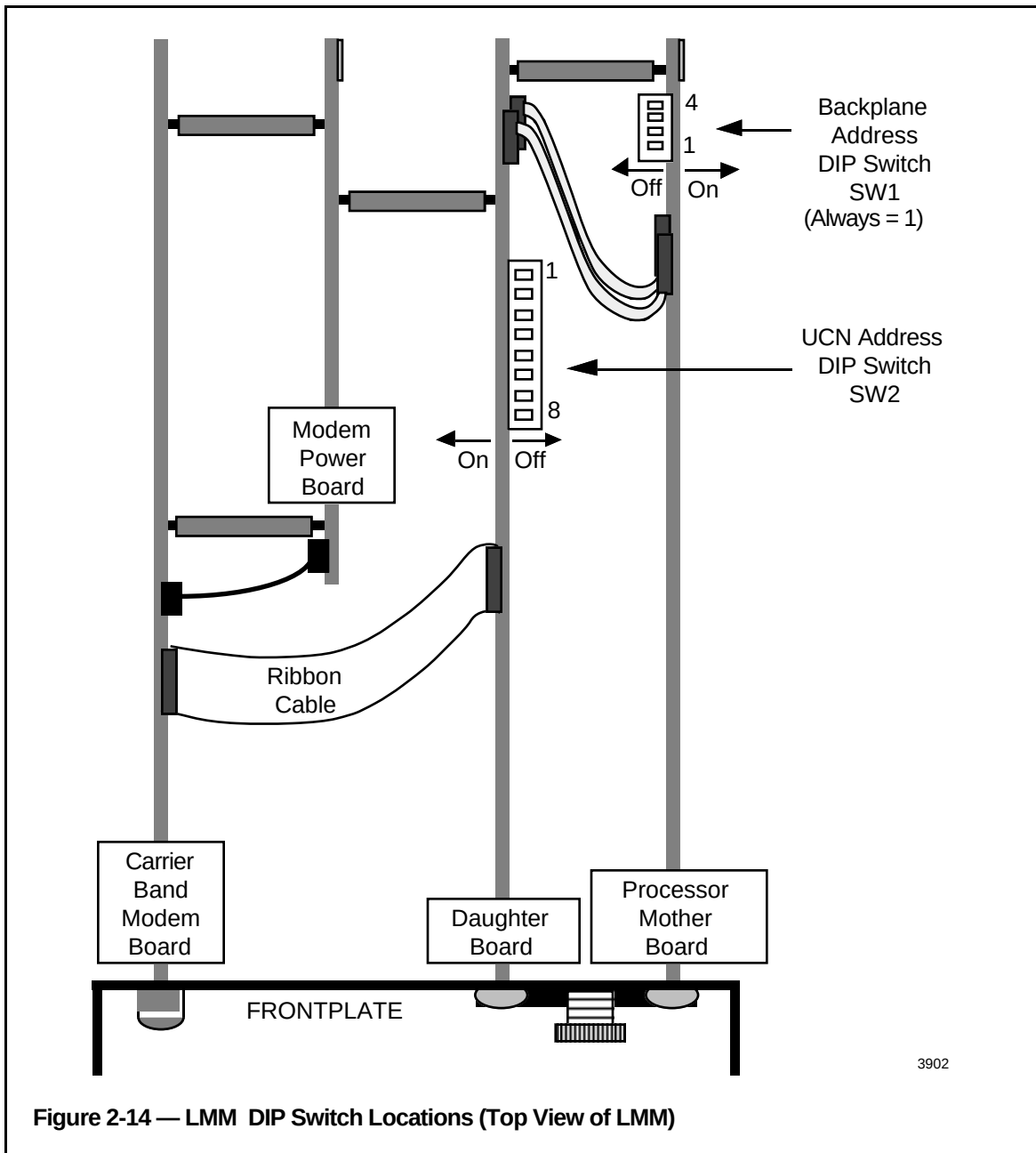
* Factory settings

The 4-position DIP switch, SW1, located on the LMM Processor Mother Board must be configured to identify the backplane address of the LMM. Only the first two positions are used. Switch positions 3 and 4 are used for factory test and must always be in the off position for proper LMM operation. Table 2-11 displays the four possible settings. The LMM should always be configured for an address of 1.

Table 2-11 — LMM Mother Board SW1 Settings

SW1 POSITION STATE		BACKPLANE ADDRESS
1	2	
Closed/On	Closed/On	0
Open/Off	Closed/On	1 *
Closed/On	Open/Off	2
Open/Off	Open/Off	3

* Factory setting



3902

Table 2-12 — LMM SW2 DIP Switch Settings For UCN Addresses

UCN ADDRESS (Redundant Pairs)	SW2 POSITION				
	6	5	4	3	2
1,2	Reserved for a NIM				
3,4	OFF	OFF	OFF	OFF	ON
5,6	OFF	OFF	OFF	ON	OFF
7,8	OFF	OFF	OFF	ON	ON
9,10	OFF	OFF	ON	OFF	OFF
11,12	OFF	OFF	ON	OFF	ON
13,14	OFF	OFF	ON	ON	OFF
15,16	OFF	OFF	ON	ON	ON
17,18	OFF	ON	OFF	OFF	OFF
19,20	OFF	ON	OFF	OFF	ON
21,22	OFF	ON	OFF	ON	OFF
23,24	OFF	ON	OFF	ON	ON
25,26	OFF	ON	ON	OFF	OFF
27,28	OFF	ON	ON	OFF	ON
29,30	OFF	ON	ON	ON	OFF
31,32	OFF	ON	ON	ON	ON
33,34	ON	OFF	OFF	OFF	OFF
35,36	ON	OFF	OFF	OFF	ON
37,38	ON	OFF	OFF	ON	OFF
39,40	ON	OFF	OFF	ON	ON

(Continued)

Table 2-12 — LMM SW2 DIP Switch Settings For UCN Addresses (Continued)

UCN ADDRESS (Redundant Pairs)	SW2 POSITION				
	6	5	4	3	2
41,42	ON	OFF	ON	OFF	OFF
43,44	ON	OFF	ON	OFF	ON
45,46	ON	OFF	ON	ON	OFF
47,48	ON	OFF	ON	ON	ON
49,50	ON	ON	OFF	OFF	OFF
51,52	ON	ON	OFF	OFF	ON
53,54	ON	ON	OFF	ON	OFF
55,56	ON	ON	OFF	ON	ON
57,58	ON	ON	ON	OFF	OFF
59,60	ON	ON	ON	OFF	ON
61,62	ON	ON	ON	ON	OFF
63,64	ON	ON	ON	ON	ON

2.3.2 Redundancy Control Module

The model 620-0059 Redundancy Control Modules (RCMs) are the primary components of the Redundant Control System and are responsible for transferring I/O data from the primary Logic Controller to the secondary Logic Controller. These modules also monitor the status of the Logic Manager Processor Modules. If this option is implemented, a module is installed in slot C of both Logic Manager Processor card files.

2.3.2.1 Front Panel

The front panel of the RCM has four LEDs:

The green **PASS** LED indicates whether the module passed its self-test. The LED remains illuminated as long as the RCM passes its ongoing test. If a fault occurs, the LED blinks, pausing in between the number of strobes (count) to indicate the type of error. See Table 2-13.

The green **BUS ACTV** LED indicates that the RCM has control of the processor card file's backplane.

The green **PORT ACTV** LED illuminates when the RCM transmits or receives data over the RCM Data Cable.

The green **LEAD UNIT** LED illuminates when the RCM's host Logic Controller is the primary unit.

In addition, the RCM's front panel includes a pushbutton and two connector ports.

The **LEAD REQUEST** pushbutton is used to manually change the primary/secondary configuration while the system is operating. When the button is pushed on the secondary RCM, leadership transfers from the primary to the secondary. When the button is pushed on the primary RCM, leadership transfers from the secondary to the primary. Position 5 of the configuration switch SW1 on the module's circuit board enables or disables the pushbutton.

PORT 1 is a 25-pin, D-type connector through which the I/O data exchange occurs. The RCM Data Cable links this port to Port 1 of the other RCM.

PORT 2 is a 3-position, detachable screw terminal that interconnects to the Parallel Bus Selector, Serial Link Selector, Serial Link, and Logic Manager Modules. This port provides the switchover control for these modules.

NOTE

Both RCMs must have the same version and revision levels.

Table 2-13 — Redundancy Control Module Fault Signals

NUMBER OF STROBES	FAULT
2	RCM checksum error
3	RAM write/read test error
4	The hardware timer is inconsistent with the software test timer or the read/write test of the control register has failed.
5	Port failure
6	RCM/Processor configuration error
7	Processor faulty
8	Primary/secondary configuration error
9	Backplane interface fault
10	Logic Manager Module fault
11	Serial link fault

2.3.2.2 Configuration

The RCM has a 5-position DIP switch module, SW1, that determines the backplane address, primary or secondary designation, primary request enable/disable, and program downloading enable/disable. See Table 2-14.

Table 2-14 — RCM SW1 DIP Switch Settings

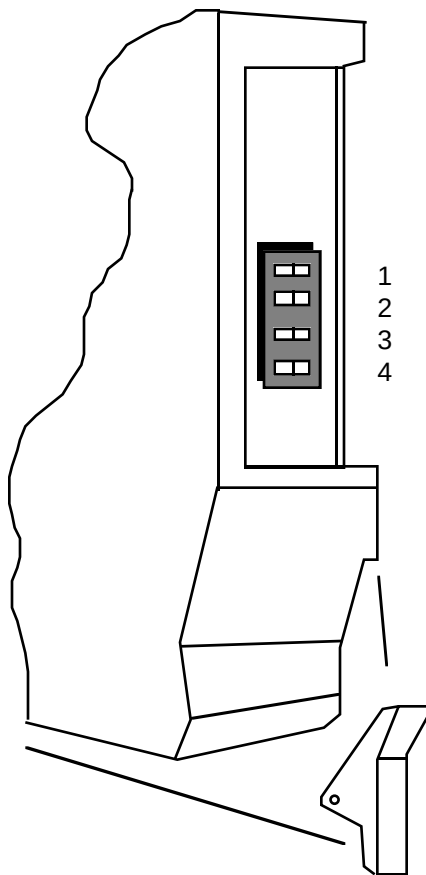
SW1 POSITION	STATE	FUNCTION
1	Closed/On	Address = 0. This is the factory setting. Only one RCM is permitted per card file. The RCM is always set to address zero.
2	Closed/On	
3	Closed/On	The RCM is designated secondary
	Open/Off	The RCM is designated primary
4	Closed/On	Prohibits the program downloading of the RCM processor
	Open/Off	Permits the program downloading of the RCM processor
5	Closed/On	Enables the LEAD REQUEST switch
	Open/Off	Disables the LEAD REQUEST switch

2.3.3 8 K Memory Module

The model 620-0027, 8 K Memory Module is an optional RAM addition to the standard 24 K Memory Module and is installed in slot D of the Logic Manager Processor card file. It contains 8 K words of memory and requires a mandatory starting memory address of 24 K. The total memory available to the Logic Manager Processor becomes 32 K words, the maximum allowed.

The memory module stores the user control program. The green status LED labeled **PASS**, located on the front panel of the module, illuminates after successful completion of the module self-test.

A 4-position DIP switch is located on the front of the module and is used to set the module's starting memory address. Set DIP switch positions 1 and 2 to Off and DIP switch positions 3 and 4 to On. Note that the DIP switch selections are additive (8 K + 16 K = 24 K). See Figure 2-15.



SWITCH POSITION	SWITCH VALUE	
	OFF	ON
1	0	2 K
2	0	4 K
3	0	8 K
4	0	16 K

Figure 2-15 — Memory Module Start Address DIP Switch Settings

3904

2.3.4 24 K Memory Module

The model 620-0024, 24 K Memory Module is installed in slot E of the Logic Manager Processor card file. It contains a total of 24 K words of memory and requires a mandatory starting memory address of zero (0).

The Memory Module stores the user control program. The green status LED labeled **PASS**, located on the front panel of the module, illuminates after successful completion of the module self-test.

A 4-position DIP switch is located on the front of the module and is used to set the module's starting memory address. All four positions on the switch are set to OFF for a starting memory address of zero (0). See Figure 2-15.

2.3.5 4 K x 4 K Register Module

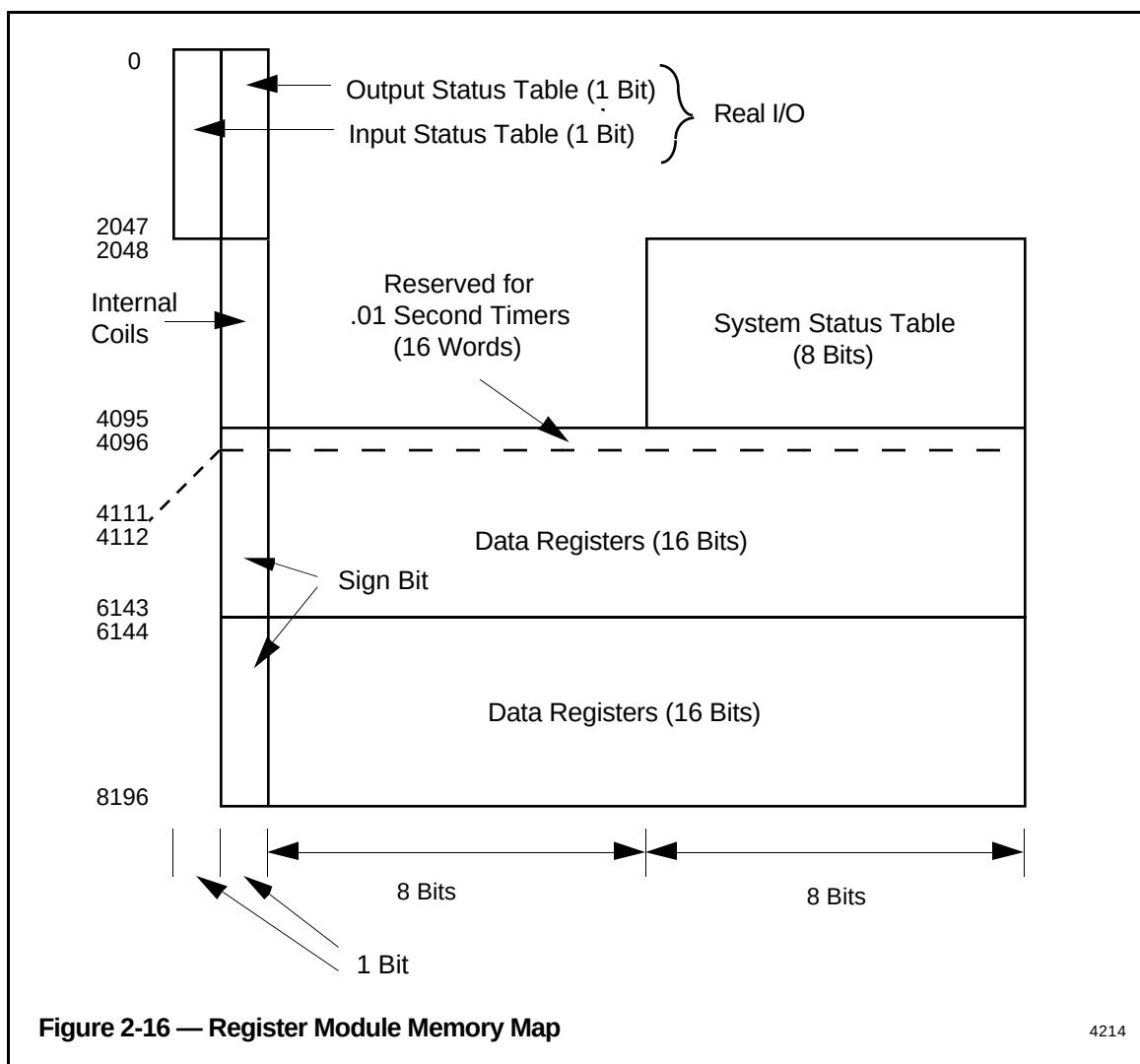
The model 620-0056, 4 K x 4 K Register Module contains the Logic Manager System's I/O Data Table and is installed in slot F of the Logic Manager card file. The Data Table is divided into three areas; the I/O Status Table, System Status Table, and Register Table. The I/O Status Table contains 4096 (4 K) continuous single bit storage locations which have a starting address of zero (0). In the 4 K x 4 K Register Module, 2048 (2 K) locations are available for real I/O status or internal coils, and an additional 2048 (2 K) locations are available for internal coils.

Timer and counter preset and accumulated values are stored in the register area. Other numerical data may be stored in the bit or register area. See Figure 2-16. The register area is 16-bits wide, plus a sign bit, and is comprised of 4096 (4 K) registers. The 17th bit is normally used to indicate the sign of the data contained in the register. In certain operations it will indicate overflow conditions.

The System Status Table stores processor diagnostic information that can be accessed by the Loader/Terminal or by the PULL instruction in the control program. Some of the information is provided in the Logic Manager Node Status Detail Display (see section 3). The System Status Table consists of memory locations that are 8-bits wide. The table is divided into three sections; System Diagnostics, System Hardware Status, and System Identification.

The Data Table is battery backed by the power supply battery to ensure data retention during system power outages. All register addresses are cleared to 0 when a total memory clear operation is executed. All I/O Status Table locations are cleared to 0 when the system enters the **RUN** mode to allow for an orderly startup of the I/O.

The green LED, labeled **PASS**, illuminates after successful completion of the module self-test.



2.3.6 System Control Module

The model 620-0054 System Control Module (SCM) is a functional extension of the Processor Module. It coordinates the interaction of all the modules that compose the processor. The SCM also contains a single bit processor that executes all relay logic in the control program.

The SCM is installed in slot G of the Logic Manager card file. A green LED indicator, located on the front panel and labeled **PASS**, illuminates after successful completion of its self-test and the on-line diagnostics.

2.3.7 Processor Module

The Processor Module (PM) executes the program stored in the memory module(s) and handles arithmetic computation and data movement instructions.

The Processor Module is installed in slot H of the Logic Manager card file. Two status indicators are located on the front panel of the PM. The green LED, labeled **PASS**, illuminates when the module passes its self-test. The red LED, labeled **TESTING**, illuminates when the series of diagnostic tests are initiated upon applying power to the system. The **TESTING** LED appears to remain illuminated as long as the processor is scanning, since a short series of diagnostics that occurs at the beginning of each scan continually illuminates it. The port connector, labeled **Loader/Terminal**, connects to the model 623-60 MS-DOS Loader and is covered by a removable cover plate.

2.3.8 I/O Control Module

The model 620-0085 Input/Output Control Module (IOCM) coordinates communication and formats the data between the Logic Manager Processor and the I/O system. It also monitors individual I/O module fault diagnostics. The IOCM works in conjunction with the Parallel Link Driver Module (PLDM) and/or the Serial Link Module (SLM) to control data flow to and from the I/O system. The IOCM is installed in slot I of the Logic Manager card file.

2.3.9 Power Supply Modules

A Power Supply Module (PSM) provides power to operate the Logic Manager card file. The dual card PSM is installed in slots J and K in the Logic Manager card file. A choice of two models are available. They are as follows:

MODEL NUMBER	INPUT VOLTAGE	AMPERAGE
620-0036	115/230 Vac	15 A (5 Vdc Output)
620-0047	24 Vdc	15 A (5 Vdc Output)

A battery compartment, accessible through a PSM front cover, contains a D size lithium battery that provides backup power to the memory and register modules in the Logic Manager Processor card file. The PSM is shipped with the battery installed.

CAUTION

An insulating wafer prevents battery discharge during shipment and storage and must be removed before the system is operated. See Figure 2-17.

WARNING

Lithium batteries are prohibited on passenger aircraft. The following conditions are potentially hazardous and must be avoided:

- Recharge
- Incineration
- Short circuit
- Forced discharge (reversal)
- Use or storage outside of the specified temperature range
- Puncturing, crushing, or disassembly



Two status indicators are located on the front panel of the PSM. A green LED labeled **POWER** indicates that power is applied to the module. A green LED labeled **BATTERY PASS** remains illuminated as long as the battery is in good condition and power is applied.

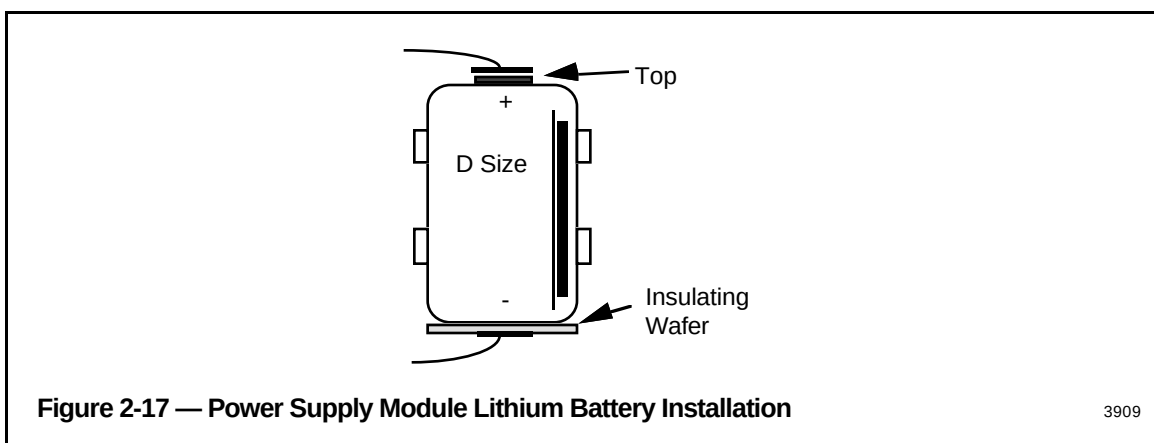


Figure 2-17 — Power Supply Module Lithium Battery Installation

3909

2.3.9.1 115/230 Vac 15 A Power Supply Module

The model 620-0036 Power Supply Module provides a maximum of 15 A of ± 5 Vdc power for use by the Logic Manager card file modules. It also supplies 300 mA of +12 Vdc for optional modules requiring the voltage.

Terminals on the front panel of the module, labeled **A** (L1 or Line), **B** (L2 or Common), and **GND** (Ground), provide ac power wiring connection points. The module operates on voltages of 115 Vac or 230 Vac. A shorting board, located inside the side cover at the top edge, selects the proper ac voltage (see Figure 2-18).

The module contains a front accessible fuse holder. It is shipped with a 2 A slow blow fuse installed for 115 Vac operation. A 1 A slow blow fuse for 230 Vac operation is also shipped with the module.

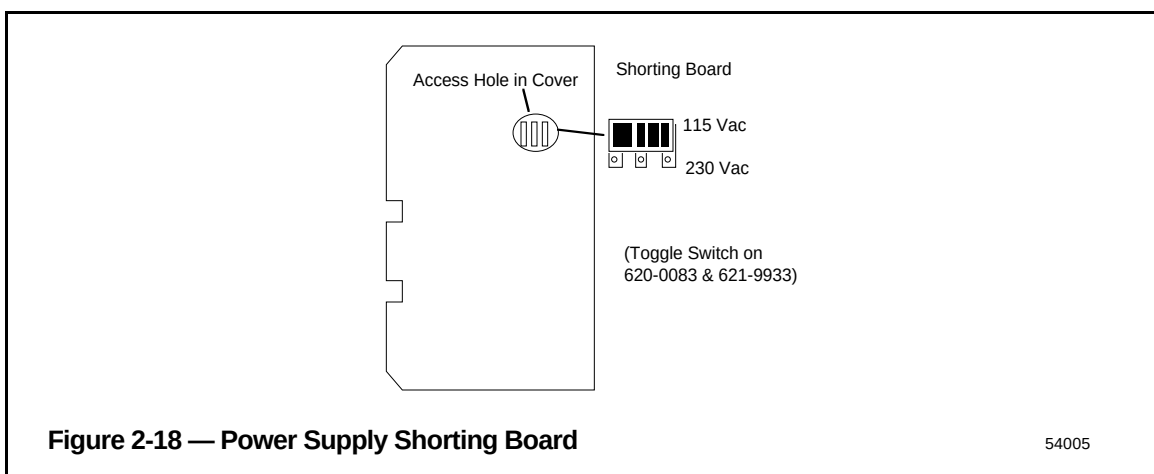


Figure 2-18 — Power Supply Shorting Board

54005

2.3.9.2 24 Vdc 15 A Power Supply Module

The model 620-0047 Power Supply Module provides a maximum of 15 A of ± 5 Vdc power for use by the Logic Manager card file modules. It also supplies 300 mA of +12 Vdc for optional modules requiring the voltage.

Input terminals on the front of the module are labeled +, -, and **GND**. The module operates with a nominal input voltage of 20 to 25 Vdc.

The module has a front accessible fuse holder and is shipped with an 8 A slow blow fuse installed.

2.3.10 Serial Link Module

The model 621-9939 Serial Link Module (SLM) controls serial I/O data transfer to and from the I/O. It is the intermediary between the Logic Manager Processor and the SIOM(s).

The SLM collects input data from the SIOM(s). The Processor reads data during its input status scan. Then, depending on its program instructions, the processor issues output commands to the SLM, which sends them to the SIOM(s). The SLM has its own microprocessor and data table and operates asynchronously to the processor.

Addresses 2040 through 2047 of the SLM provide serial I/O status information to the processor. This information includes the SIOM off-line flag, status of SIOM(s) on the redundant channel, the starting address of the off-line SIOM(s), and the number of SIOMs on-line for each channel.

Each SLM can control two serial channels. Therefore, a Logic Manager Processor card file with one SLM installed in slot L, supports a maximum of two channels. A Logic Manager Processor card file with one SLM installed in slot L and one SLM installed in slot M supports a maximum of four channels. Each channel operates independently of the other serial channels in the Logic Manager System.

A channel accommodates 1 to 16 drops (SIOMs), or a maximum of 2040 I/O. However, the total number of I/O points must stay within the I/O capacity of the Logic Manager System.

A kit of terminating resistors is provided with the SLM. These resistors are used for terminating the multidrop serial I/O link, and for terminating the control switchover link in a redundant configuration. Figure 2-19 shows the locations where terminating resistors may be installed on the SLM if required. If an SLM serial I/O communications channel is at one end of the communications link, a resistor must be installed for that channel (R31 and/or R32). Similarly, if an SLM is at the end of a redundant system control switchover link, a 100 ohm terminating resistor must be installed at location R35 of the SLM. Refer to Figures 2-3, 2-4, and 2-6. Also, additional information on wiring and installation of terminating resistors may be found in the installation section of binder 2070.

2.3.10.1 Front Panel

Figure 2-20 is a view of the Serial Link Module front panel.

The front panel of the SLM is permanently attached and contains a 19-contact terminal block. The terminal block includes a two piece connector to which channels 1 and 2 and RCM control are wired. The terminal block is removable so that channel wiring can be easily disconnected from the module if the module requires servicing.

The SLM front panel has five LEDs that indicate the status of the module. The operating state of the SLM is shown by the status of the front panel LEDs.

A green **PASS** LED indicates the SLM has successfully completed its self-test.

The green **ACTV** (Active) LED indicates that data is being transmitted properly. The LED is normally illuminated during transmission. There is one LED for each channel.

Each channel has a yellow **LINK FAULT** LED. It indicates that a card file fault or a communication fault has occurred on the channel. The normal state of the LED is off. When a Link Fault occurs, the LED is illuminated. The LED remains illuminated as long as the problem exists. Corrective action for resetting serial link faults depends on the type of fault, system configuration, and the state of the controlled process.

After the SLM has been reset, or had power cycled, the serial system reinitializes. This causes the serial drops to be taken off-line, except those that have been manually powered down because of the fault. The **LINK FAULT** LED is extinguished when the fault is corrected.

2.3.10.2 Configuration

An 8-position DIP switch module, SW1, located near the top edge of the SLM circuit board, is used to configure the SLM. See Figure 2-19.

Positions 1 through 4 determine the SLM's shutdown response (positions 1 and 2 for channel 1 and positions 3 and 4 for channel 2).

Position 5 selects the number of PUSH/PULL operations that are updated per I/O scan. Position 6 determines if one or two SLMs will be configured in the Logic Manager Processor card file. Position 7 selects a redundant or nonredundant serial I/O configuration. Position 8 selects normal operation or factory test of the serial I/O system. Factory test is a diagnostic tool provided for bench test procedures. It is not applicable to routine system operation. When Factory test is selected, the operation of switch positions 1 through 7 is redefined.

See Table 2-15 for the SLM SW1 DIP switch module functions.

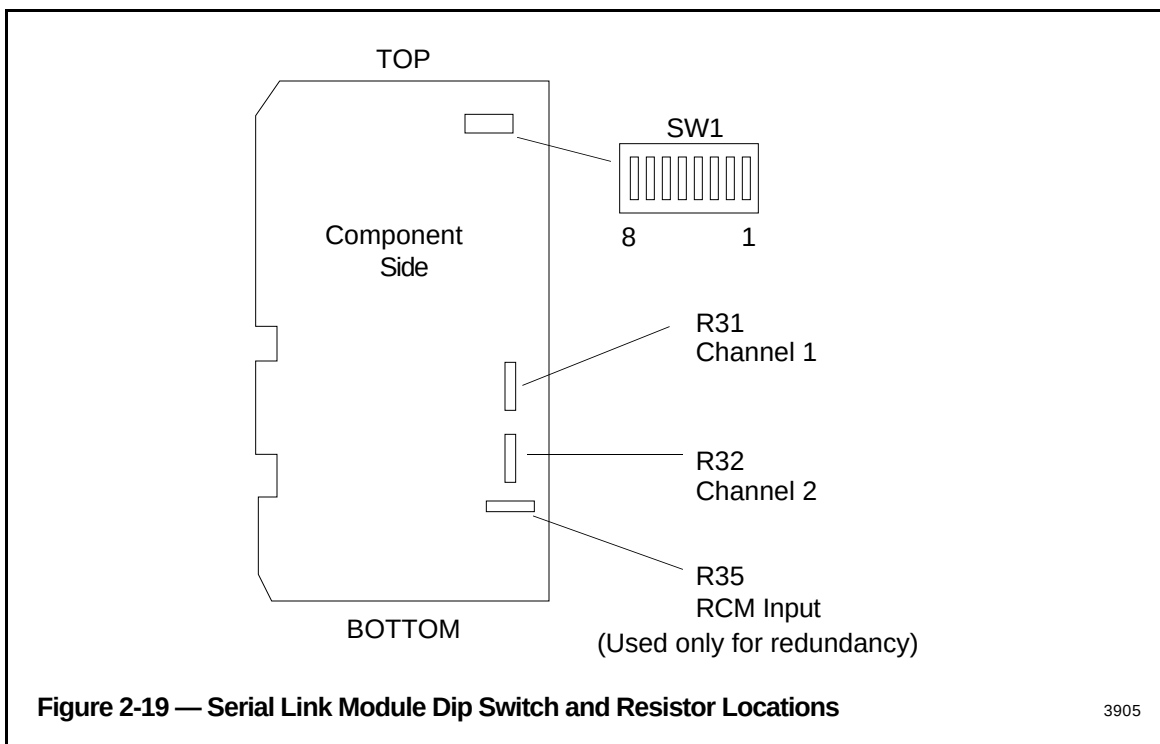


Table 2-15 — Serial Link Module DIP Switch Settings

SW1 POSITION STATE		FUNCTION (SHUTDOWN RESPONSE)
1 and 3*	2 and 4*	
Open/Off**	Open/Off**	The channel continues to operate if a channel fault occurs.***
Closed/On	Closed/On	The channel ceases operation if a channel fault occurs.***
Open/Off	Closed/On	Both channels on SLM cease operation if a channel fault occurs.***
Closed/On	Open/Off	Both the SLM channels shutdown, the CLROL line on the I/O bus is enabled, resetting the parallel I/O and causing the second SLM, if present, to clear or freeze data in its SIOMs, depending on the configuration of the SIOM DIP switch settings. <u>Do not use for the preferred lead processor of a redundant system.</u> ***

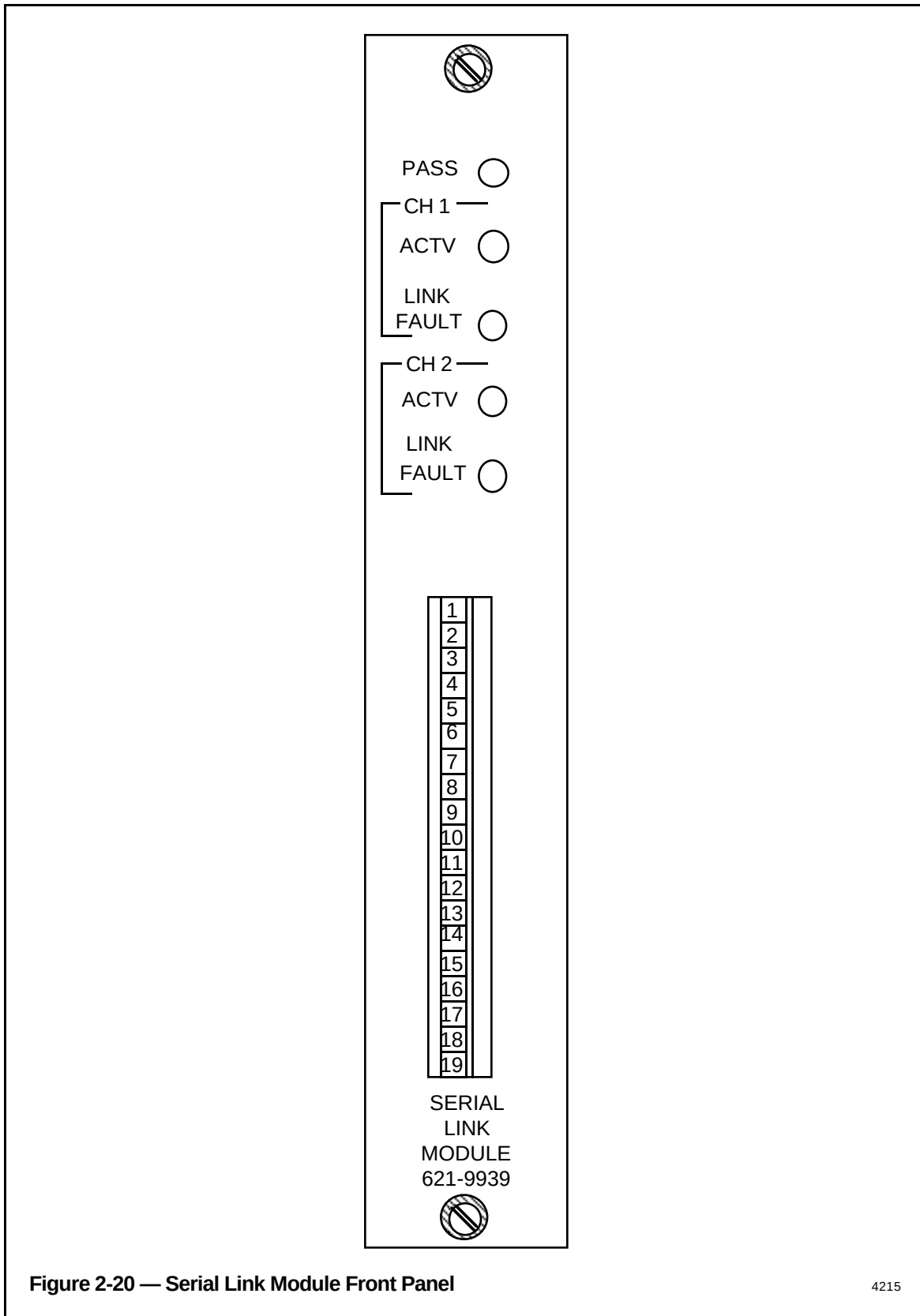
SW1 POSITION	STATE	FUNCTION
5	Closed/On	Service one PUSH/PULL card per I/O rack each scan.
	Open/Off**	Service all PUSH/PULL cards per I/O rack each scan.
6	Closed/On	Identifies the SLM as the second SLM in the card file.
	Open/Off**	The first or only SLM in the card file requires this setting.
7	Closed/On	Redundant serial I/O system
	Open/Off**	Nonredundant serial I/O system
8	Closed/On	Factory test enabled
	Open/Off**	Factory test disabled (normal operation)

*Switches 1 and 2 apply to channel 1; Switches 3 and 4 apply to channel 2.

The switches are set in pairs as shown in the table.

**Factory setting

***A channel fault is defined as an SIOM being taken off-line because of power failure, serial link timeout, or two consecutive serial data errors received.



4215

2.3.11 Parallel Link Driver Module

The model 620-0088 Parallel Link Driver Module (PLDM) operates with the I/O Control Module (IOCM) to coordinate communications between the parallel or serial I/O and the processor. The PLDM is installed in slot N of the Logic Manager Processor card file.

2.3.11.1 Front Panel

The PLDM controls the operation of the processor with a 4-position Mode Keyswitch. The positions are labeled; **PROGRAM**, **DISABLE**, **RUN/PROG**, and **RUN**.

A D-type connector on the PLDM front panel interfaces to the PIOM in the first parallel I/O card file.

The front panel of the PLDM also has three LED status indicators.

The green LED, labeled **RUN**, illuminates while the processor is scanning the ladder program and is operating correctly.

The red LED, labeled **FORCE**, illuminates when at least one address in the program is in the forced state.

The green LED, labeled **PASS**, illuminates when the PLDM has passed its self-test.

2.3.11.2 Configuration

Two DIP switch modules, labeled SW1 and SW2, located on the front edge of the PLDM circuit board under the module side cover, are used to select the I/O response to system faults and to set various operating parameters. They can be set without removing the module's side cover. The Closed/On setting is selected by pressing the switch toward the circuit board. See Figure 2-21 and Table 2-16 for the location and function of each PLDM DIP switch module.

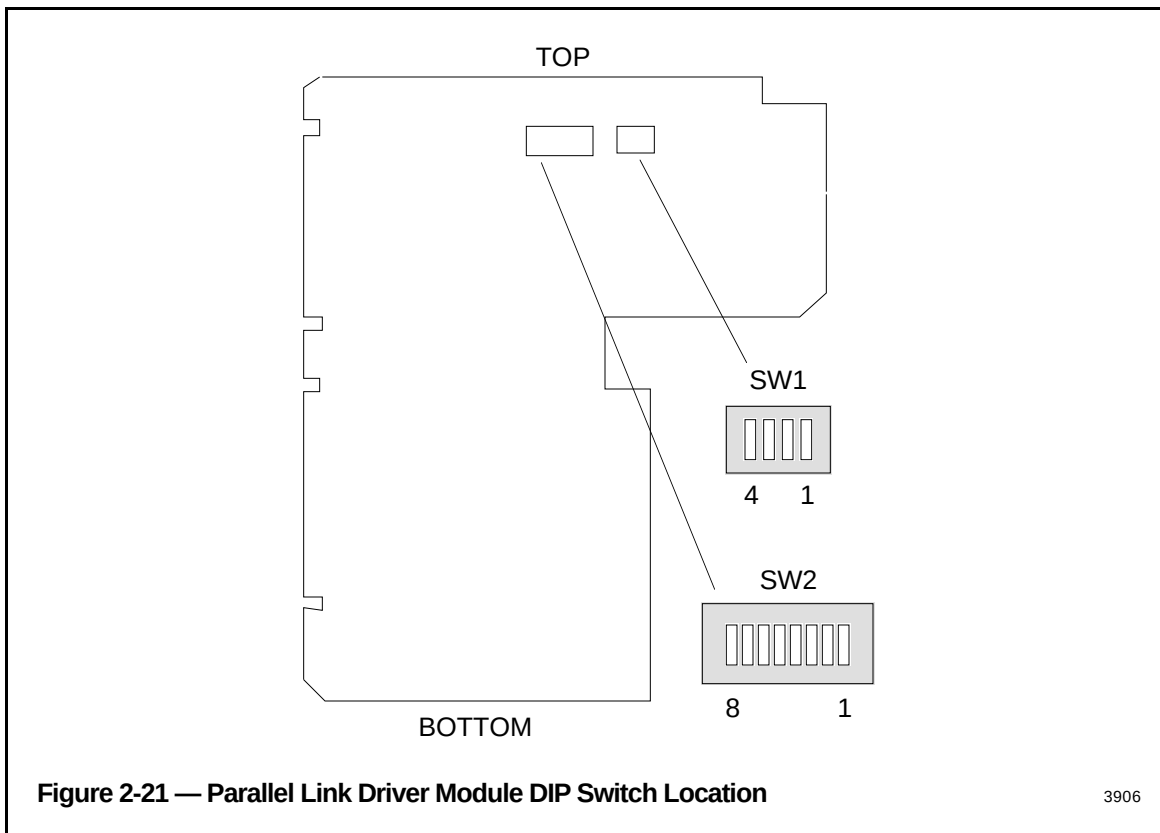


Figure 2-21 — Parallel Link Driver Module DIP Switch Location

3906

Table 2-16 — Parallel Link Driver Module DIP Switch Settings

SW1 POSITION	STATE	FUNCTION	
1	Closed/On	Disables the processor scan function	
	Open/Off*	Enables the processor scan function	
2	Closed/On*	Adds 20 ms	Set to 60 ms in excess of processor scan time. The module is factory set to the maximum value
	Open/Off	Adds 0 ms	
3	Closed/On*	Adds 40 ms	
	Open/Off	Adds 0 ms	
4	Closed/On*	Adds 80 ms	
	Open/Off	Adds 0 ms	

(Continued)

*Factory setting

Table 2-16 — Parallel Link Driver Module DIP Switch Settings (Continued)

SW2 POSITION	STATE	FUNCTION
1	Closed/On*	Must be set to the Closed/On position.
2	Closed/On*	Must be set to the Closed/On position.
3**	Closed/On	I/O card files clear or freeze outputs according to PIOM or SIOM freeze/clear setting when in the PROGRAM mode is used.
	Open/Off	When in the PROGRAM mode, all outputs remain in their last state regardless of the PIOM or SIOM freeze/clear setting.
4	Closed/On*	Enables the Force function
	Open/Off	Disables the Force function
5	Closed/On*	Enables the data changes to the RM Table values and Sequencer Register constant values in the RUN mode.
	Open/Off	Disables the above data changes in the RUN mode
6	Closed/On*	Enables on-line programming in the RUN/PROGRAM mode.
	Open/Off	Disables on-line programming in the RUN/PROGRAM mode.
7		Not used

*Factory setting

Ordinary coils (those not retentive or latched) are cleared when the processor is switched to the **RUN mode.

2.4 I/O CARD FILE MODULES

The Serial and Parallel I/O card files use modules that are standard or optionally installed in both I/O card files. In particular, the power supply, discrete, analog, and special function process I/O modules are shared. See subsections 2.2.2.1 through 2.2.2.4 for the acceptable power supply and interface modules. In subsection 2.4.7, the discrete, analog, and special function process I/O modules are referenced.

2.4.1 Enhanced Diagnostic Module

The model 621-0021R Enhanced Diagnostic Module (EDM) provides the Logic Manager System with additional system monitoring capability such as the detection of problems with data and control lines on the I/O bus. It is installed in slot A of both the serial and parallel I/O card files. The EDM can also detect a scan loss with its watchdog timer. Upon detecting a system malfunction, the EDM deenergizes an internal relay that can be used to remove power from the system's process output modules. The relay provides an effective method of halting the controlled process. Access to the relay is by a front panel mounted terminal block.

The Enhanced Diagnostic Module must be assigned a quantity of eight I/O addresses and must always be installed in slot A of a Serial or Parallel I/O card file.

2.4.1.1 Front Panel

A three pole terminal is mounted on the front panel. It connects to the contacts of the form C relay that deenergizes when a system fault is detected.

A green **SCAN** LED illuminates when the system is operating properly.

2.4.1.2 Configuration

A four-position DIP switch module located at the top left edge of the EDM board sets operating parameters. Positions 1 and 2 select the Watchdog Timer, and position 3 selects Parallel or Serial I/O operation. Position 4 selects the reaction of the EDM to a fault. See Table 2-17 for the EDM DIP switch selections.

NOTE

The EDM must be set to 4.5 seconds, if used in Redundant Logic Managers.

Table 2-17 — Enhanced Diagnostic Module DIP Switch Module Settings

SWITCH POSITION STATE		FUNCTION
1	2	
Open/Off	Open/Off	1.74 sec. time out
Closed/On*	Open/Off*	200 ms time out
Open/Off	Closed/On**	4.5 sec. time out for redundant LMs
Closed/On	Closed/On	490 ms time out

SWITCH POSITION	STATE	FUNCTION
3	Closed/On**	Identifies serial I/O operation
	Open/Off	Identifies parallel I/O operation
4	Closed/On*	A data bus or control line fault deenergizes the fault relay immediately.
	Open/Off	Signal from SIOM, PIOM or LM deenergizes the relay.

* Factory setting for nonredundant systems

** Factory setting for redundant systems

2.4.2 Serial Link Selector Module

The model 621-9928R Serial Link Selector Module (SLS) is a double wide module that is installed in the I/O slots of a Serial or Parallel I/O card file in a redundant Logic Manager System. The SLS serves as a general purpose switching device and switches serial communication devices shared by the redundant Logic Manager processors.

The SLS's only connections to the Serial I/O card file backplane are the +5 Vdc supply, ground, and the power fail signal. All other connections are made through the front panel connectors.

2.4.2.1 Front Panel

The state of the SLS's bus selector circuitry is controlled by the Redundancy Control Modules (RCMs), which send switch select output signals to the SLS through the two 3-contact front panel connectors.

Two 12-contact detachable screw terminals, labeled **A** and **B**, are used to connect to an optional communications module in each redundant Processor card file, A and B, respectively. A third 12-contact terminal, labeled **OUT**, connects to a serial device such as a host computer or Loader/Terminal.

Two **ON-LINE** LEDs, labeled **A** and **B**, indicate which Logic Manager processor, A or B, is connected to the serial I/O system.

2.4.2.2 Configuration

A 2-position DIP switch module, SW1, on the SLS circuit board sets operating parameters. See Table 2-18 for SLS DIP switch module selections.

Table 2-18 — Serial Link Selector Module DIP Switch Settings

SW1 POSITION	STATE	FUNCTION
1	Closed/On	A 100 ohm resistor is placed across the A switchover control wiring connection.
	Open/Off*	No resistor is placed across the A switchover control wiring connection.
2	Closed/On	A 100 ohm resistor is placed across the B switchover control wiring connection.
	Open/Off*	No resistor is placed across the B switchover control wiring connection.

*Factory setting

NOTE

A terminating resistor is required if the SLS is the last module on the switchover control wiring connection. Refer to Section 5 of the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070 for additional information regarding the installation of terminating resistors on the switchover control link used in redundant serial I/O configurations.

2.4.3 Parallel Bus Selector Module

The model 621-9929 Parallel Bus Selector Module (PBS) is a double wide module installed in slots D and E of one I/O card file per Parallel I/O system in a redundant configuration only. A model 621-9928R Serial Link Selector (SLS) must also be installed in slots B and C of the same card file. This is a “Special System” because redundant parallel I/O systems are not recommended. The PBS connects the primary Logic Manager Processor to the Parallel I/O system and switches the Parallel I/O system, upon request, from the primary (lead) to the secondary (backup) Logic Manager Processor.

The PBS’s only connections to the I/O card file backplane are the +5 Vdc supply, ground, and the power fail signal. All other connections are made through the front panel connectors.

2.4.3.1 Front Panel

The state of the PBS’s bus selector circuitry is controlled by the Redundant Control Modules (RCMs), which send a switch select output signal to the PBS through the two 3-contact front panel connectors.

Two 50-pin D-type connectors, labeled **A** and **B**, interface with the Parallel Link Driver Modules (PLDMs) in the Logic Manager Processor card files, A and B, respectively. The third 50-pin D-type connector, labeled **OUT**, interfaces with the first Parallel I/O Module (PIOM) in the parallel I/O system.

Two **ON-LINE** LEDs, labeled **A** and **B**, indicate which Logic Manager Processor, A or B, has control of the I/O system.

2.4.3.2 Configuration

A 2-position DIP switch module, labeled SW1, on the PBS circuit board sets operating parameters. See Table 2-19 for the PBS DIP switch module selections.

Table 2-19 — Parallel Bus Selector Module DIP Switch Settings

SW1 POSITION	STATE	FUNCTION
1	Closed/On	A 100 ohm resistor is placed across the A switch control input.
	Open/Off*	No resistor is placed across the A switch control input.
2	Closed/On	A 100 ohm resistor is placed across the B switch control input.
	Open/Off*	No resistor is placed across the B switch control input.

* Factory setting

2.4.4 Power Supply Modules

A Power Supply Module (PSM) provides power to operate the I/O card file. A choice of three models is available. They are as follows:

MODEL NUMBER	INPUT VOLTAGE	AMPERAGE
621-9932	24 Vdc	8 A (5 Vdc Output)
621-9933	115/230 Vac	15 A (5 Vdc Output)
621-9934	115/230 Vac	8 A (5 Vdc Output)

2.4.4.1 I/O 24 Vdc 8 A Power Supply Module

The model 621-9932 single card I/O Power Supply Module provides 8 A of +5 Vdc for the I/O logic circuitry in the card file. It also supplies 600 mA of ± 15 Vdc power. The module occupies slot M in the full size I/O card file or slot G in the half size card file. The maximum power consumption of the module is 96 VA.

Terminals on the front of the module are labeled +, –, and **GND**. The module operates within a input voltage range of 20 to 28 Vdc. The power fail level is 19 Vdc with a power fail delay of 4 ms. A green LED, labeled **5 VDC**, illuminates when 5 Vdc is present. The module has a front panel accessible fuse holder and is shipped with a 8 A fast blow fuse.

2.4.4.2 I/O 115/230 Vac 15 A Power Supply Module

The model 621-9933 I/O Power Supply Module is a double wide module that provides 10 to 15 A of +5 Vdc for the logic circuitry in the I/O card file. It also supplies 1.2 to 2 A of ± 15 Vdc. The graph in Figure 2-22 shows the relationship between current loads for the ± 15 Vdc and the ± 5 Vdc sources. If, for example, the ± 15 Vdc load requires 2 A, the ± 5 Vdc load can not exceed 10 A. If the ± 15 Vdc load requires 1.2 A, the ± 5 Vdc load can not exceed 15 A. The maximum power consumption of the module is 110 VA.

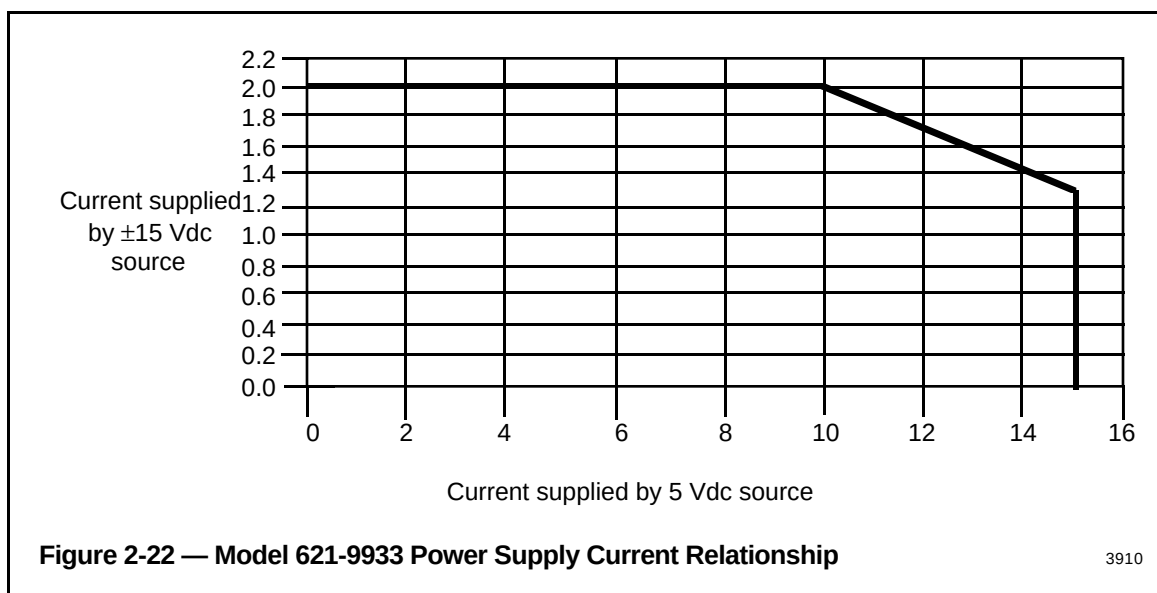
The Power Supply Module is installed in slots L and M of the full size I/O card file or slots F and G of the half size card file.

The input is selectable for 115 or 230 Vac operation by the position of the toggle switch located under the component cover of the module. The 115 Vac selection allows a voltage range of 85 to 132 Vac. The 230 Vac selection allows a voltage range of 170 to 250 Vac operation. The frequency for both ranges is 47 to 63 Hz. The power fail level is 83 volts for 115 Vac and 166 volts for 230 Vac. The power fail delay is 11.5 milliseconds.

The Power Supply Module has a front accessible fuse holder. It is shipped with a 4 A slow blow fuse installed for 115 Vac operation. A 2 A slow blow fuse for 230 Vac operation is also shipped with the module. A green LED, labeled **5 VDC**, is illuminated when the 5 Vdc power is present.

WARNING

Be sure to match the 115/230 Vac toggle switch position with the proper fuse selection before applying power to the card file.



2.4.4.3 I/O 115/230 Vac 8 A Power Supply Module

The model 621-9934 I/O Power Supply Module provides 8 A of +5 Vdc for the I/O logic circuitry in the I/O card file. It also supplies 600 mA of ± 15 Vdc power for special function I/O and option module operation. The maximum power consumption of the module is 95 VA.

The model 621-9934 Power Supply Module is installed in slot M of the full size I/O card file or slot G of the half size card file.

The voltage input is selectable for 115 or 230 Vac by the position of a selection board located under the component cover of the Power Supply Module. The 115 Vac selection allows a voltage range of 85 to 140 Vac. The 230 Vac selection allows 170 to 250 Vac operation. The frequency for both voltage ranges is 47 to 63 Hz.

The Power Supply Module offers a front accessible fuse holder. It is shipped with a 1.6 A slow blow fuse installed for 115 Vac operation. A 0.8 A slow blow fuse for 230 Vac operation is also shipped with the module.

The power fail level is 83 volts for 115 Vac and 166 volts for 230 Vac operation. The power fail delay is 11.5 ms.

A green LED, labeled **5 VDC**, is illuminated when the 5 Vdc power is present.

WARNING

Be sure to match the 115/230 Vac voltage selection board position with the proper fuse selection before applying power to the card file.

2.4.5 Serial I/O Modules

The model 621-9938R or model 621-9940 Serial I/O Modules (SIOM) are installed in slot N of the Serial I/O card file. They control the Serial I/O card file and interface with a model 621-9939 SLM in the Logic Manager Processor card file.

The model 621-9940 SIOM is used in nonredundant Logic Manager Serial I/O Systems, while the model 621-9938R SIOM is used in a redundant serial I/O configuration, but can also be used for nonredundant applications.

The model 621-9938R has two serial ports and the model 621-9940 has one. The model 621-9938R is used in redundant applications which require communication with redundant processors.

2.4.5.1 Front Panel

Figure 2-23 is a view of the model 621-9938R and model 621-9940 Serial I/O Module front panels.

The model 621-9938R front panel has six LEDs to display the module status. The model 621-9940 has three LEDs.

The green **ACTV** (Active) LED illuminates when the associated port is transmitting and receiving data properly. The model 621-9938R has an **ACTV** LED for each port. The normal state of the **ACTV** LED is flashing brightly for the model 621-9940 and the lead (primary) port of the model 621-9938R. The **ACTV** LED will flash dimly for the backup (secondary) port of the model 621-9938R.

When illuminated, the green **PASS** LED indicates the SIOM has successfully completed its self-test, which is initiated power is applied to the equipment or if the SIOM reset terminals are closed. The LED is normally illuminated.

The yellow **RACK FAULT** LED indicates that an output module data fault has occurred. The LED is normally extinguished.

The green **LEAD** LED indicates that the port is connected to the primary (lead) Logic Manager Processor in a redundant configuration and is successfully communicating with the primary Logic Manager Processor's SLM. The model 621-9938R has two **LEAD** LEDs. The LED for the port connected to the primary processor is illuminated and extinguished for the port connected to the secondary (backup) processor. Only one LED should be illuminated at a given time.

2.4.5.2 Configuration

Two terminating resistors are required in an SIOM communications port if the port is at the end (either end) of a serial I/O communications link. The resistors are supplied in a kit that comes with an SLM. Use the 200 ohm resistors from the kit for installations using Belden 9729 cable, and use the 300 ohm resistors, also supplied in the SLM resistor kit, for installations using Belden 9182 cable. (See Note.)

Refer to Figures 2-3 and 2-4 for examples showing terminating resistor requirements. Also refer to Section 5 of the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070 for additional information on wiring and termination of serial I/O links. Figure 2-24 identifies the location for terminating resistors, R36 through R39, on the SIOM circuit board. R36 and R38 terminate Port A, while R37 and R39 terminate Port B on the model 621-9938R SIOM. The model 621-9940 SIOM has only R36 and R38 locations, because it is a single port module.

The SIOM is configured by using the four 8-position and one 4-position DIP switch modules located near the top edge of the SIOM circuit board. These DIP switches select the module starting address, card fault recognition, communication fault responses, normal or test mode operation, and the number of process I/O points located in each I/O card file.

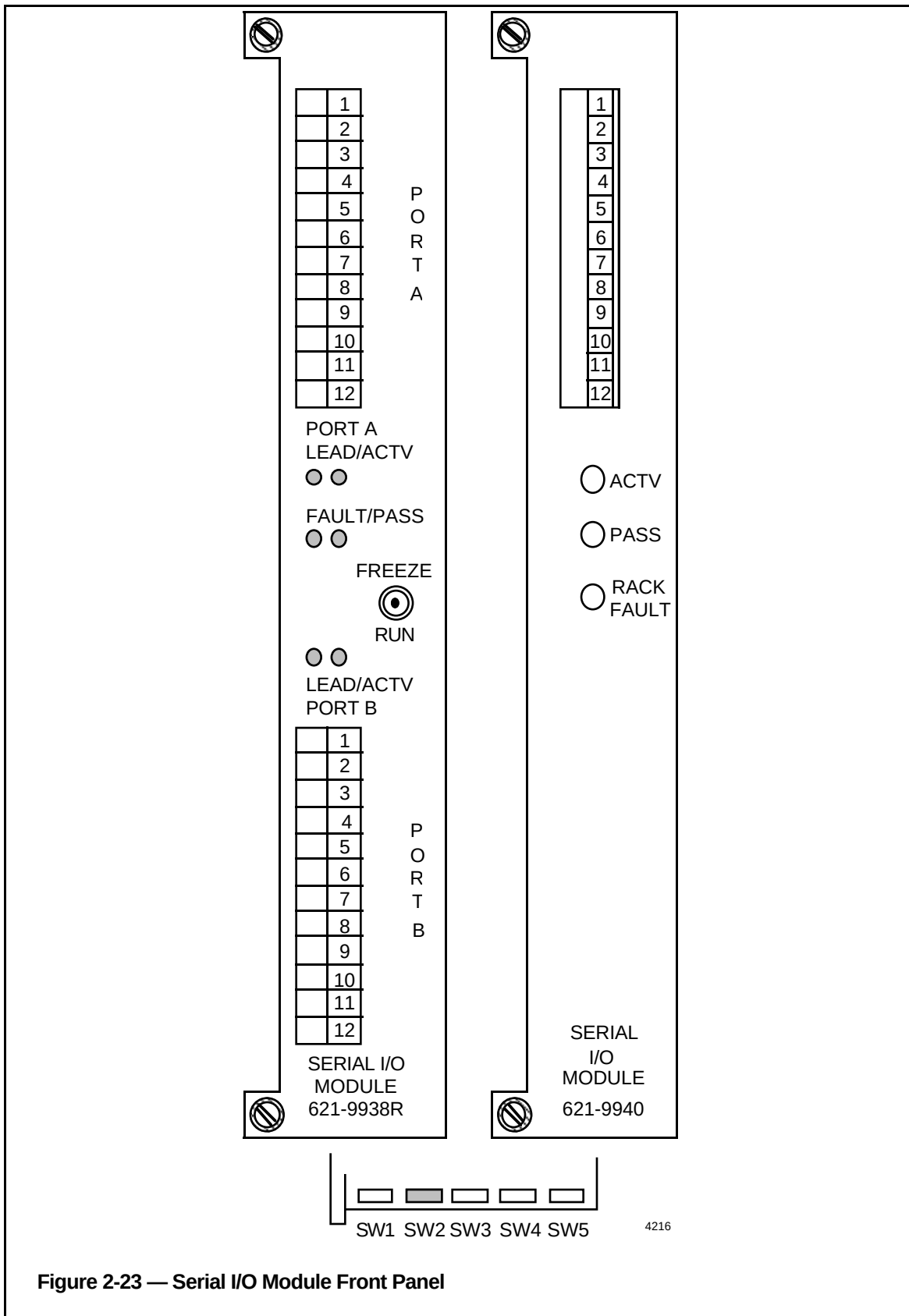
DIP switch module SW1 selects the absolute starting address for each card file. The starting address is selected from a range of 0 through 2032 in increments of eight. During startup, the SLM queries all connected SIOMs for their starting addresses and the number of allocated addresses in the card file. The SLM also determines the sequence in which the I/O card files on the channel are accessed based on the absolute starting address of each SIOM. Overlapping SIOM starting addresses, overlapping I/O addresses, and addresses higher than 2032 are prohibited.

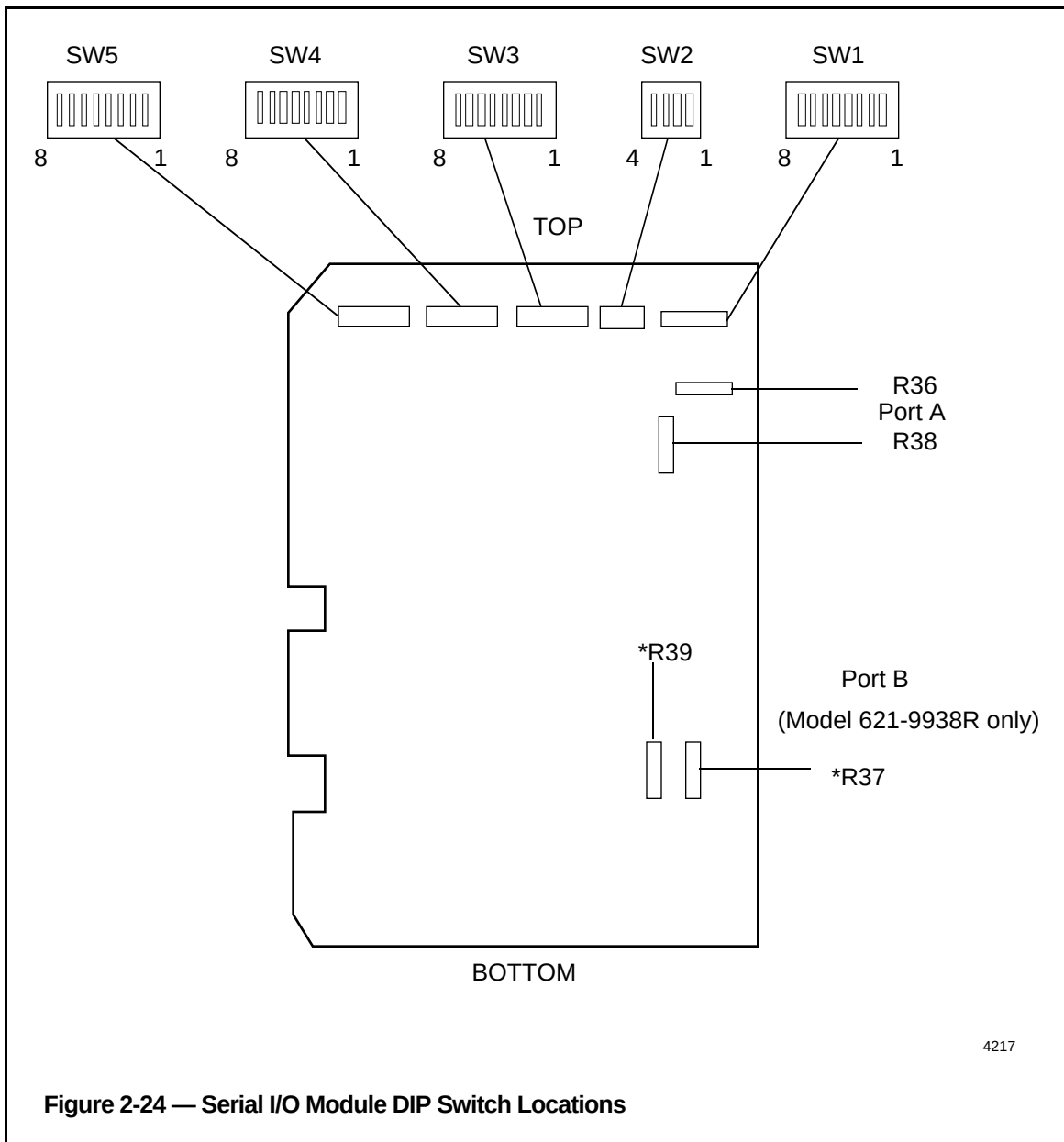
DIP switch module SW2 determines card file output states in case of a fault and permits the selection of factory test or normal operation. DIP switch modules SW3, SW4, and SW5 determine the number of process I/O points per card file slot.

Refer to Figure 2-24 and Table 2-20 for the SIOM DIP switch module locations and selections.

NOTE

Use of cables other than those noted must have Honeywell Engineering approval.





* 621-9940 does not have receptacles for R37 or R39. These resistors are only used on the 621-9938R.

Table 2-20 — Serial I/O Module DIP Switch Settings

SW1 STATE	SWITCH POSITION ADDRESS VALUE							
	1	2	3	4	5	6	7	8
Open/Off*	0	0	0	0	0	0	0	0
Closed/On	8	16	32	64	128	256	512	1024

SW2 POSITION	STATE	FUNCTION
1	Closed/On	Clear outputs
	Open/Off*	Freeze outputs
2	Closed/On	Recognize I/O card fault
	Open/Off*	Ignore card faults
3	Open/Off *	Enabled - Freeze/run switch on 621-9938R; Freeze enabled not used on 621-9940
4	Closed On	Not used
	Open/Off*	Not used

(Continued)

* Factory setting

Table 2-20 — Serial I/O Module DIP Switch Settings (Continued)

CONTROLLING SWITCH			SELECTION	NUMBER OF I/O POINTS PER SLOT			
SW5	SW4	SW3	SWITCH**	0	8	16*	32
I/O SLOT				SWITCH STATE			
D	H	L	1 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			2 (1)	Closed/On	Open/Off	Closed/On	Open/Off
C	G	K	3 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			4 (1)	Closed/On	Open/Off	Closed/On	Open/Off
B	F	J	5 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			6 (1)	Closed/On	Open/Off	Closed/On	Open/Off
A	E	I	7 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			8 (1)	Closed/On	Open/Off	Closed/On	Open/Off

*Factory setting

**Switch selection in parentheses denotes the silk-screen labeling on the SIOM printed circuit board.

NOTE

I/O card files are not allowed to have overlapping addresses.

2.4.6 Parallel I/O Module

The model 621-9937 Parallel I/O Module (PIOM) is the interface to the other Parallel I/O card files or the Parallel Link Driver Module in the Logic Manager Processor card file. The PIOM is installed in slot N of the full size Parallel I/O card file or slot H of the half size Parallel I/O card file.

The PIOM allows any combination of 8, 16, or 32 point I/O modules in the I/O card file, providing the number of I/O points stay within the boundaries set for each Logic Manager System.

2.4.6.1 Front Panel

The front panel of the PIOM has two 50-pin D-type connectors. The **IN** connector (male) is at the upper portion of the modules front panel, and the **OUT** connector (female) is at the lower portion.

A green **ACTIVE** LED illuminates when the module is communicating properly with a preceding card file.

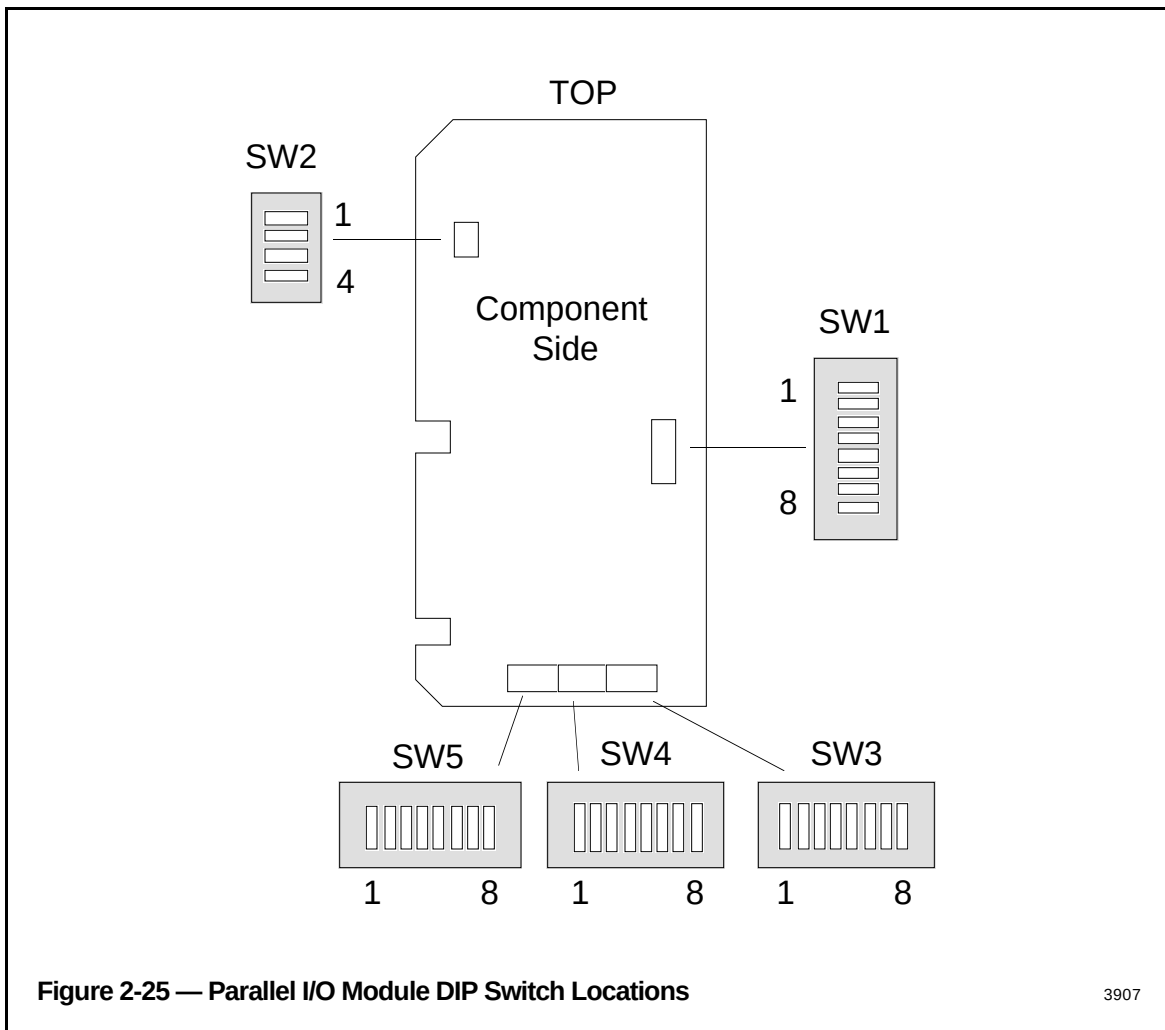
2.4.6.2 Configuration

The Parallel I/O Module has five DIP switch modules.

DIP switch modules SW1 and SW2 should be set before installing the PIOM. SW1 determines the starting address of the Parallel I/O card file and SW2, positions 1 and 2, determine card file output states if an output fault occurs.

DIP switch modules SW3, SW4 and SW5 are used to set the number of I/O addressed in slots A through L of the Parallel I/O card file. A slot also may be configured for 0 if it is occupied by a multi-slot special function module.

See Figures 2-25 and 2-26 and Table 2-21 for the location and a further explanation of the PIOM DIP switch selections.



3907

Table 2-21 — Parallel I/O Module Switch Settings

SW1 POSITION	STATE	SWITCH VALUE	
1	Closed/On	8	Determines the starting address for the parallel I/O card file containing this module. For example, closing positions 2, 4, and 6 yields a starting address of: $16 + 64 + 256 = 336$. NOTE: No two card files are allowed to have overlapping I/O addresses.
	Open/Off*	0	
2	Closed/On	16	
	Open/Off*	0	
3	Closed/On	32	
	Open/Off*	0	
4	Closed/On	64	
	Open/Off*	0	
5	Closed/On	128	
	Open/Off*	0	
6	Closed/On	256	
	Open/Off*	0	
7	Closed/On	512	
	Open/Off*	0	
8	Closed/On	1024	
	Open/Off*	0	

(Continued)

* Factory setting

Table 2-21 — Parallel I/O Module Switch Settings (Continued)

SW2 POSITION	STATE	FUNCTION
1	Closed/On	Clears the outputs when; (1) SW2 Position 2 is closed and an I/O fault occurs, (2) The external cable is disconnected, (3) An external power failure, and (4) The processor is in PROGRAM or DISABLE modes.
	Open/Off*	Outputs remain the same with the above conditions.
2	Closed/On	Recognize an output module fault
	Open/Off*	Ignore an output module fault
3	Not used	
4	Not used	

* Factory setting

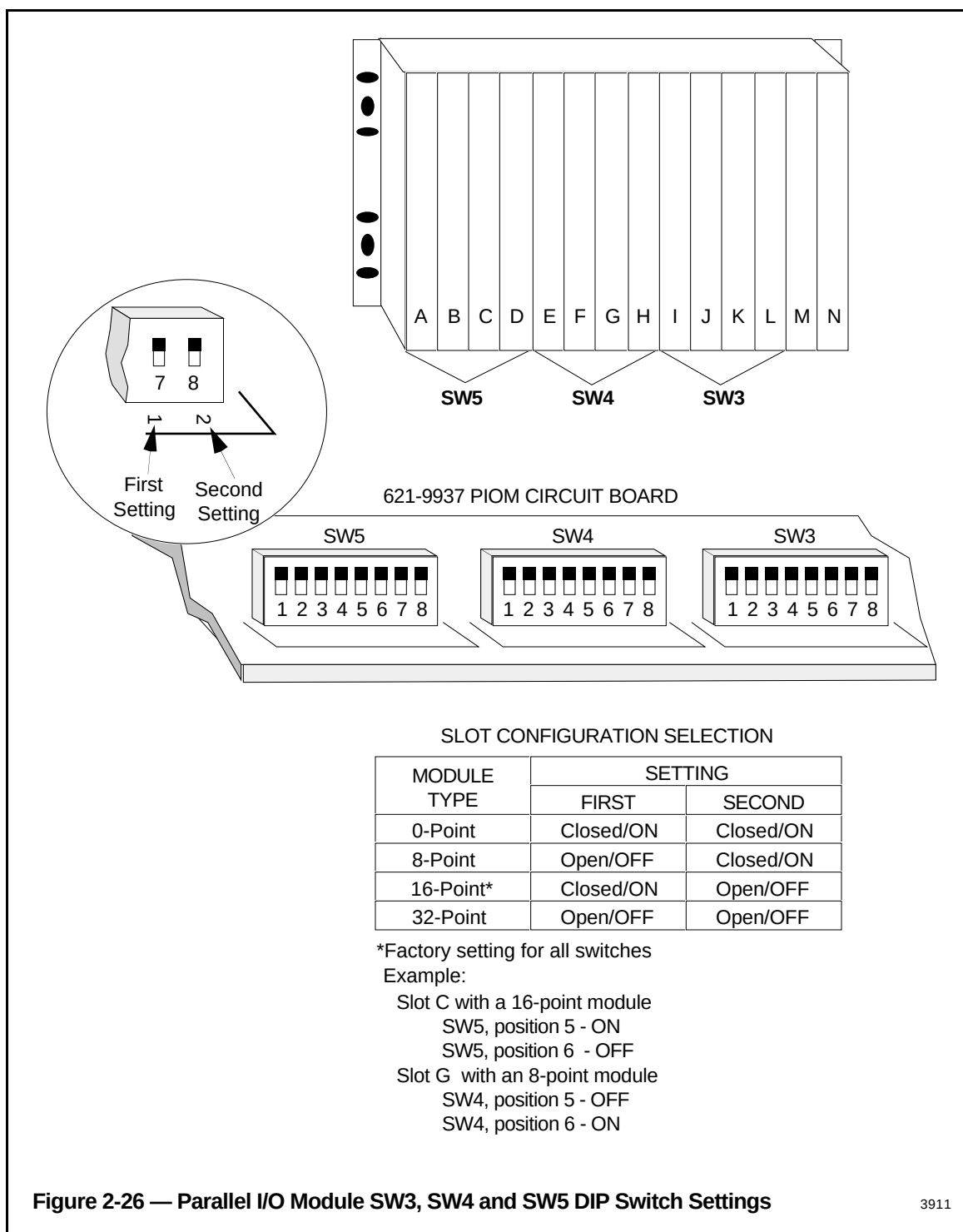


Figure 2-26 — Parallel I/O Module SW3, SW4 and SW5 DIP Switch Settings

3911

2.4.7 Process I/O Modules

This section provides information required for the servicing of the process I/O modules that reside in the serial and parallel I/O card files. However, it is difficult to separate installation from service when presenting module information because the two subjects are closely related. Therefore, the reader is advised to reference the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070 for any additional information needed to properly service a particular I/O module. Topics of interest include:

- Terminal block set selection and installation
- Address label selection and installation
- Input module addressing considerations
- Output module addressing considerations
- Module jumper and DIP switch settings if applicable
- Cable selection and routing

The module types listed in this section, in order, include discrete input, discrete output, analog input, analog output, and special function modules.

2.4.7.1 Discrete Input Modules

Discrete input modules sense On or Off voltage levels of various input devices. Among the more common devices are limit switches, proximity sensors, and pushbuttons. Once this On/Off status is obtained, the input modules convert it into logic level data to be used by the Logic Manager Processor.

LED indication for each input circuit on 6-, 8-, and 16-point modules is on the field device side of the input module circuitry. The 32-point input modules have the LED indication for each circuit on the 5 V logic side of the module. All discrete inputs are optically isolated to 2500 Vdc.

These modules are two piece units. A printed circuit board with a permanent front cover and label to identify each particular module, and a set of 8-, 16-, or 32-point terminal blocks make up each complete I/O module. The printed circuit board is inserted into the desired slot in the I/O card file. An offset backplane connector makes it impossible to install a module upside down.

The removable 8- and 16-point double swing terminal blocks allow for easy module insertion, removal, and wiring. They are installed by firmly seating the edge connector of each terminal block onto the front edge of the circuit board and securing the thumbscrew fastener. The terminal blocks are keyed for the circuit board. The 32-point terminal blocks are inserted into block headers located on the front of the module. Field wiring connects to the terminal block from the front of the module and can be accessed without removing the terminal block.

Each DC input module is either a sink or a source module. A sink input module provides the ground potential for the input device when the input device turns on. Current flows from the power supply through the field sensor input device (when it turns on) to the sink input module (ground). A source input module supplies the +Vdc potential to the field sensor input device when it turns on. Current flows from the power supply to the source input module through the field input sensor device to ground.

Allowable Leakage Current:

Each discrete input module has a specification called “Allowable Leakage Current.” This specification is for the user who wants to connect a discrete output or an external sensor (such as a proximity switch) to a discrete input.

A small amount of current usually leaks from any discrete output or external sensor before the output energizes. Although this current, called “Off-State Leakage Current,” is usually less than 5 milliamps, it can cause an input to recognize an On condition before the output actually turns on.

To prevent such a situation, you can drain the leakage current by connecting a resistor from the discrete input to ground. Use the following procedure to determine the value and wattage of the resistor:

1. Subtract the Allowable Leakage Current value for the discrete input module from the Off-State Leakage Current value of the discrete output module (or the external sensor).
2. Divide the Logic 0 voltage of the discrete input by the value determined in step 1. This will give the value for the resistor to be used.

$$R_v = \frac{LOV}{OSLC - ALC}$$

where: R_v = resistor value
 LOV = Logic 0 value
 $OSLC$ = Off State Leakage Current
 ALC = Allowable Leakage Current

3. Determine the wattage by dividing the squared maximum output voltage by the value determined in step 2.

$$Watt = \frac{(VO_{max})^2}{RV}$$

where: $Watt$ = wattage of R_v
 $(VO_{max})^2$ = maximum output voltage squared

Typically the wattage should be 200 percent of the calculated value so that the resistor does not degrade over time.

Table 2-22 is a list of the discrete input modules available for use. The number of point inputs are 8, 16, or 32. The proper cable class rules for each module must be applied for each module model. Refer to the *Logic Manager Installation* manual for an explanation of the cable classes. Module specifications, a wiring diagram, and related information are also found in the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070.

Table 2-22 — Discrete Input Modules

MODULE DESCRIPTION	INPUTS	CABLE CLASS	MODEL NUMBER
115 Vac/Vdc Discrete Input	8	DATA 3	621-1100R
115 Vac/Vdc Isolated Discrete Input	6	DATA 3	621-1101R
115 Vac Discrete Input	16	DATA 3	621-1160R
115 Vac Discrete Input	32	DATA 3	621-1180R
230 Vac/Vdc Discrete Input	8	DATA 3	621-1200R
230 Vac/Vdc Isolated Discrete Input	6	DATA 3	621-1201R
240 Vac Discrete Input	16	DATA 3	621-1250R
240 Vac Isolated Discrete Input	16	DATA 3	621-1251R
24 Vac/Vdc Discrete Input	8	DATA 2	621-1500R
24 Vac/Vdc Discrete Input	16	DATA 2	621-1550R
5 Vdc Sink Discrete Input	8	DATA 1	621-3300R
12 Vdc Sink Discrete Input	16	DATA 1	621-3450R
12-24 Vdc Sink Discrete Input	8	DATA 1, 2*	621-3500R
12-24 Vdc Sink Fast Response Discrete Input	8	DATA 1, 2*	621-3502R
24 Vdc Sink Fast Response Discrete Input	16	DATA 2	621-3552R
24 Vdc Sink Discrete Input	16	DATA 2	621-3560R
24 Vdc Sink Discrete Input	32	DATA 2	621-3580R
48 Vdc Sink Discrete Input	8	DATA 2	621-3600R
48 Vdc Sink Discrete Input	16	DATA 2	621-3660R
5 Vdc Source Discrete Input	8	DATA 1	621-4300R
5 Vdc TTL Discrete Input	16	DATA 1	621-4350R
12-24 Vdc Source Discrete Input	8	DATA 1, 2*	621-4500R
12-24 Vdc Source Fast Response Input	8	DATA 1, 2*	621-4502R
24 Vdc Source Discrete Input	16	DATA 2	621-4550R

*The IEEE 518 wiring class is application dependent and can vary with the voltage and current present in the wiring.

2.4.7.2 Analog Input Modules

An analog input module accepts analog signals from field devices connected to the module's terminals and converts the signal data to 12-bit binary values (0 to 4095) via an analog-to-digital converter contained in the module's circuitry. The digital values are stored in a holding register in the module and can be accessed by the Logic Manager control processor via a PULL instruction. The PULL instruction can read from one to eight 16-bit data words at a time.

Specific connectors or terminal block sets are required and must be ordered separately for each module, unless otherwise noted. See the *Logic Manager Installation* manual for information.

Table 2-23 is a list of analog input modules available for use. Cable class DATA 1 rules for the module must be observed. See the *Logic Manager Installation* manual for an explanation. Module specifications, a wiring diagram, and related information are also found in the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070.

Table 2-23 — Analog Input Modules

MODULE DESCRIPTION	INPUTS	CABLE CLASS	MODEL NUMBER
Universal Analog Input	16	DATA 1	621-0020R
Isolated 0-20 mA Analog Input	8	DATA 1	621-0022-AR
Isolated 0-10 V Analog Input	8	DATA 1	621-0022-VR

2.4.7.3 Discrete Output Modules

The typical discrete output module offers the ability to control the On and Off state of its end devices. Among the common output devices are motor starters, solenoid valves, and indicator lamps.

All dc output modules are source type modules. This means they provide the +V potential to the field output device when the circuit is turned on. Current flows from the external power supply to the source output module, to the field output device.

All AC output modules incorporate zero crossing. The output is turned On at the zero crossing of the ac wave form.

A LED indicator for each output circuit on the 8- and 16-point modules is on the load side of the output (except isolated outputs and reed relays). The 32-point output modules have the LED indication for each circuit on the 5 Vdc logic side of the module. All discrete outputs are optically isolated to 2500 Vdc.

These output modules provide self-diagnostic assurance, since each module's front plate has a LED to indicate a fault, thus reducing an otherwise lengthy troubleshooting procedure.

Each discrete output module consists of three components: a printed circuit board with a permanent front plate, address labels, and a set of 8-, 16-, or 32-point terminal blocks.

The printed circuit board is inserted into the desired slot in the Logic Manager I/O card file. Offset backplane connectors prevent installing a module upside down. The removable 8- and 16-point double swing terminal blocks allow easy module insertion, removal, and wiring. They are installed by firmly seating the edge connector of each terminal block onto the front edge of the circuit board and securing the thumbscrew fastener. The terminal blocks are keyed for the particular circuit board. The 32-point terminal blocks are inserted into block headers located on the front of the module. Field wiring connects to the 32-point terminal block from the front of the module and can be accessed without removing the terminal block.

Discrete 8-point modules use terminals T1 and B1 to supply voltage connections for circuits 1 through 4, and terminals T2 and B2 to supply voltage connections for circuits 5 through 8. Separate voltage sources may power circuits 1 through 4 and 5 through 8 by removing the factory installed jumpers across terminals T1 and T2 and terminals B1 and B2. Therefore, the 8-point modules have the potential for two common returns per module.

All discrete 16-point modules provide one common for all 16 outputs, requiring that all 16 circuits be powered from the same voltage source. The factory installed jumpers across terminals T1, T2, and T3 and terminals B1, B2, and B3 distribute the load current to all 16 outputs. For proper 16-point operation, do not remove these jumpers.

Table 2-24 is a list of the discrete output modules available. Module specifications, a wiring diagram, and related information are found in the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070.

Table 2-24 — Discrete Output Modules

MODULE DESCRIPTION	OUTPUTS	CABLE CLASS	MODEL NUMBER
Reed Relay Discrete Output	6	DATA 1, 2, 3	621-0007R
115 Vac Discrete Output	8	DATA 3	621-2100R
115 Vac Isolated Discrete Output	6	DATA 3	621-2101R
115 Vac Self-Protected Discrete Output	8	DATA 3	621-2102R
115 Vac Discrete Output	16	DATA 3	621-2150R
115 Vac Discrete Output	32	DATA 3	621-2175R
230 Vac Discrete Output	8	DATA 3	621-2200R
230 Vac Isolated Discrete Output	6	DATA 3	621-2201R
24 Vac Discrete Output	8	DATA 2	621-2500R
24 Vac Discrete Output	16	DATA 2	621-2550R
5 Vdc Source Discrete Output	8	DATA 1	621-6300R
5 Vdc TTL Source Discrete Output	16	DATA 1	621-6350R
12 Vdc Source Discrete Output	16	DATA 1	621-6450R
12-24 Vdc Source Discrete Output	8	DATA 1, 2*	621-6500R
12-24 Vdc Source Self-Protected Discrete Output	8	DATA 1, 2*	621-6503R
24 Vdc Source Discrete Output	16	DATA 2	621-6550R
24 Vdc Source Low Power Discrete Output	16	DATA 2	621-6551R
24 Vdc Source Discrete Output	32	DATA 2	621-6575R
48 Vdc Source Discrete Output	8	DATA 2	621-6600R
48 Vdc Source Discrete Output	16	DATA 2	621-6650R
120 Vdc Source 0.5 A Discrete Output	8	DATA 3	621-6700R
120 Vdc Source 2 A Discrete Output	8	DATA 3	621-6701R

*The IEEE 518 wiring class is application dependent and can vary with the voltage and current present in the wiring.

2.4.7.4 Analog Output Modules

An analog output module accepts numerical data (digital values) from the Logic Manager Processor and converts the data into equivalent analog voltage and current, providing control of field devices (requiring analog signals) connected to the module's output points. The digital values are translated via 12-bit digital-to-analog converters contained in the module's circuitry.

The converters are addressed by the Logic Manager Processor via a PUSH instruction. Once addressed, a converter can translate the digital value to a corresponding analog voltage or current level within microseconds.

Note that specific connectors or terminal block sets are required and must be ordered separately for each module unless otherwise stated.

Table 2-25 is a list of the analog output modules available. Module specifications, a wiring diagram, and related information are found in the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070.

Table 2-25 — Analog Output Modules

MODULE DESCRIPTION	OUTPUTS	CABLE CLASS	MODEL NUMBER
Analog 4-20 mA Output	4	DATA 1	621-0010-AR
Analog 0-10 V Output	4	DATA 1	621-0010-VR

2.4.7.5 Special Function Modules

The Logic Manager Special Function I/O Modules provide a variety of control and communication functions for the Logic Manager I/O System. The special function modules use the I/O bus for data transfer with the Logic Manager Processor and rely on PUSH/PULL instructions for communications. The PUSH/PULL interface allows these modules to send or receive up to eight 16-bit registers of data at one time.

The modules are listed in numerical order in Table 2-26. Module specifications, a wiring diagram, special instructions, and related information are found in the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070.

Table 2-26 — Special Function Modules

MODULE DESCRIPTION	CABLE CLASS	MODEL NUMBER
BCD Conversion Module	N/A	621-0006R
Pulse Input Module	DATA 1	621-0008 *
Input Simulator Module	N/A	621-0009R
ASCII Communication Module	DATA 1	621-0012R
Absolute Encoder Module	DATA 1	621-0018R
Pulse/Quadrature Input Module	DATA 1	621-0019R
Pulse Input Module	DATA 1	621-0024R
Resistance Temperature Detector Module	DATA1	621-0025R
High-Speed Counter Module	DATA 1	621-0307R
24 Vdc 24-Sink Input, 8-Source Output Module	DATA 2	621-0576R
3-Port MiniCOP Module	DATA 1	627-1002R
4-Port MiniCOP Module	DATA 1	627-1003R

* Superseded by 621-0024R

FAULT ISOLATION

Section 3

This section provides a procedure for isolating faults in the Logic Manager System. It also provides the user with a description of the fault and a recommended action for correcting the fault.

3.1 FAILURE TYPES

Logic Manager System soft failure codes for the Logic Controller (LC) and the Logic Manager Module (LMM) are found in the Logic Manager Node Detail Status Display. See Figure 3-1 and Table 3-1.

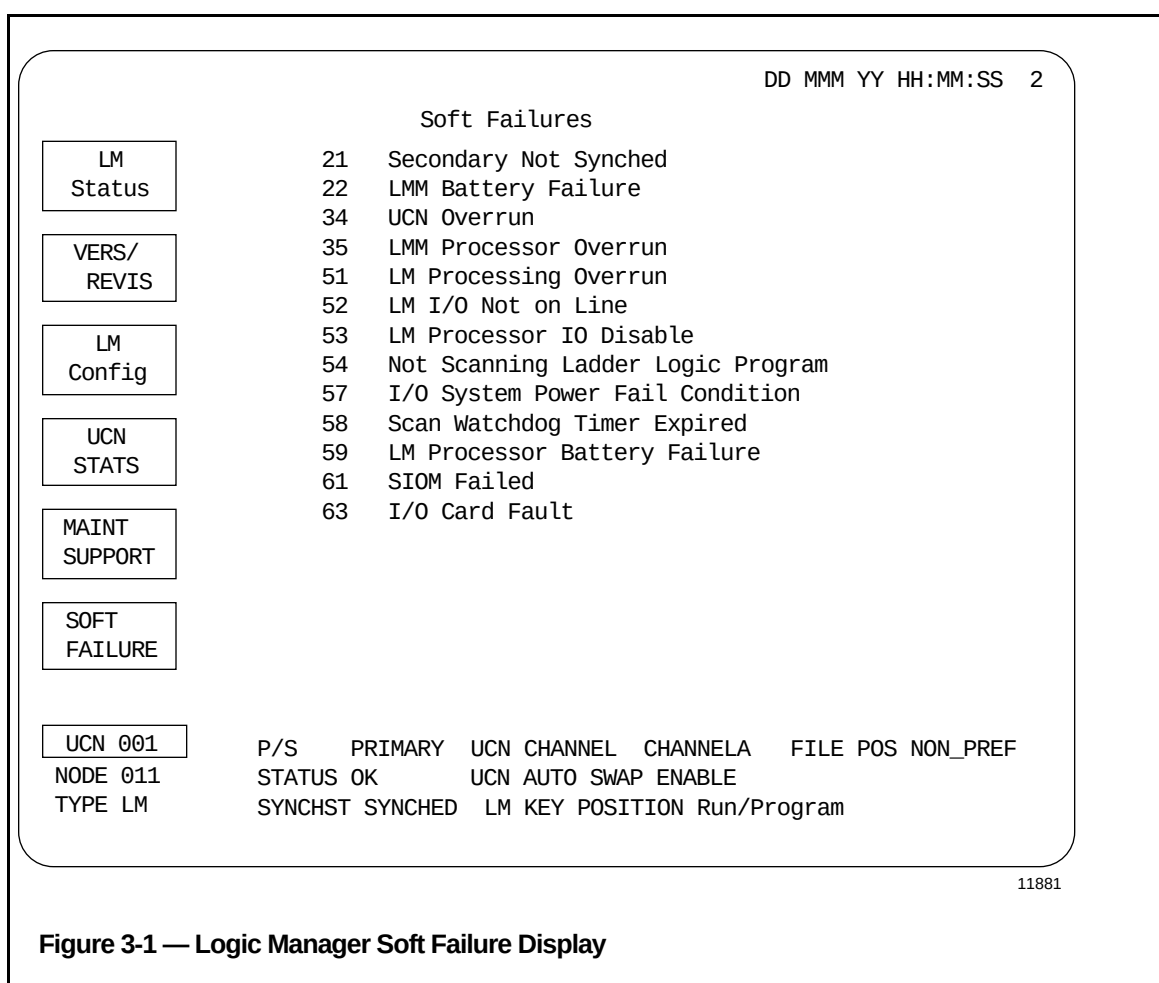


Table 3-1 — Logic Manager Soft Failure Codes

FAILURE CODE	DESCRIPTION	CORRECTIVE ACTION
21	Secondary Not Synched. When the secondary LM comes on line, this softfail is set for $\approx$ 30 seconds. If it persists, the secondary is not ready for a failover. The LC is ready for failover, but the LMM database was never transferred.	If communications are good, reissue a STARTUP or IDLE command to the primary LM. This will initiate three more attempts at synchronization. If this is unsuccessful, it is likely that network communication is not the problem. The status of the UCN and RCM communication is available via the Node Info Status on the Maintenance Information Display.
22	LMM Battery Failure	Replace battery
34	UCN Overrun. Another UCN node on the network is not available or is overloaded. This problem is the result of a failed peer-to-peer communication via logic points.	Correct the problem in the other node(s).
35	LMM Processor Overrun. The LM point database is consistently taking longer than 500 milliseconds to process. Usually this is a result of overloading the LM with too many parameter requests.	Cut back on peer-to-peer and/or LCN access.
51	LM Processing Overrun. The LMM access to the LC has been cut off. The LM should go to the FAILED state if communication is lost. This condition should never exist.	Call the Honeywell Technical Assistance Center (TAC).
52	LMM I/O Not On Line. Ladder logic must exist in the LC to monitor the status of serial I/O (must also be inserted for parallel I/O systems). When this softfail exists, the serial I/O system is not being scanned.	Load ladder logic or correct the serial I/O problem.
53	LM Processor I/O Disable. Indicates the Keyswitch is in the DISABLE mode. On a redundant system, setting the LC in DISABLE would cause a failover before the softfail is asserted.	If ready, change the Keyswitch to another mode.

(Continued)

Table 3-1 — Logic Manager Soft Failure Codes (Continued)

FAILURE CODE	DESCRIPTION	CORRECTIVE ACTION
54	Not Scanning Ladder Logic Program. The LM Processor is in the Software Program Mode.	Check the keyswitch on the LM. Make sure it is in the Run/Program position. Use the MSDOS Loader and clear the Software Program Mode in the LM Processor.
57	I/O System Power Fail Condition. A power failure condition has been detected by the LM and the LM has shut down.	For parallel I/O systems, check for correct input and output of power on the I/O rack power supplies. For serial I/O systems, check for link faults and clear them. After clearing the fault(s), cycle the input power on the LM to reset the fault condition in the SLM. (Note: This soft failure will only occur if the dipswitch setting on the SLM is set to shut down all serial links <u>and</u> to send a CLEAR OUTPUT signal to the I/O upon detection of a link fault.)
58	Scan Watchdog Timer Expired Logic Controller Ladder Logic Scan Tmer has expired.	The Logic Controller's scan time has exceeded 140ms. Either disable the watchdog timer or reduce the amount of relay ladder logic in the Logic Controller.
59	LM Processor Battery Failure.	Replace battery in LM Processor.
61	SIOM Failed. The LM has detected anSIOM failure and depending on the SLM dip switch settings, may have caused a switchover.	Check wiring and connectors for breaks or loose connections. Also check power on failed card file. If no problems are found, replace suspected faulty SIOM. Restart the Serial I/O by cycling power on the LM or by PUSHING a restart value to the SLM.
63	I/O Card Fault. An output type module has failed or certain analog type cards have gone into overcurrent mode.	For hot insertion modules, place the Freeze/Run switch on the SIOM to Freeze. Replace faulty I/O module or correct overcurrent condition. Returning the Freeze/Run switch to Run will then clear the fault. For SIOMs without the Freeze/Run switch, disconnect power, replace faulty module, and then reconnect power.

UCN EXERCISER Section 4

This section explains the operation of the UCN Exerciser program for isolating communications problems to a UCN Node.

4.1 OVERVIEW

This section covers the operation of the UCN Exerciser program that is used for isolating communications problems in a UCN node.

CAUTION

The UCN Exerciser is an Off-process program intended to provide a level of confidence for the connectivity and communication capabilities of an installed network. This test should not be used in an on-process system because the added communication and CPU loads induced by the test may cause throughput and response time degradation of the control system. The test works best if the physical node states of the nodes configured to participate in the test do not change during the test duration period.

Each node must be configured in the NIM to participate in either test.

More than one logical UCN can normally be configured for a single physical UCN. The UCN Exerciser program must be configured to include, for test, only NIMs (and their redundant partners) from the same logical UCN. Configuring NIMs from more than one logical UCN will cause false errors to be indicated for some of the NIMs. This restriction does not apply for other UCN node types, such as a Logic Manager, Advanced Process Manager, or Process Manager.

4.2 APPLICATIONS

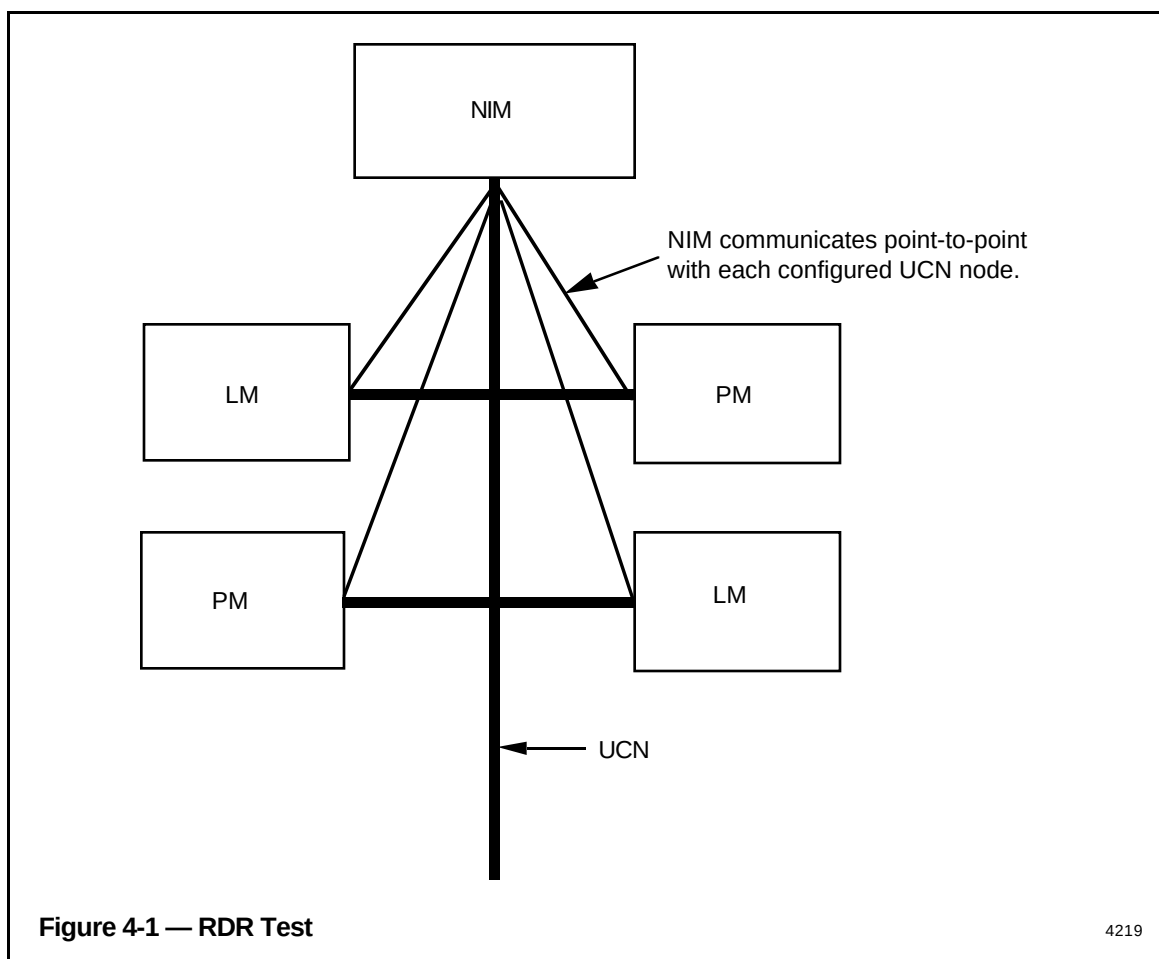
The purposes of the UCN Exerciser is:

- To provide the user with confidence that the UCN is operating properly by sending messages between UCN Nodes.
- To test peer-to-peer communications. The status of peer-to-peer communications between Logic Managers is not shown on the Universal Station (US) display.
- To provide a means of isolating communication problems to a certain LMs by observing the gross level of errors detected.

The test operates in two parts, an RDR test and a Triangulation test. They are performed in succession repeatedly until the test is halted.

4.3 REQUEST DATA RESPONSE (RDR) TEST

In the RDR test, the Test Master (a Network Interface Module) attempts to communicate on a point-to-point basis with every UCN node configured to participate in the test. A node may be ALIVE, BUT NOT TOKEN PASSING and still participate in this test, using the "shadow address" numbered 65-128. See Figure 4-1. This test is to determine with which members of the test the Master can communicate.



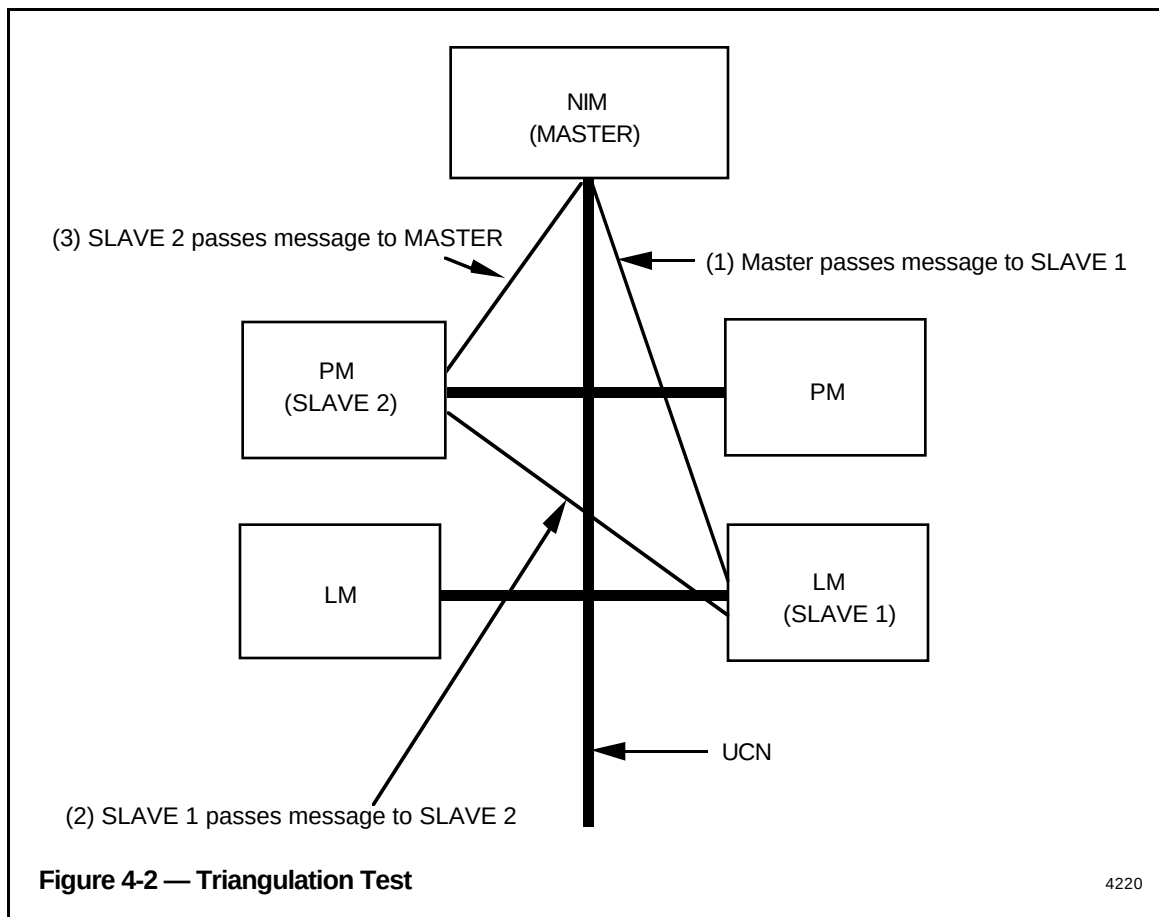
The communication protocol used for this part of the test is the Request-Data-Response (RDR) protocol. All nodes that have power applied, and that are capable of receiving and transmitting messages, should respond to the RDR from the master. A node that does not respond is assumed to be off-network and will not be requested to participate in the triangulation part of the test. A node that does respond, but whose status indicates that it is not capable of token passing, also will not be requested to participate in the triangulation part of the test.

4.4 TRIANGULATION TEST

NOTE

This test requires a minimum of one NIM and one server (other UCN nodes). With only two nodes, the NIM talks point-to-point. Three points are required for true triangulation.

In the triangulation test, only nodes that are configured in the test and that are capable of token passing participate. For example, Logic Managers (LMs) that have not been loaded cannot pass tokens and therefore do not participate. For those nodes that do participate, the master issues a message to another node in the test and requests that the message be forwarded to a third node. The third node in the test then sends a message back to the master, which verifies that the path has completed successfully. See Figure 4-2. This passing of the message checks the following communication path: Master (NIM) to Slave 1 to Slave 2 to Master.



Because a message may be lost, or an error may occur, the Master assumes a reasonable time-out policy for receiving the response back from Slave 2. The time-out policy assumes that each slave in the test is serving only one master. This assumption means that responses that arrive late are considered errors. By examining several paths together, it is usually possible to determine which nodes are experiencing communications problems.

4.5 TEST OPERATION

The operator at the Universal Station (US) can control the operation of the UCN Exerciser. Use the following procedure to perform the UCN Exerciser tests. The display used in the test is explained below. Use the procedure in Table 4-1 to run the UCN Exerciser.

NOTE

If the display does not appear after completing Step 1 above, the Path Name Catalog must be reconstituted, adding the search path **NET>&DSY**.

4.5.1 Starting the UCN Exerciser

The UCN Exerciser is started by pressing the <**SCHEM**> key, typing in **HPNXOVR**, and pressing the <**ENTER**> key.

Page 1 of the UCN Exerciser Overview display will appear as shown in Figure 4-3. The display shows the status of the UCN Exerciser for the LCN nodes 1 through 64 that are NIMs on the LCN. If no status is indicated for the node address, the node is not a NIM.

PAGE Target

Selecting the **PAGE** target will invoke Page 2 of the UCN Exerciser Overview display as shown in Figure 4-4. LCN nodes 65 through 96 are represented.

HELP Target

Selecting the **HELP** target will invoke the UCN Exerciser Help display as shown in Figure 4-5. Selecting the **UCN EX OVERVIEW** target provides a return to the UCN Exerciser Overview display.

4.5.2 UCN Exerciser Overview Displays

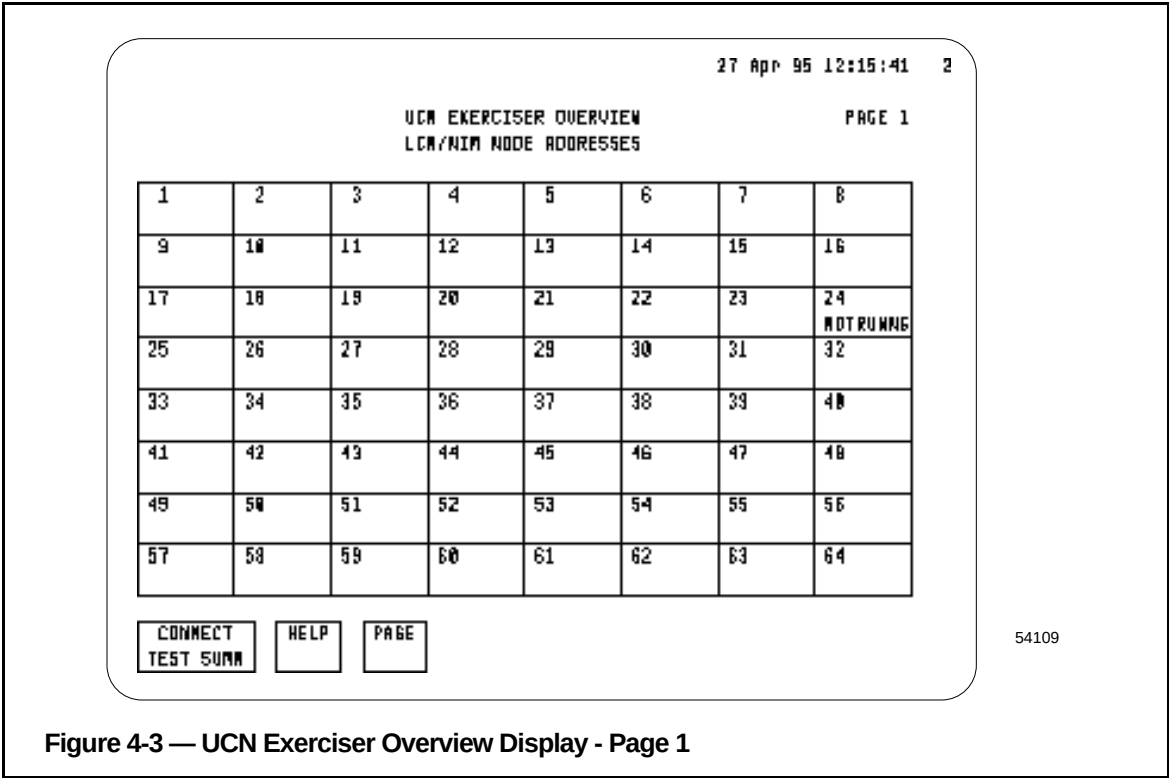


Figure 4-3 — UCN Exerciser Overview Display - Page 1

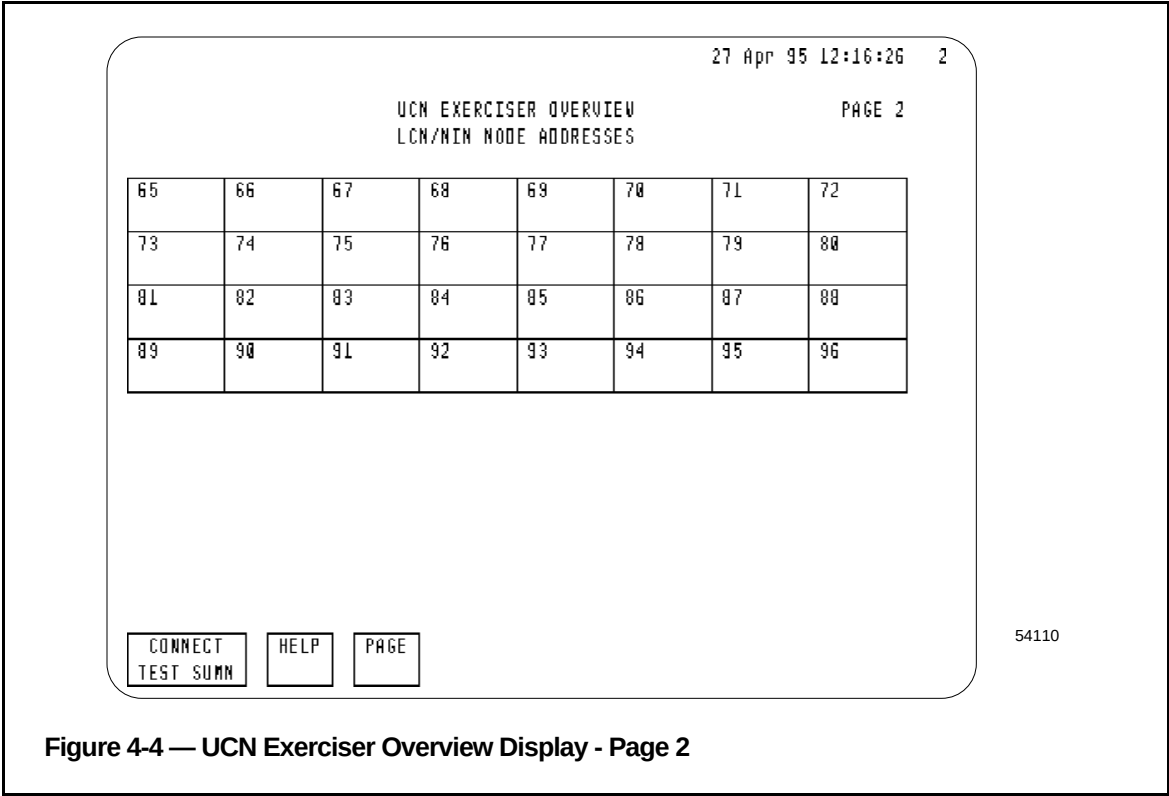
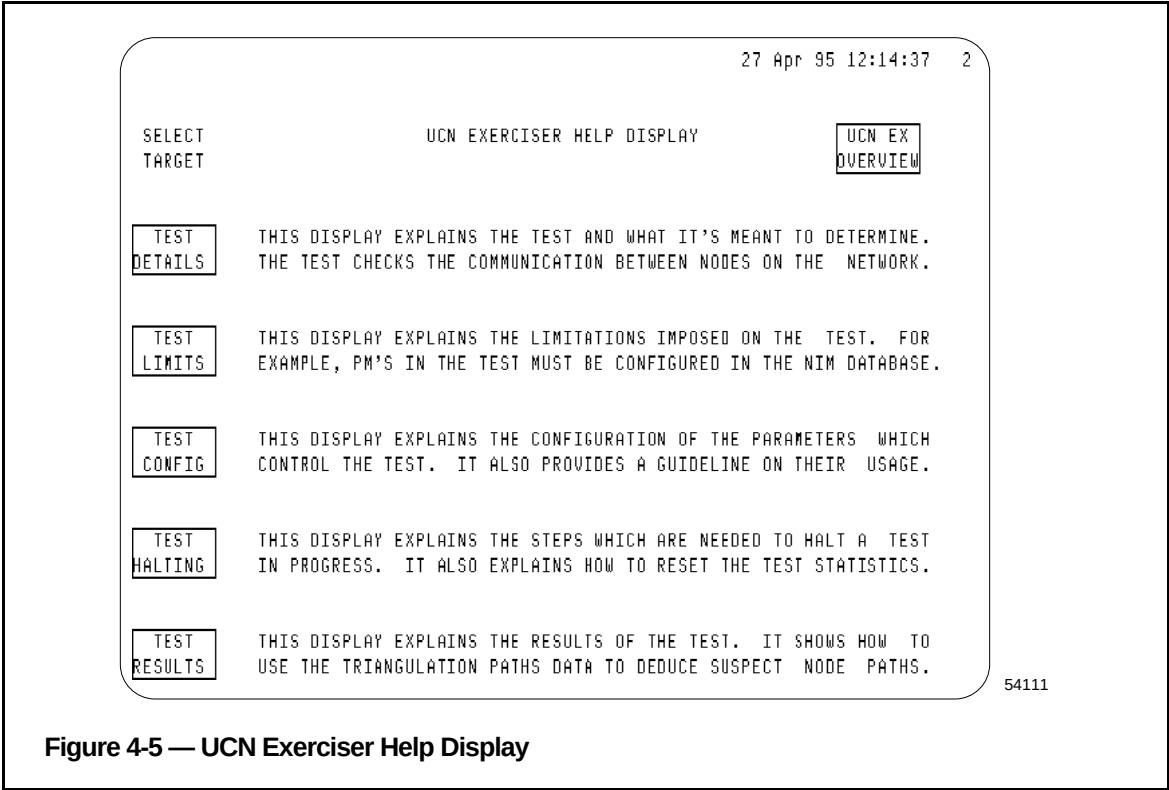


Figure 4-4 — UCN Exerciser Overview Display - Page 2

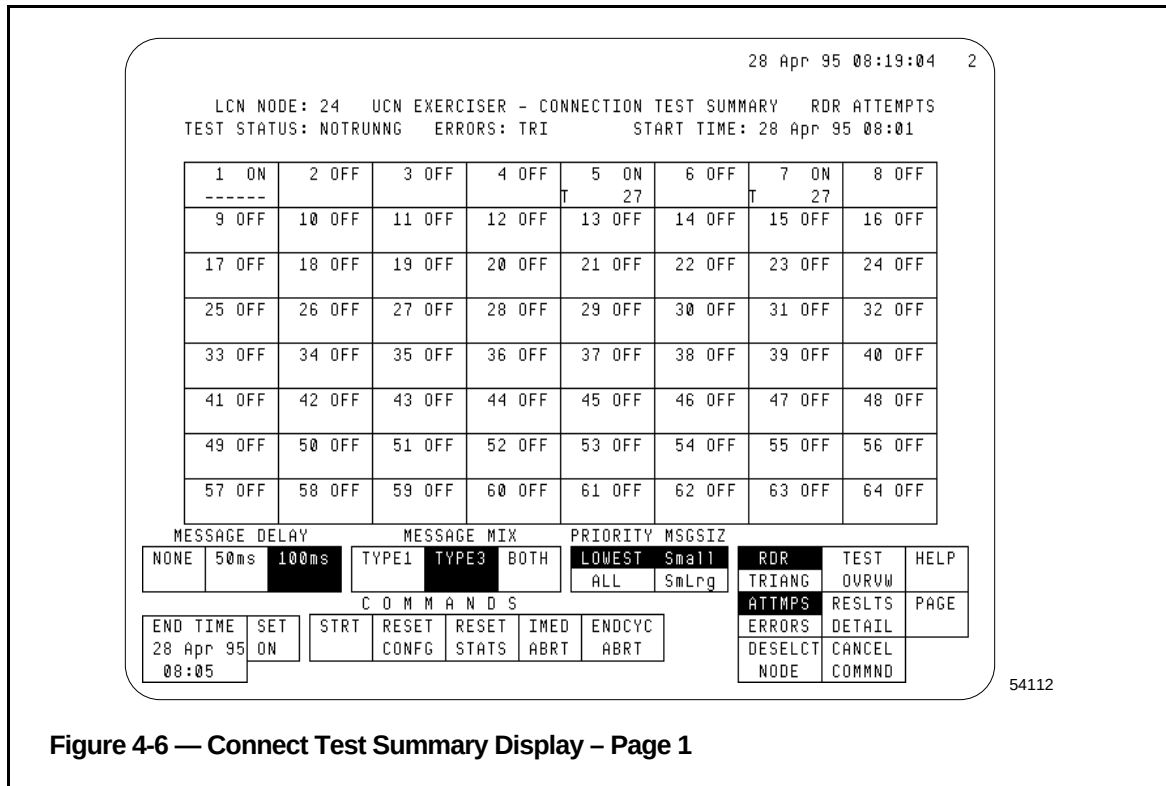
4.5.3 UCN Exerciser Help Display



4.5.4 Connection Test Summary

Each box in the grid on the UCN Exerciser Overview is a target, that when chosen and followed by selection of the **CONNECT TEST SUMM** target, invokes Page 1 of the Connection Test Summary display or the NIM node.

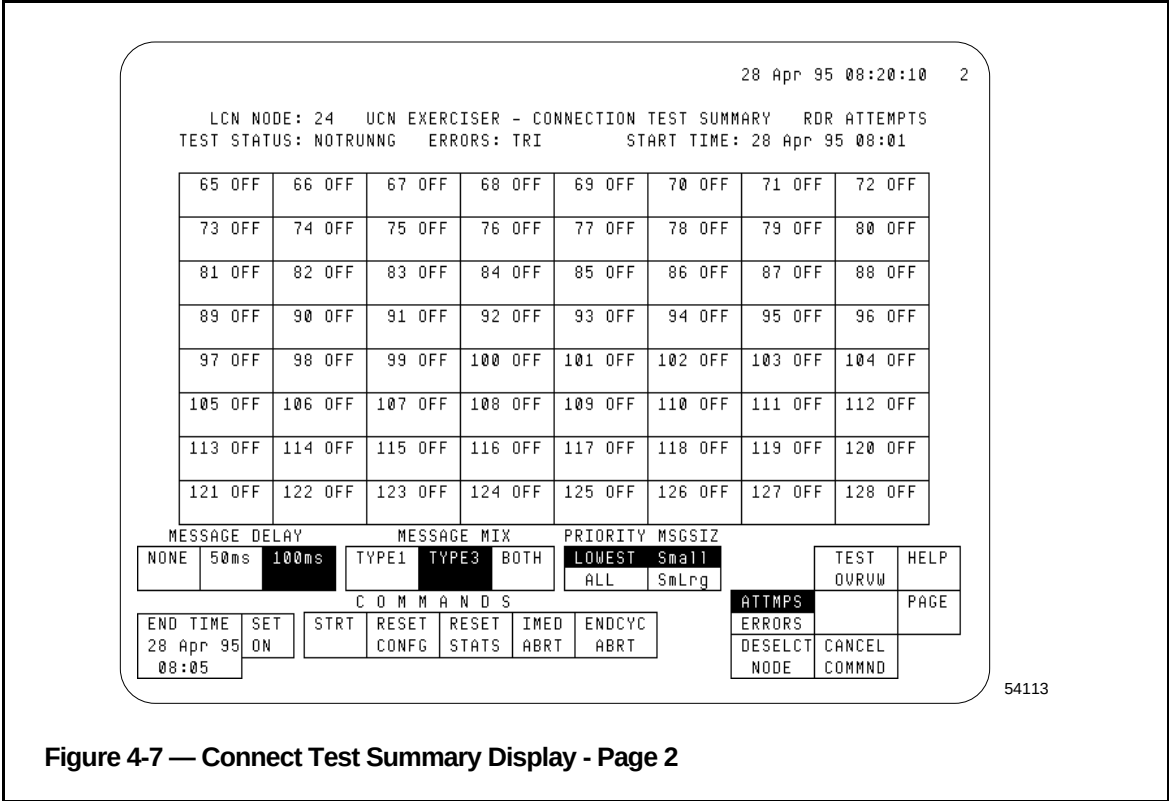
The display is shown in Figure 4-6. Nodes 1 through 64 are represented.



4.5

Selecting the **PAGE** target invokes Page 2 of Connection Test Summary display. Nodes 65 through 128 are represented. The display is shown in Figure 4-7.

Selecting the **PAGE** target provides a return to Page 1 of the Connection Test Summary display.



4.6 UCN Exerciser Test Procedure

Follow the procedure to test the UCN.

Step	Action
1	Start the UCN Exerciser by pressing the < SCHEM > key, type in HPNXOVR , and press the < ENTER > key.
2	Each box in the grid on the UCN Exerciser Overview display is a target, that when chosen and followed by selection of the CONNECT TEST SUMM target, invokes Page 1 of the Connection Test Summary display for the NIM node. Nodes 1 through 64 are represented. Selecting the PAGE target invokes Page 2 of Connection Test Summary display. Nodes 65 through 128 are represented. Make a selection.
3	Select each node on the UCN that is to participate in the test. This is accomplished by choosing the node in the grid and then the SET ON target. Repeat this procedure until all desired nodes are on. For example, Figures 4-6 and 4-7 show nodes 1, 5, and 7 configured for the test. If a node is selected by mistake, select it again and the SET ON target will toggle to SET OFF. Selecting SET OFF deselects that node.
4	Configure the MESSAGE DELAY parameter by selecting the NONE , 50 ms , or 100 ms target. For example, Figures 4-6 and 4-7 show 100 ms selected.
5	Configure the MESSAGE MIX parameter by selecting one of the three targets • TYPE 1 Send only Type 1 messages. These are unacknowledged connectionless messages. • TYPE 2 Send only Type 3 messages. These are acknowledged messages. • BOTH Send both Type 1 and Type 3 messages. Figures 5-6 and 5-7 show TYPE 3 selected.
6	Configure the PRIORITY parameter by selecting one of two targets. • LOWEST Send messages at the lowest priority. • ALL Send messages at each of four priorities.
7	Configure the MSGSZ (MESSAGE SIZE) parameter by selecting one of two targets. • SMALL Send 50 word messages. • SMLRG Send 50 and 100 word messages.
8	Configure the end time date parameter by selecting the END TIME target and entering the date in the format shown. Press < ENTER >. Enter the time parameter in the format shown and press < ENTER >.

continued...

Step	Action
9	Select the STRT (START) target followed by the EXEC (EXECUTE) target to start the test. The EXEC target appears to the right of the ENDCYC (END CYCLE) target when the STRT target has been selected. There are five commands shown in Figures 4-6 and 4-7. To execute the chosen command, the EXEC target must also be selected. The following commands are available. • STRT Starts the RDR and Triangulation test. • RESET CONFIG Resets the configuration to off for all nodes. • RESET STATS Resets the statistics to zero. • IMED ABRT Aborts the test wherever it is in the procedure. • ENDCYC ABRT Aborts the test at the end of the current test cycle.
10	Select the TEST OVRVW (TEST OVERVIEW) target to return to the UCN Exerciser Overview display. See Figures 4-6 and 4-7. Wait for the test to complete. The NOTRUNNG (NOT RUNNING) status appears on the UCN Exerciser Overview display as shown in Figure 4-3.

4.7 ANALYZING RESULTS

To see the results, chose the node of interest and select the **RESLTS DETAIL** (RESULTS DETAIL) target on the Connect Test Summary display. This invokes the Connection Test Detail display as shown in Figure 4-8.

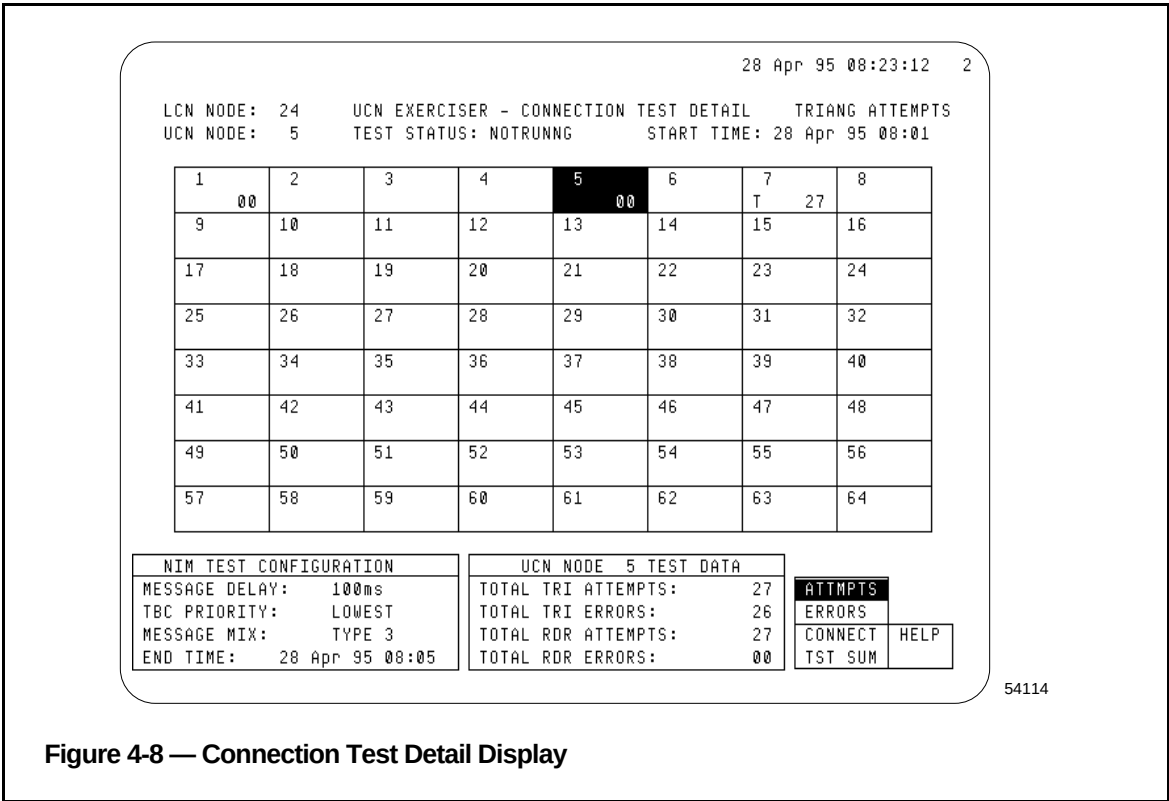


Figure 4-8 — Connection Test Detail Display

Check the boxes in the grid target for nodes 0, 5, and 7. The number in the lower right corner of the box reports the number of attempts if the **ATTMPS** (ATTEMPTS) target is backlighted, or the number of errors if the **ERRORS** target is backlighted on the Connect Test Summary display. The display is selected with the **CONNECT TST SUM** (Connect Test Summary) target.

The node that has the highest number of errors is likely to be the node causing a problem. The problem should be corrected by replacing the **Modem** card, **Advanced Commun** (**Commun**) card, or by checking the UCN cable connections. If this does not correct the problem, call the Technical Assistance Center (TAC) for assistance.

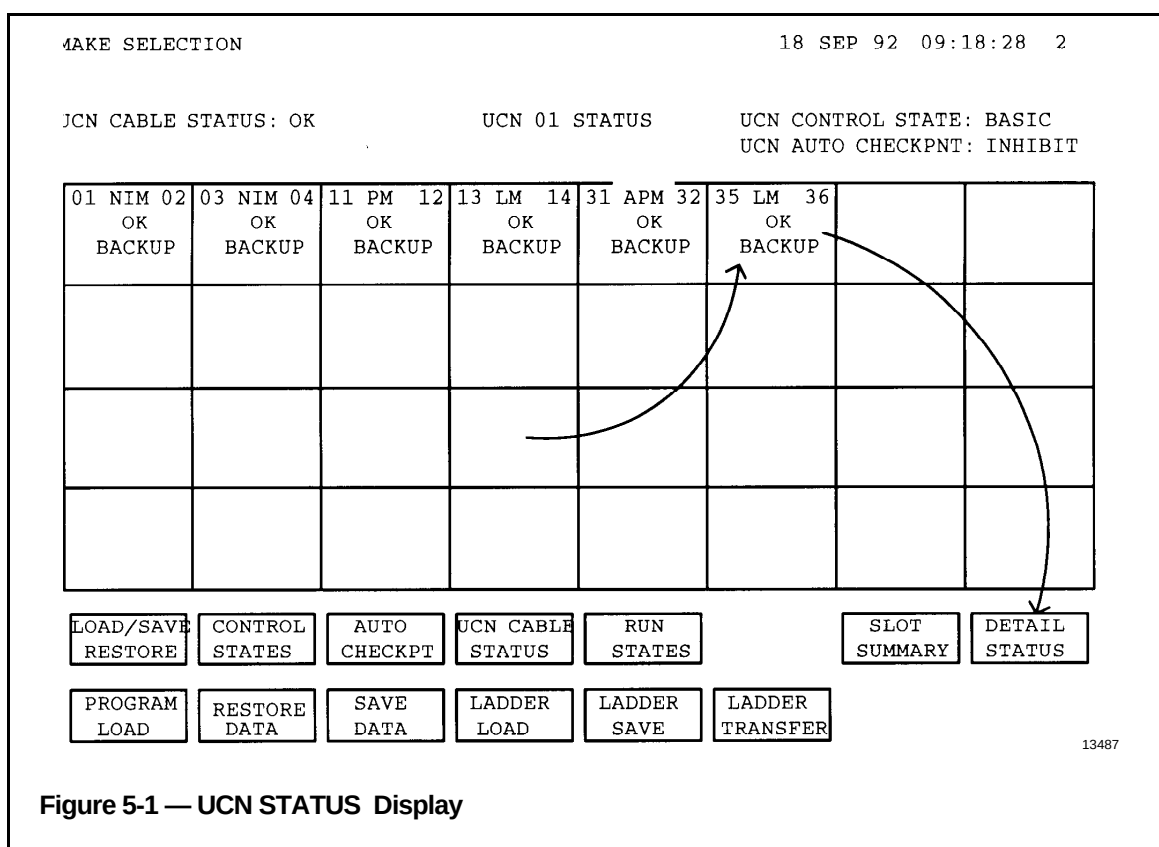
LM/UCN STATISTICS DISPLAYS

Section 5

This section describes the statistics shown on the LM/UCN Statistics displays, their probable cause, the acceptable accumulative total, and system recovery, when applicable.

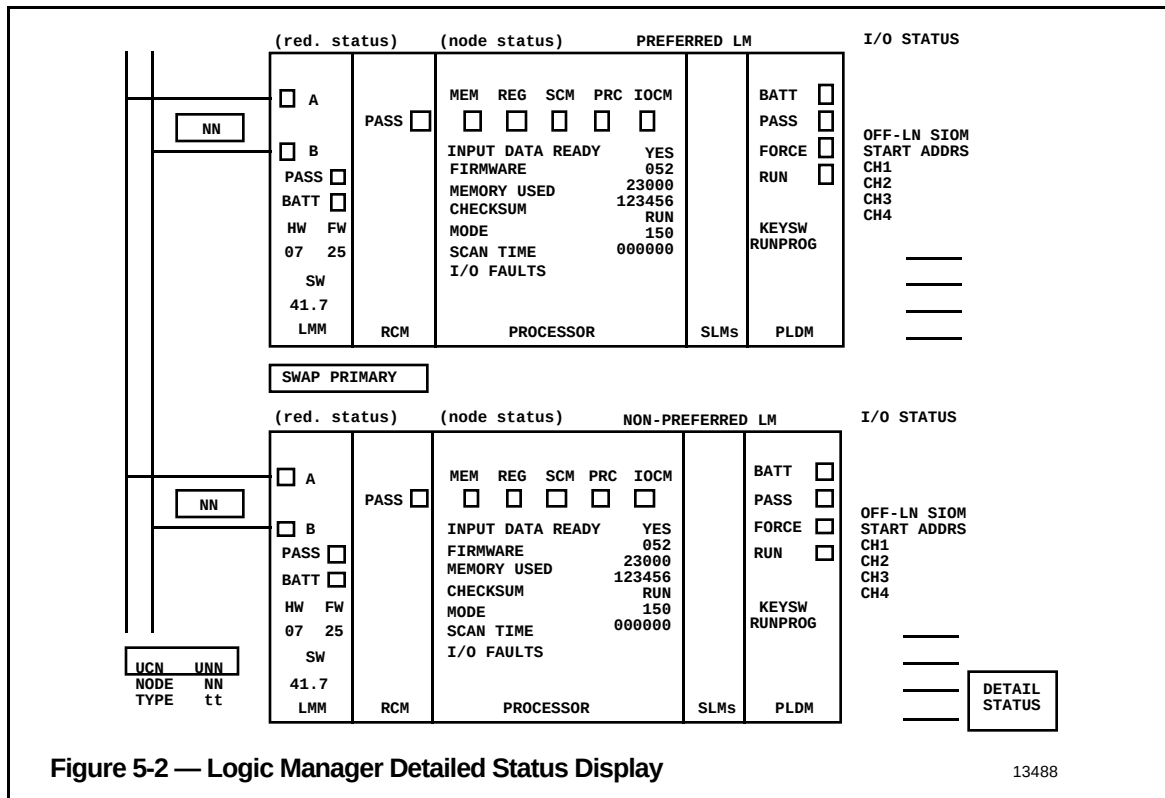
5.1 UCN STATUS DISPLAY

The Logic Manager detailed status display can only be accessed via the UCN STATUS display. The operator “picks” the node for which diagnostic displays are desired, then “picks” the **DETAIL STATUS** target. Refer to figure 5-1.



5.2 LOGIC MANAGER DETAILED STATUS DISPLAY

This display is called up by selecting the node and the **DETAIL STATUS** target at the UCN STATUS Display. From here, the various diagnostic displays can be accessed via the **DETAIL STATUS** target on the **lower right side** of the screen.



Targets:

Swap Primary

This target provides the operator with a mechanism for requesting a redundancy switchover. It is functionally equivalent to a target on the UCN Status Display. It is visible when NODESTS is not equal to OFFNET for both chassis. An ENTER is exposed to the right of this target that completes the request.

System Parameter: PMMCMD

UCN nn - Displays UCN number and acts as a target for calling up (returning to) UCN STATUS DISPLAY.

System Parameter: NTWKNUM

Range: 1 - 20

NN - Identifies node number (1-64) and acts as a target to select the chassis for calling additional detailed status displays.

Detail Status - Acts in conjunction with the NN targets for calling up additional detailed status displays.

Node Status fields:

NODE nn - Node number of the LMM.

System Parameter: NODENUM

Range: 1 - 64

TYPE tt - UCN Node Type.

System Parameter: NODE Typ

Range: NotConfig, LM

(red.status) - Current redundancy status of the LMM.

System Parameter: PMMOPER

Range: NON-REDUNDANT, PRIMARY, SECONDARY

(node status) - Current node status of the LMM.

System Parameter: NODESTS

Range: ALIVE, IDLE, OK, PARTFAIL

LMM Fields (Logic Manager Module)

A, B LEDs - Indicates the two UCN channels, with the small box used to indicate the active channel.

System Parameter: UCNRECHN

PASS LED - Indicates the current self test status of the LMM.

System Parameter: NODESTS (If NODESTS = Fail, Red block, else Green block)

Range:

- 0 = OK
- 2 = PF_IDLE
- 5 = PARTFAIL
- 6 = FAILED
- 7 = ALIVE
- 8 = LOADING
- 9 = POWERON
- 10 = OFFNET
- 12 = NOTCONFIG
- 14 = UNAVAIL

BATT LED - Indicates the current LMM battery status.

System Parameter: PMMSFST[22] - LMM battery softfail

Range:

- 0 = OK (Green)
- 1 = BAD (Red)

HW - Current version of LMM hardware. Visible if the LMM is on-line.

System Parameter: COMHWREV

Color: Cyan

FW - Current version of LMM firmware. Visible if the LMM is on-line.

System Parameter: COMFWREV

Color: Cyan

SW - Current version of LMM software. Visible if the LMM is on-line.
 System Parameter: CONVERS, COMREV
 Color: Cyan (Dot and Strings)

RCM

PASS LED - Indicates the current self test-status of the RCM, valid (visible) only when an RCM is installed and functional.
 Color: Green

Processor Fields

MEM LED - Indicates the current self test status of the Logic Processor's memory module.

System Parameter: MEMMODL

Range:	0 = NPF (not present/failed)	Color: Red
	1 = OK (not present/functioning)	Color: Green
	2 = NF (not functioning)	Color: Red
	3 = NA (not applicable)	Color: Red
	4 = NT (not tested)	Color: Red
	5 = NP (not present)	Color: Red
	6 = IA (inactive)	Color: Red

REG LED - Indicates the current self test status of the Logic Processor's Register module.

System Parameter: REGMODL

Range:	0 = NPF (not present/failed)	Color: Red
	1 = OK (present/functioning)	Color: Green
	2 = NF (not functioning)	Color: Red
	3 = NA (not applicable)	Color: Red
	4 = NT (not tested)	Color: Red
	5 = NP (not present)	Color: Red
	6 = IA (inactive)	Color: Red

SCM LED - Indicates the current self test status of the Logic Processor's System Control Module.

System Parameter: SYCTMODL

Range:	0 = NPF (not present/failed)	Color: Red
	1 = OK (present/functioning)	Color: Green
	2 = NF (not functioning)	Color: Red
	3 = NA (not applicable)	Color: Red
	4 = NT (not tested)	Color: Red
	5 = NP (not present)	Color: Red
	6 = IA (inactive)	Color: Red

PRC LED - Indicates the current self test status of the Logic Processor Module.

System Parameter: PRCSMODL

Range:	0 = NPF (not present/failed)	Color: Red
	1 = OK (present/functioning)	Color: Green
	2 = NF (not functioning)	Color: Red
	3 = NA (not applicable)	Color: Red
	4 = NT (not tested)	Color: Red
	5 = NP (not present)	Color: Red
	6 = IA (inactive)	Color: Red

IOCM LED - Indicates the current self test status of the Logic Processor's System Control Module.

System Parameter: IOCTMODL

Range:	0 = NPF (not present/failed)	Color: Red
	1 = OK (present/functioning)	Color: Green
	2 = NF (not functioning)	Color: Red
	3 = NA (not applicable)	Color: Red
	4 = NT (not tested)	Color: Red
	5 = NP (not present)	Color: Red
	6 = IA (inactive)	Color: Red

INPUT DATA READY - Current status of the I/O PULL DATA READY address (2045).

System Parameter: PMMSFST[52] - I/O Not Ready Softfail

Range:	0 = YES
	1 = NO

FIRMWARE - Current Logic Processor's firmware release number.

System Parameter: PLCREV

MEMORY USED - Indicates the amount of memory used by the relay ladder logic program. A value of 2 (or less) indicates a cleared/empty memory.

System Parameter: LHMEMUSE

CHECKSUM - Current relay ladder logic program checksum.

System Parameter: PMEMCKSM

??? - Visible when the Logic Processor's checksum parameters do not match.

System Parameter: PMEMCKSM

MODE - Indicates the current status of the Logic Processor's mode of operation.

System Parameter: EXECMODE

Range: 0 = PROG (program)
 1 = DSBL (disable)
 2 = RPRG (run/program)
 3 = RUN

SCAN TIME - Indicates the current scan time of the Logic Processor in milliseconds.

System Parameter: SCANTIME

I/O FAULTS - Indicates the current accumulation of I/O faults.

System Parameter: IOMFAULT

PLDM Fields (Parallel Link Driver Module)

BATT LED - Indicates the current status of the Logic Processor's battery.

System Parameter: PMMSFST[59]

Range: 0 = OK Color: Green
 1 = FAILED Color: Red

PASS LED - Indicates the current self test status of the Logic Processor.

System Parameter: SELFTEST

Range: 0 = OK Color: Green
 1 = FAILED Color: Red

FORCE LED - Indicates whether or not the Logic Processor's relay ladder logic contains any forced elements.

System Parameter: FOCEFNC

Range: 0 = NO FORCES Color: Green
 1 = FORCES Color: Red

RUN LED - Indicates the current position of the Logic Processor's keyswitch.

System Parameter: EXECMODE

Range: 0 = PROG (program) Color: Red
 1 = DSBL (disable) Color: Red
 2 = RPRG (run/program) Color: Green
 3 = RUN Color: Green

I/O Status Fields

The following become visible beneath the “I/O Status” label when the indicated softfail occurs.

DISABLED - Visible when a I/O disabled softfail is detected.

System Parameter: softfail parameter PMMSFST[53] = TRUE

POWER FAIL - Visible when an I/O power fail softfail is detected.

System Parameter: softfail parameter PMMSFST[57] = TRUE

CARD FAULT - Visible when an I/O card fault softfail is detected.

System Parameter: softfail parameter PMMSFST[63] = TRUE

SIOM FAULT - Visible when a SIOM fault softfail is detected.

System Parameter: softfail parameter PMMSFST[61] = TRUE

OFF LINE SIOM STARTING ADDRESS

This table lists the starting addresses of I/O racks that have gone off line. The four fields apply to the four channels that exist in the I/O system (2 SLMs per Logic Processor, 2 channels per SLM). On any given channel, if more than one are off line, then only the first posted is listed in the respective field. As racks go off line and back on line, they are listed in order of the first posted, first listed.

Dashes (when value = 9999) - indicate no off-line SIOMS visible when NODESTS <> OFFNET.

System Parameter: ALMOLSIO[4]

Range: Default = 9999 = No off-line SIOMS; else 0-2040

Conditionally Visible Warnings

Swap Primary Field

WARNING: NO CONTROL PROGRAM - A warning message visible when either the Primary or Secondary memory used count (LHMEMUSE) is less than 3.

Color: Red

WARNING: LC CHECKSUM MISMATCH - A warning message visible when the Primary and Secondary PMEMCKSM parameters mismatch.

Color: Red

WARNING: FORCE COUNT MISMATCH - A warning message visible when the Primary and Secondary FRCCOUNT parameters mismatch.

Color: Red

WARNING: LMM PROGRAM MISMATCH - A warning message visible when the Primary and Secondary CONVERS and COMREV parameters mismatch.
Color: Red

LMM Field

(flt) - A warning message related to LMM status. Applies with precedence in the order shown:

No Prg - Informs the Operator there is no personality loaded in the LMM. Visible when NODESTS = ALIVE or FAIL.
Color: Red

No DB - Informs the operator that the LMM database memory is scrubbed. Visible when the Point Mix parameters are all zero.
Color: Red

5.3 LM SELF TEST DISPLAY

This display will be the first display shown when the **DETAIL STATUS** target is selected at the Logic Manager Detailed Status Display. The **I/O SYSTEM STATUS** display or the **HARDWARE STATUS** display can be chosen from this display.

The **SELF TEST** display is chosen by selecting **DETAIL STATUS** from the target at the lower right side of any LM Detailed Status Display.

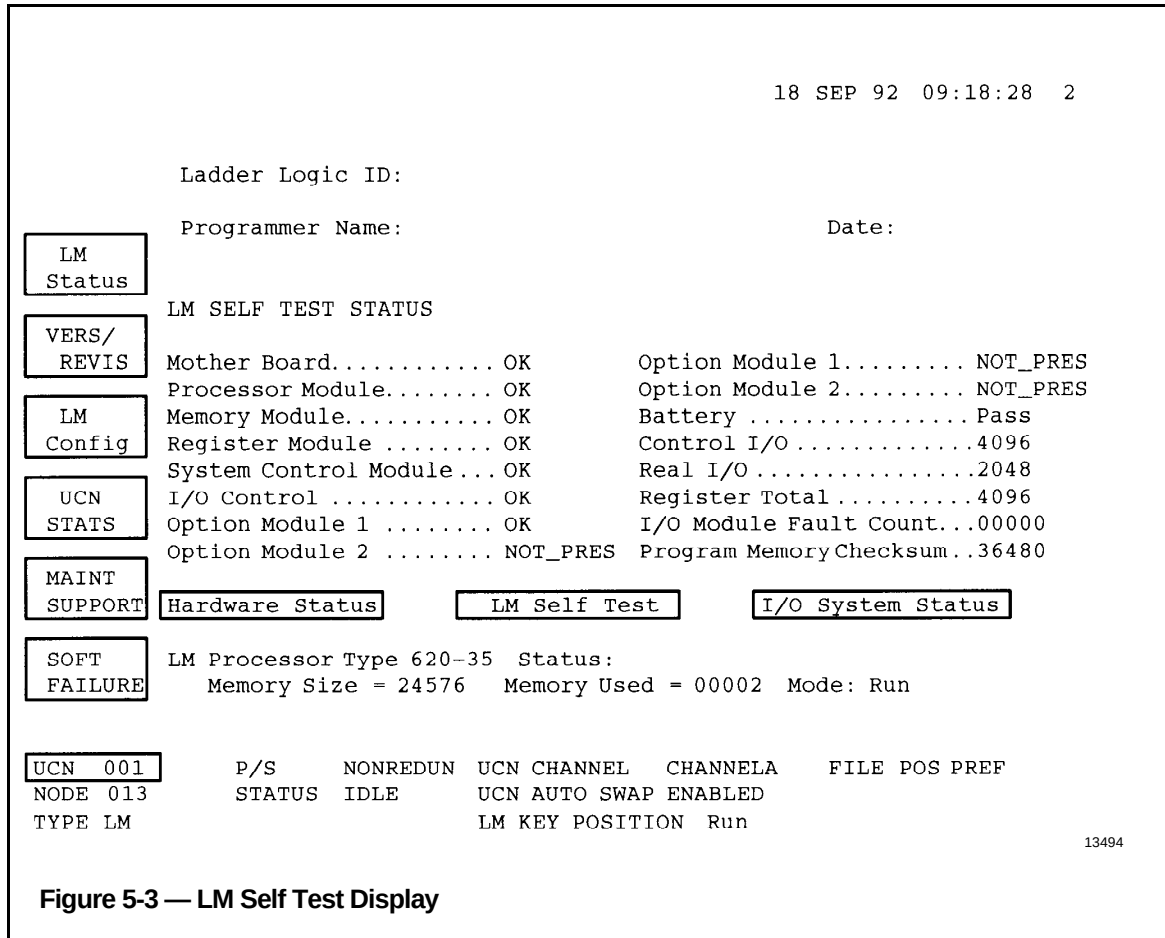


Figure 5-3 — LM Self Test Display

The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display represents an overview of LM's 620 PLC hardware status. The LMM pulls this information from the 620 LCN System Status Table. Similar data is available via the 620 Loader Diagnostic Displays.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFTFAILURE, UCN xxx - see section on LM Blank Status Display.

Hardware Status - selects this same display.

LM Self Test - selects the LM Self Test page of the LM STATUS Displays.

I/O System Status - selects the I/O System Status page of the LM STATUS Display.

Fields:

Ladder Logic ID - reflects the 620 Processor program name information.

Programmer Name - reflects the 620 Processor programmer name information.

Date - reflect the 620 Processor Program date information.

Motherboard - reflects result of 620 Processor self-testing of its motherboard (OK, FAIL).

Processor Module - reflects result of 620 Processor self-testing of its Processor Module (OK, FAIL).

Memory Module - reflects result of 620 Processor self-testing of its Memory Module (OK, FAIL).

Register Module - reflects result of 620 Processor self-testing of its Register Module (OK, FAIL).

System Control Module - reflects result of 620 Processor self-testing of its System Control (OK, FAIL).

I/O Control - reflects result of 620 Processor self-testing of its I/O Control Module (OK, FAIL).

Option Module (n) - reflects status of up to four Option Modules (OK, Not Present). When following the Installation Guidelines, Option Module 1 represents the RCM (if redundant) or the LMM itself. If redundant, Option Module 2 would represent the LMM.

The 620 Processor fills in these four locations as it finds option modules. These four numbers do not correspond to option module backplane addresses. For example, if two option modules are installed with backplane addresses of 1 and 3, they will show up on this display as Option Module 1 and 2, with the device with the lowest address listed first (Optional Module 1). If an option module is removed, the Processor, when it restarts, will shift any card with higher addresses into the lower slots.

Battery - reflects result of 620 Processor self-testing of its battery (PASS, FAIL).

Control I/O - size of the 620 processor's I/O Status Table (2048,4096).

Real I/O - size of the 620 Processor's Real I/O space (1024, 4096).

Register Total - size of the 620 Processor's Register Table (2048, 4096).

I/O Module Fault Count - number of I/O Module Fault Counts recorded within the 620 Processor System Status Table.

Program Memory Checksum - a checksum value calculated by the 620 Processor against its program memory.

This value will, of course, change if the program changes. So as not to cause a major increase in scan time (while calculating this number), the 620 Processor distributes the task over multiple scans. Specifically, the calculation proceeds at a rate of twelve (12) program memory words per 620 Processor scan, e.g., a 1200 word program would take 100 scans. If the scan time was 100 msec, this would amount to 10 seconds.

IMPORTANT - DO NOT ATTEMPT REDUNDANCY SWITCHOVERS WHILE THIS VALUE IS BEING CALCULATED, I.E., JUST AFTER AN ON-LINE PROGRAM CHANGE OCCURS. Use this field to monitor the checksum during such an operation.

LM Processor Type - indicates which of the family of 620 LCS processors is hosting the LM. Limited to the 620-35.

Status - not used (blank).

Memory Size - size of 620 Processor program memory. Usually 24K.

Memory Used - amount of program memory current used. A cleared program memory should indicate a value of 2.

Mode - 620 Processor mode (Program, Run).

5.4 I/O SYSTEM STATUS DISPLAY

This display is called:

by selecting the **LM SELF TEST** target from either the **HARDWARE STATUS** or **I/O SYSTEM STATUS** displays.

18 SEP 92 09:18:28 2

Ladder Logic ID:

Programmer Name:

Date:

I/O SYSTEM STATUS

I/O Module Fault Count: 0000

Most Significant Address of Faulted Module

		Off Line SIOMs Starting Address
1.	00000	1. 00000
2.	00000	2. 00000
3.	00000	3. 00000
4.	00000	4. 00000
5.	00000	
6.	00000	
7.	00000	
8.	00000	

LM
Status

VERS/
REVIS

LM
Config

UCN
STATS

MAINT
SUPPORT

SOFT
FAILURE

Hardware Status

LM Self Test

I/O System Status

LM Processor Type 620-35 Status:

Memory Size = 24576 Memory Used = 00002 Mode: Run

UCN 001
NODE 013
TYPE LM

P/S	NONREDUN	UCN CHANNEL	CHANNELA	FILE POS PREF
STATUS	IDLE	UCN AUTO SWAP	ENABLED	
		LM KEY POSITION	Run	

13495

Figure 5-4 — I/O System Status Display

The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display represents an overview of LM's 620 PLC hardware status. The LMM pulls this information from the 620 LCS System Status Table. A similar data is available via the 620 Loader Diagnostic Displays.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxx - see section on LM Blank Status Display.

Hardware Status - selects this same display.

LM Self Test - selects the LM Self Test page of the LM STATUS Displays.

I/O System Status - selects the I/O System Status page of the LM STATUS Display.

Fields:

Ladder Logic ID - reflects the 620 Processor program name information.

Programmer Name - reflects the 620 Processor programmer name information.

Date - reflects the 620 Processor program date information.

I/O Module Fault Count - number of I/O Module Fault Counts recorded within the 620 Processor System Status Table.

Most Significant Address of Faulted Module - Up to eight (8) card address collected by the SLM(s). These addresses pertain to I/O cards that have reported faults.

Off Line SIOMs Starting Address - This table lists the starting addresses of I/O racks that have gone off-line. The four fields apply to the four channels that can exist in the I/O System (2 SLMs per Processor, 2 channels per SLM).

On any given channel, if more than one are off-line, then only the first posted is listed in the respective field. As racks go off and back on line, they are listed in order of first posted, first listed.

LM Processor Type - indicates which of the family of 620 LCS processors is hosting the LM. Limited to the 620-35.

Status - not used (blank).

Memory Size - size of 620 Processor program memory. Usually 24K.

Memory Used - amount of program memory current used. A cleared program memory should indicate a value of 2.

Mode - 620 Processor mode (Program, Run).

5.5 LM HARDWARE STATUS DISPLAY

This display is called upon:

by selecting the **HARDWARE STATUS** target from either the **LM SELF TEST** or **I/O SYSTEM STATUS** displays.

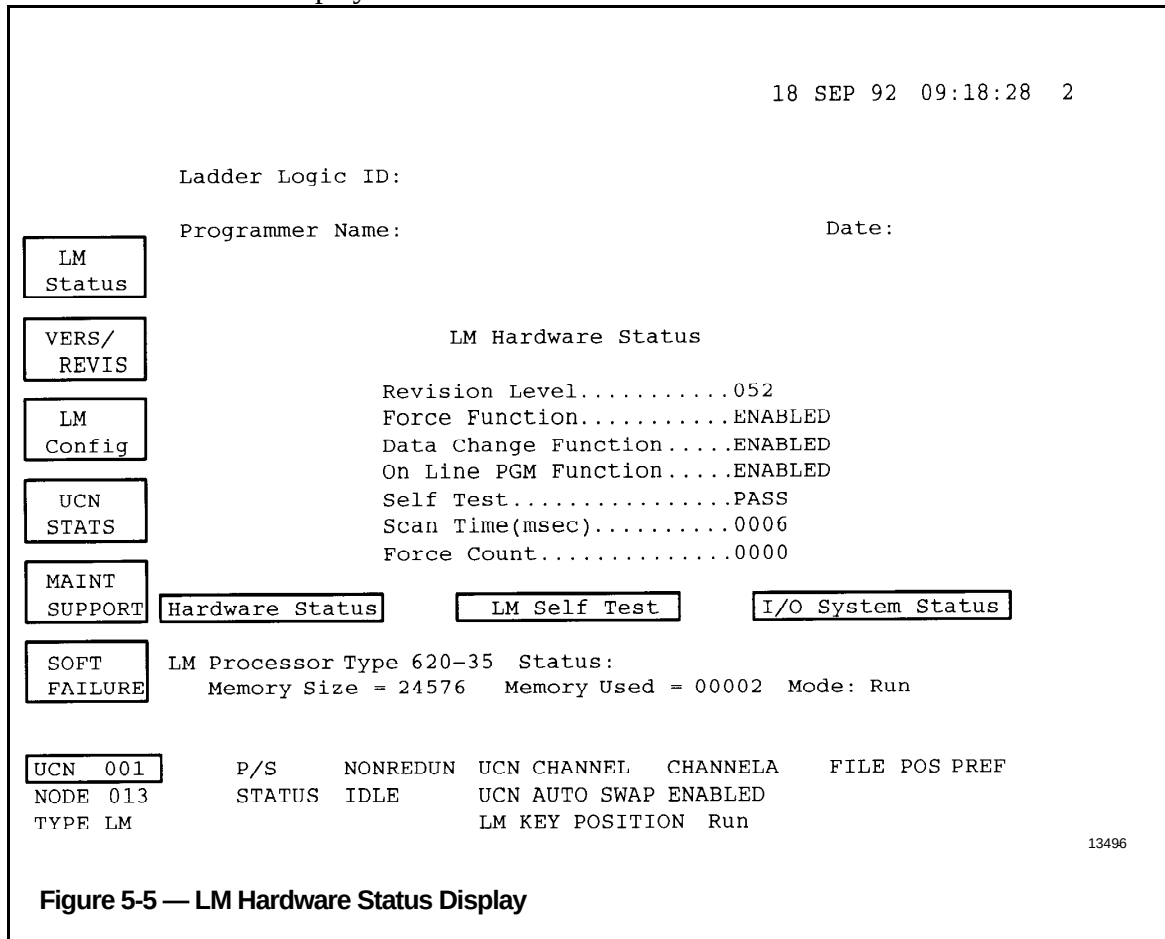


Figure 5-5 — LM Hardware Status Display

The target along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display represents an overview of LM's 620 PLC hardware status. The LMM pulls this information from the 620 LCS System Status Table. Similar data is available via the 620 Loader Diagnostic Displays.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxx - see section on LM Blank Status Display.

Hardware Status - selects this same display.

LM Self Test - selects the LM Self Test page of the LM STATUS Displays.

I/O System Status - selects the I/O System Status page of the LM STATUS Displays.

Fields:

Ladder Logic ID - reflects the 620 Processor program name information.

Programmer Name - reflects the 620 Processor programmer name information.

Date - reflects the 620 Processor program date information.

Revision Level - 620 Processor firmware revision level.

Force Function - 620 Processor force function status (ENABLED, DISABLED).

Data Change Function - 620 Processor data change function (ENABLED, DISABLED).

On Line PGM Function - 620 Processor on-line programming function (ENABLED, DISABLED).

Self Test - 620 Processor self-test status (PASS, FAIL).

Scan Time (msec) - 620 Processor scan time in milliseconds.

Force Count - 620 Processor force count.

LM Processor Type - indicates which of the family of 620 LCS processors is hosting the LM. Limited to the 620-35.

Status - not used (blank).

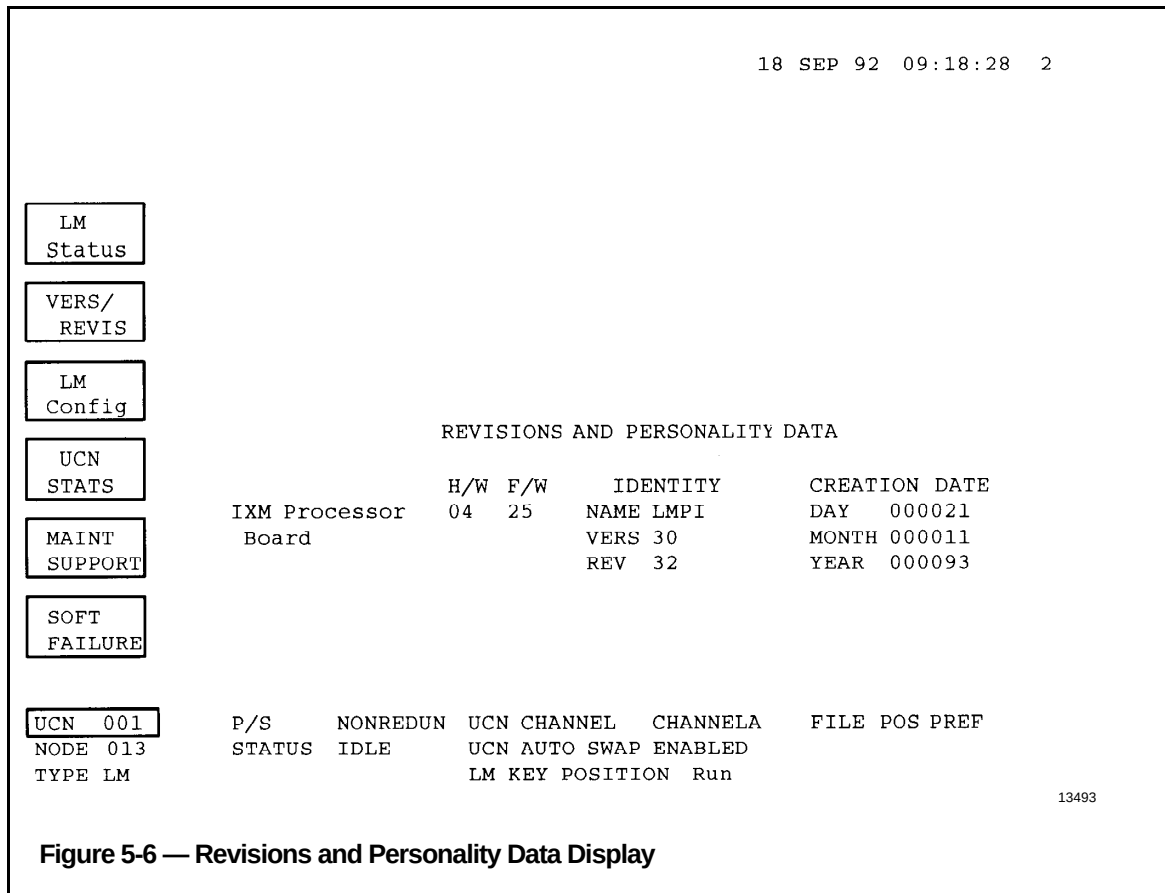
Memory Size - size of 620 Processor program memory. Usually 24K.

Memory Used - amount of program memory current used. A cleared program memory should indicate a value of 2.

Mode - 620 Processor mode (Program, Run).

5.6 REVISIONS AND PERSONALITY DISPLAY

This display is called up from the **VERS/REVIS** target of any of the LM Detailed Status Displays except figure 5-1, the Logic Manager Detailed Status display.



The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display presents identification information for the LMM hardware, firmware, and Personality.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAIN SUPPORT, SOFT FAILURE, UCN xxx - see section on LM Blank Status Display.

Fields:

H/W - LMM Processor Board hardware version.

F/W - LMM Processor Board firmware version.

NAME - Personality name (always LMPI).

VERS - Personality version.

REV - Personality revision.

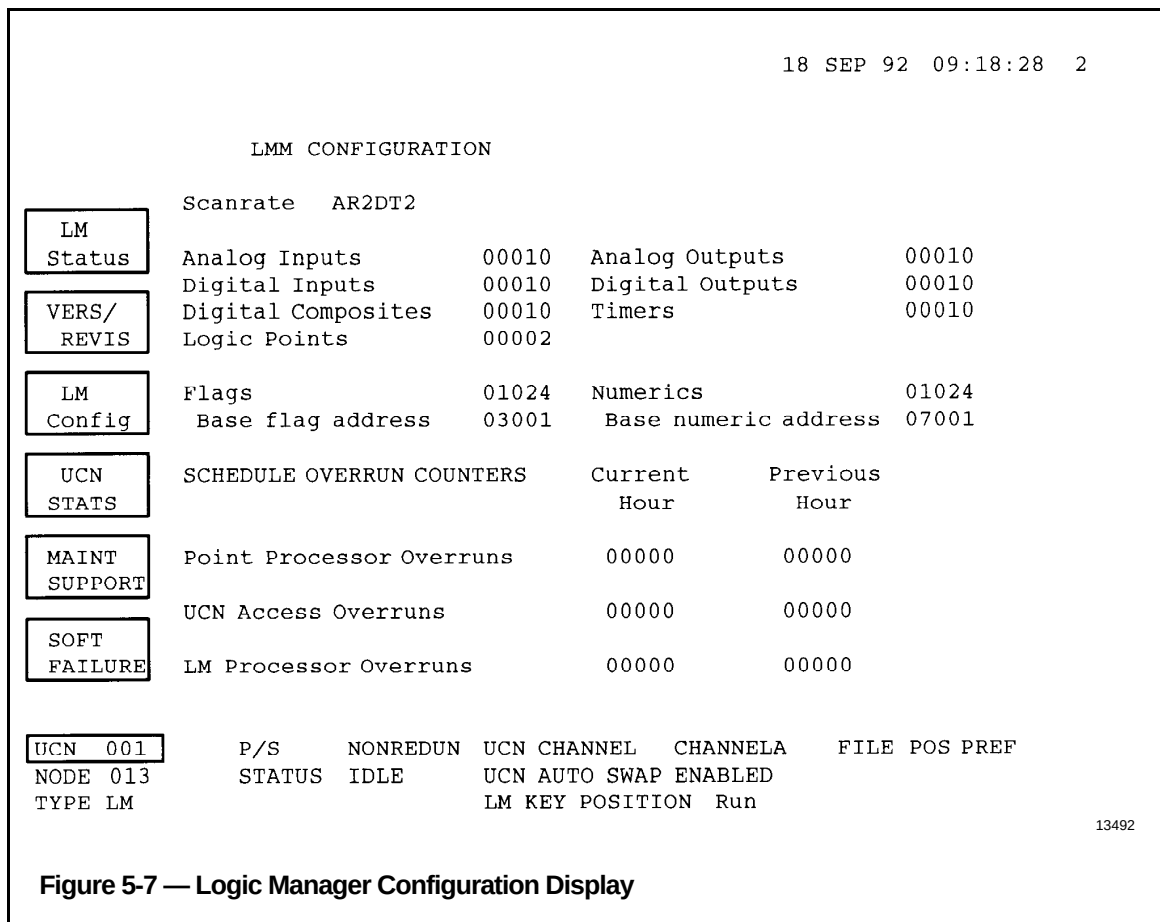
DAY - Personality creation day.

MONTH - Personality creation month.

YEAR - Personality creation year.

5.7 LOGIC MANAGER CONFIGURATION DISPLAY

This display is called up from the **LM Config** target of any LM Detailed Status Display.



The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display summarizes the LMM's Point Mix and the data access overrun statistics.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxx - see section on LM Blank Status Display.

Fields:

Scanrate - LMM Point Processor scan rate AR represents analog point scans, with the number following being 1 or 2 (once or twice per second). DT represents digital timer, and logic point scans, with only the value of 2 available for LM.

System parameter: SCANRATE

Range: AR1DT2, AR2DT2

Analog Inputs - Number of AI points available.

System Parameter: NAISLOT

Range: 0-127/254

Analog Outputs - Number of AO points available.

System parameter: NAOSLOT

Range: 0-482/965

Digital Inputs - Number of DI points available.

System parameter: NDISLOT

Range: 0-1866

Digital Outputs - Number of DO points available.

System parameter: NDOSLOT

Range: 0-4000

Digital Composites - Number of DC points available.

System parameter: NDCSLOT

Range: 0-304

Timers - Number of Timer points available.

System parameter: NTIMER

Range: 0-700

Logic Points - Number of Logic points available.

System parameter: NLOGSLOT

Range: 0-14

Flags - Number of Flag points available.

System parameter: NFLAG

Range: 0-1024

Base Flag address - Starting address (least significant), within the logic controller of the block of flags.

System parameter: FLLSBA

Range: -1 (not configured), 0-4095

Numerics - Number of Numeric points available.

System parameter: NNUMERICS

Range: 0-1024

Base numeric address - Starting address (least significant), within the logic controller, of the block of numerics.

System parameter: NNLSBA

Range: -1 (not configured), 4096 to 8191

Point Processor Overruns - Number of times LMM was unable to complete a full scan of its points within the allotted 1/2 or 1 second. Such overruns are usually caused by a burst of activity (e.g., alarms or parameter requests) and should only be a cause for concern if they occur frequently. If this number accumulates steadily, it usually means the LMM is receiving an excessive amount of parameter requests.

System parameter: CRPPXORN, LSPPXORN

Range:

UCN Access Overrun - Number of times LMM was unable to collect UCN data in time to start its next scan. This involves Logic Point inputs connected to points on other UCN nodes. Such overruns are usually caused by a burst of activity (e.g., alarms or parameter requests) and should only be a cause for concern if they occur frequently. Excessive loading at the LMM or at one or more remote nodes could cause frequent overruns, a situation which should be corrected.

System parameter: CRUCNORN, LSUCNORN

Range:

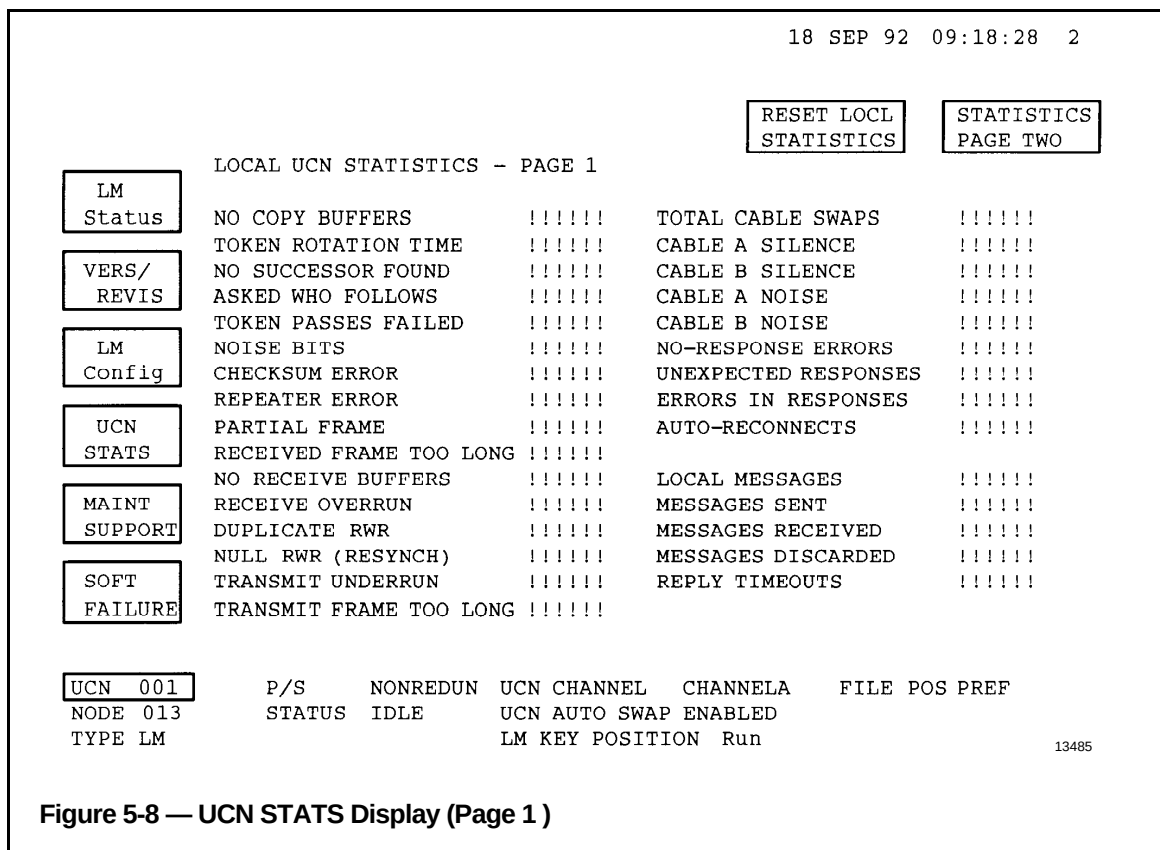
LM Processor Overruns - Number of times LMM was unable to collect fresh Logic Controller data in time to start its next scan. Such overruns are usually caused by a burst of activity (e.g., alarms or parameter requests) and should only be a cause for concern if they occur frequently. Excessive logic controller scan times (approaching LMM scan time) could cause such overruns.

System parameter:

Range:

5.8 UCN STATS DISPLAY

This display is called up from the **UCN STATS** target of any LM Detailed Status Display. The display may also be called up from the **UCN STATS** display, Page 2, by selecting the **STATISTICS PAGE ONE** target.



The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

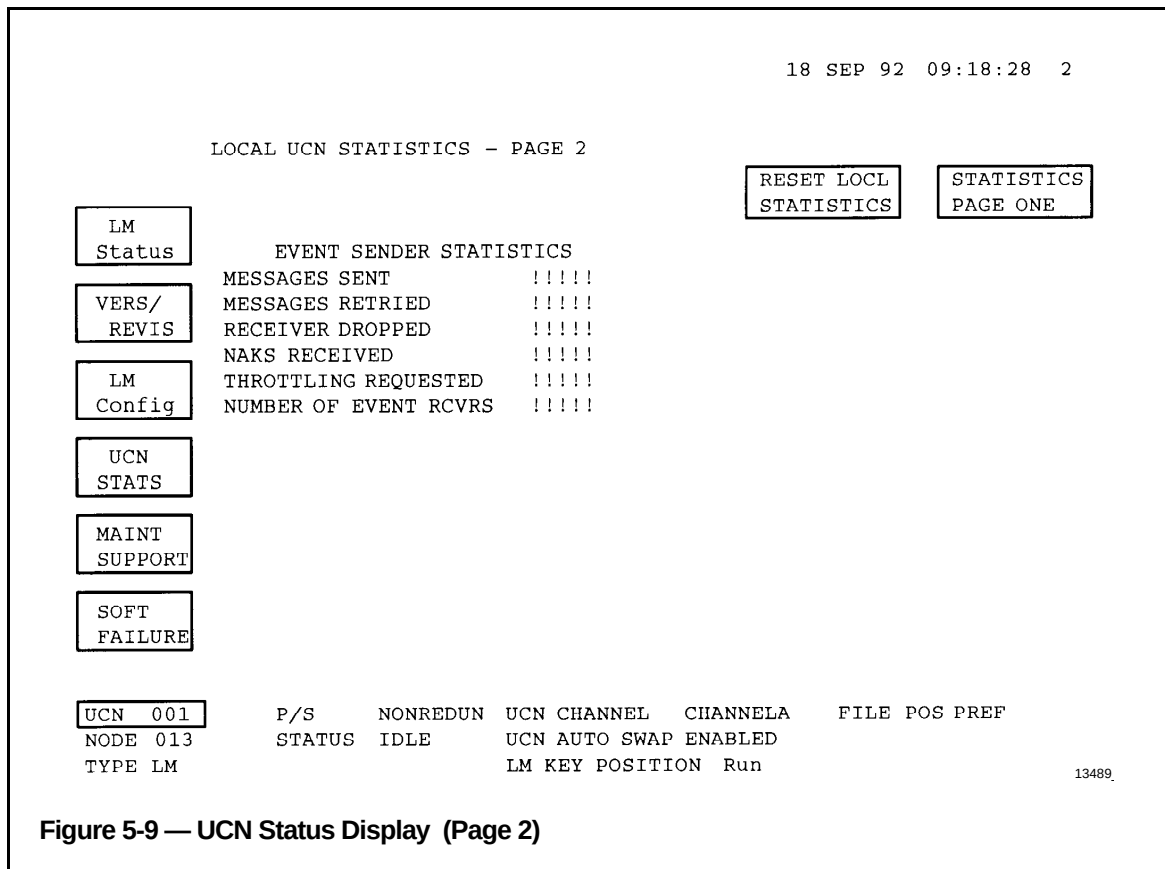
The display, in conjunction with PAGE 2, summarizes the LMM's UCN Error statistics.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxxx - see section on LM Blank Status Display.

Page 2 is selected by touching the **STATISTICS PAGE TWO** target.

This display is called up from the **UCN STATUS** target of any LM Detailed Status Display, followed by the selection of the **STATISTICS PAGE TWO** target.



The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display, in conjunction with PAGE 1, summarizes the LMM's UCN error statistics.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxx - see section of LM Blank Status Display.

Descriptions of the LM Local UCN Statistics Display's statistics are found in Tables 5-1 and 5-2.

Table 5-1 — Local UCN Statistics Descriptions

Statistic	Description
No Copy Buffers	The number of times no buffers on the LCN side of PNI/EPNI for copying received messages (NIM only) were available. The count is normally zero. Increasing counts indicate extreme NIM congestion. The LLC increments the count. The situation can lead to a “No Receive Buffers” count.
Token Rotation Time	NIM only. Sampled, the averaged token rotation time in 0.1 millisecond units. This is not an absolute measurement, and only roughly correlates to the Concord Token Scope. A 2-node UCN with no traffic will be in the 0.1 millisecond range, while a moderately-loaded 64-node UCN will be in the 4-6 millisecond range. OFFNET nodes and heavy traffic will increase the observed token rotation time. The nominal token rotation time for the system should be recorded when there are no errors and the load is moderate. Deviations from the count noted in a smoothly operating system should be investigated. An abnormally slow token rotation time may be caused by a level of trunk noise not quite high enough to cause a cable swap.
No Successor Found	The ring collapsed, and token-passing was lost. The count is incremented once in most nodes for a ring collapse. It is preceded by an “Asked Who Follows” count in the node that had the token. The count is incremented by the TBC.
Asked Who Follows	The number of times the successor node dropped out of the ring, temporarily or permanently (shutdown, failure, failover). It does not change in a system that is running normally. The count is incremented by the TBC.
Token Passes Failed	The number of times a token pass to successor node was retried. The count is normally zero. This does not change in a smoothly operating system. The cause can show up as noise, checksum error, or frame fragment count in the successor node. The count is incremented by the TBC.
Noise Bits	Noise periods or bursts are detected by the TBC. This may be caused by physical network problems. It can occur without the loss of any messages. A count of zero is expected, but low counts are acceptable. A burst of noise, frame fragment counts over 25, three successive 300 millisecond periods of noise counts of 3, and/or partial frames of 2, will cause noise to be reported, the “Cable A/B Noise” count to be incremented, and the cable to be swapped. These low thresholds are empirically derived from introducing various types of trunk faults.

(Continued)

Table 5-1 — Local UCN Statistics Descriptions (Continued)

Statistic	Description
Checksum Error	The number of times message corruption was detected by the TBC. The theoretical bit error rate for errors not detected by the modem and noted as noise or frame fragments is $1 \times 10E-9$. This works out to approximately three checksum errors per hour, per network. The observed rate is much less, and because messages are tokens, real messages are seldom lost. If frequent checksum errors occur, there may be a physical network or modem problem. If the predecessor to a node with checksum errors indicates a corresponding increase in the "Token Pass Failed" count, the problem is likely to be in the node with the checksum errors.
Repeater Error	The number of times the TBC detected that the error bit in the message end delimiter was set, indicating that a repeater received a message with a bad checksum, then retransmitted it. Because the UCN does not use repeaters, this means corruption in the end delimiter of the message. A count of zero is expected, but occasional counts are acceptable.
Partial Frame	The number of times message corruption was detected by the TBC. This may be caused by physical network problems. It can occur without loss of messages. See the "Noise" description.
Received Frame Too Long	The number of times the received message exceeded the 8 Kbyte IEEE 802.4 limit. Note that UCN messages are limited to 1 Kbyte in length. A count of zero is expected. The count is incremented by the TBC.
No Receive Buffers	The number of times no buffers were available for the TBC to store received messages. Zero is the expected count, but an occasional count under continuous, very heavy demand is acceptable. The count is incremented by the TBC.
Receive Overrun	The number of times there was insufficient local processor DMA bandwidth for the TBC to copy a received message into memory. A count of zero is expected. The count is incremented by the LLC based on the transmit status.
Duplicate RWR	The number of times a duplicate Type 3 message was received. This could be caused by the TBC retrying a message because an ACK was lost. A count of zero is expected, but a small number is acceptable. A count here can sometimes be explained by an increase in one of the other error statistics. The count is incremented by the LLC based on the transmit status.
Null RWR (Resynch)	The number of times the Null RWR messages were used by other nodes to resynchronize with this node upon startup or after an error. A node that leaves and reenters a running UCN will probably show and cause some counts. The count is incremented by the LLC based on the transmit status.
Transmit Underrun	The number of times there was insufficient local processor DMA bandwidth for the TBC to transmit a message. A count of zero is expected. The count is incremented by the LLC based on the transmit status.
Transmit Frame Too Long	The number of times there was a discrepancy between the frame length and the sum of the data block lengths given to the TBC. A count of zero is expected. The count is incremented by the LLC based on the transmit status.

Table 5-1 — Local UCN Statistics Descriptions (Continued)

Statistic	Description
Total Cable Swaps	A count of the operator, periodic, or fault-induced cable swaps. The periodic swap is every 15 minutes.
Cable A Silence	The number of times Cable A was found silent. A count of zero is expected. Normally, a silent cable is a broken or disconnected drop cable, or a bad tap. Certain types of trunk cable problems, such as a short in the middle of a trunk cable segment, can also cause silence to be reported, perhaps by multiple nodes.
Cable B Silence	The number of times Cable B was found silent. A count of zero is expected.
Cable A Noise	The number of times Cable A was found noisy. A count of zero is expected.
Cable B Noise	The number of times Cable B was found noisy. A count of zero is expected.
No-Response Errors	The number of times one or more nodes did not respond to RDR messages. One or more nodes can be OFFNET or can be temporarily overloaded. The count is normally zero, but will increase when access to an OFFNET node is attempted. The count is incremented by the LLC based on the transmit status. The count is not incremented if the TBC is successful on its automatic retry.
Unexpected Responses	The number of times MAC control was not correct for an RWR response message (such as the wrong node number in a response). This indicates network contention or corruption within a node. A count of zero is expected. It also indicates the number of times SAPs were not as expected. The count is incremented by the LLC based on the transmit status and/or message header contents.
Errors In Responses	The number of times the LLC part of an RWR response was incorrect. This can indicate corruption in a node, or network contention. A count of zero is expected. The count is incremented by the LLC based on the transmit status and/or message size, or header contents.
Auto-Reconnects	The number of times this node attempted to reconnect to the UCN after a serious UCN communication fault.
Local Messages	The number of messages between tasks in this node. The count is always incrementing in a NIM, zero in a LM.
Messages Sent	The number of messages sent from this node. The count does not include automatic TBC retries or driver retries.
Messages Received	The number of messages received by this node.
Messages Discarded	The number of messages discarded by this node. Normally zero, but may occasionally count occurrences of: 1) a reply received after the reply timeout period expired; 2) a duplicate reply message caused by retry because the immediate ACK of a reply message was not received by the node sending the reply message; 3) a bad protocol version; 4) a message for an inactive function; 5) a message from a nonconfigured node (NIM only).

(Continued)

Table 5-1 — Local UCN Statistics Descriptions (Continued)

Statistic	Description
Reply Timeouts	The number of times a reply was not received during the user-specified timeout interval. If a Type 3 request, the request was received and ACK'd, but the reply was not received. It may be caused by receive buffer overload in the local node, or the remote node failed after the ACK but before sending the reply.

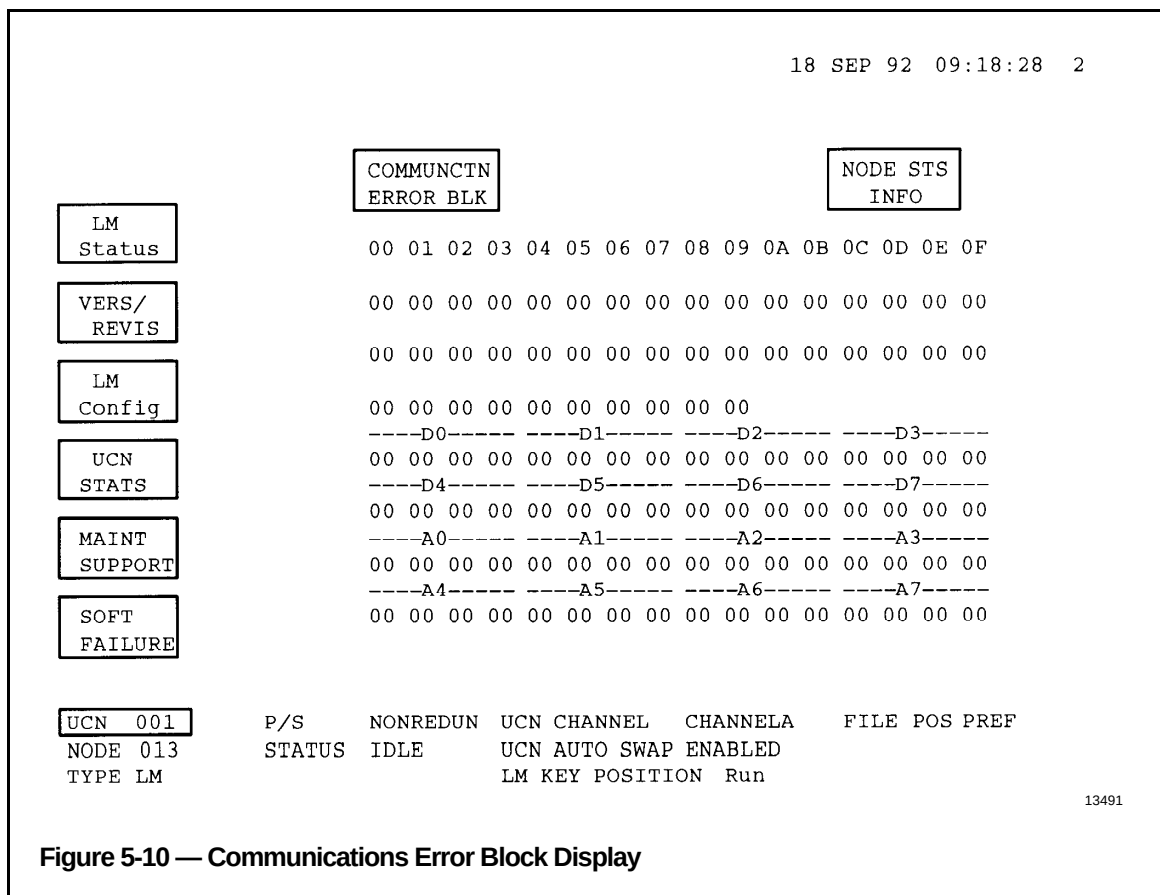
Table 5-2 — Event Sender Statistics Descriptions

Statistic	Description
Messages Sent	The number of messages sent, not including retries. When an idle or running event sender has no new events to send, it resends the last message every 10 seconds so that the NIM can watchdog event delivery.
Messages Retried	The number of messages retried caused by no response, lost ACK, or NAK. The count is normally zero unless event overload has occurred. The count normally correlates with the "NAKs Received" count in the event senders.
Receiver Dropped	The number of times any event receiver failed to respond to a message after retries, and thus was removed from the retry list. Retries are no longer performed to nonresponding event receivers. The count is normally zero in NIMs. A single count in each event sender can accompany a NIM failure (primary and secondary).
NAKs Received	The number of times a message was temporarily not accepted by an event receiver. Retries are performed after a delay. The count is normally zero, except under heavy event traffic.
Throttling Requested	The number of times this node was requested to delay before sending another message. A count of zero is normal, except under heavy event traffic. The count corresponds to the "At Node Throttle Threshold" statistic in the event senders.
Number Of Event Rcvrs	The number of event receivers that are currently acknowledging events from this event sender. In LMs this count is normally the number of primary NIMs.

5.9 MAINTENANCE SUPPORT DISPLAYS

Communications Error Block Display

This display is called up from the **MAINT SUPPORT** target of any LM Detailed Status Display, or by selecting the **COMMUNCTN ERROR BLK** target within the **NODE STS INFO** display.



The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display presents the content of an error buffer within the LMM. This display is normally used to assist product support personnel during fault situations. Consult TAC or Design Engineering for details.

Targets:

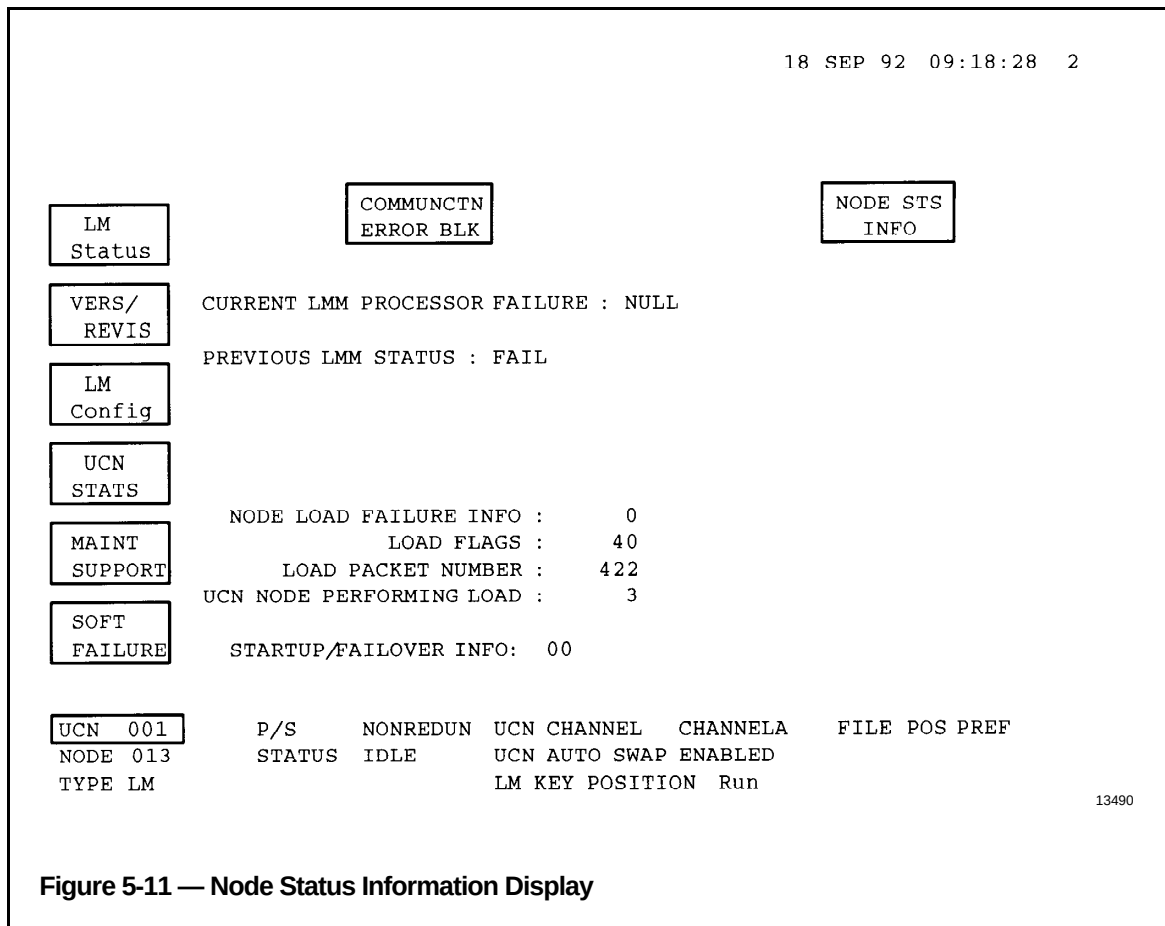
LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxx, see section on LM Blank Status Display.

COMMUNCTN ERROR BLK - target for this display.

NODE STS INFO - target for NODE STATUS (NIM version) display.

5.9.1 Node Status Information Display

This display is called up from within the **MAINT SUPPORT** target of any LM Detailed Status Display, by selecting the **NODE STST INFO** target within that display.



The target along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display presents additional failure information and a few statistics related to Personality download. Although pertaining to the LMM, the data resides in the NIM.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxx - see section on LM Blank Status Display.

COMMUNCTN ERROR BLK - target for COMM ERROR BLOCK display.

NOTE STS NODE - target for this display.

Fields:

See section on LM Blank Status Display for a description of the fields along the bottom.

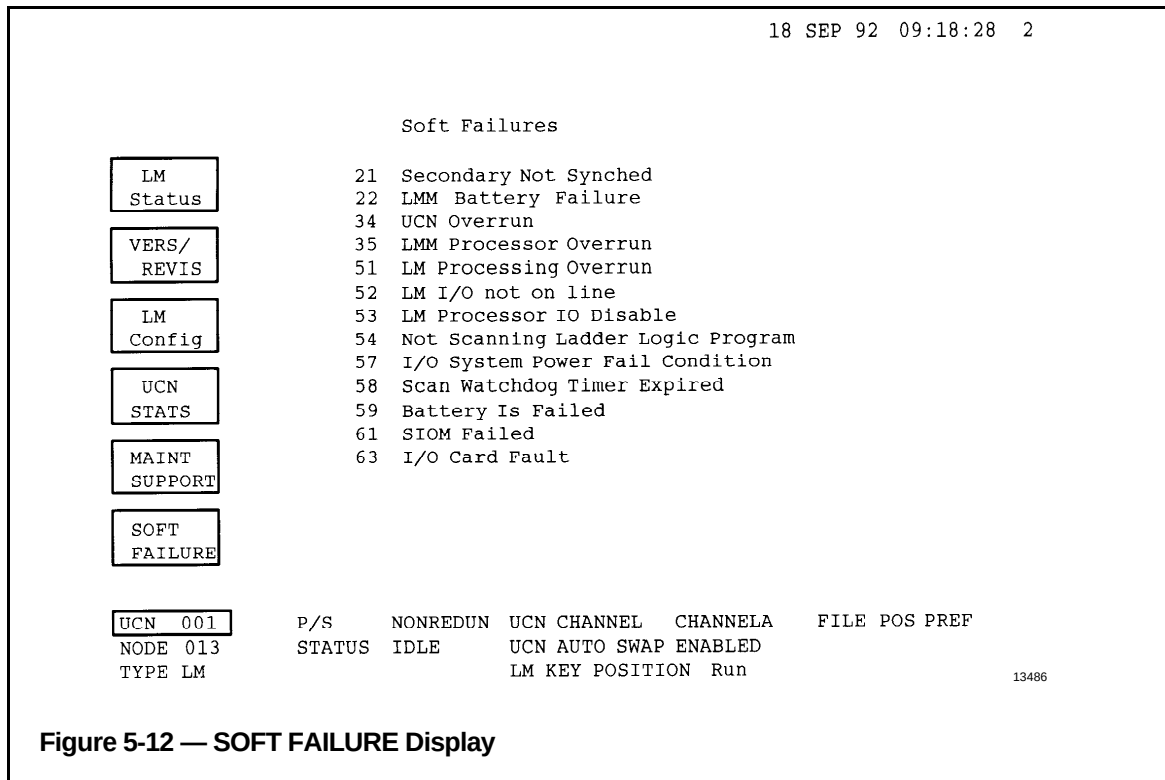
CURRENT LMM PROCESSOR FAILURE - Per NIM, current LMM status.

System parameter: NODESTS

Range: CONFGMIS, FAIL, NOTSYNC, OK, PARTFAIL, OFFNET

5.10 SOFT FAILURE DISPLAY

This display is called up from the **SOFT FAILURE** target of any LM Detailed Status Display.



The targets along the left side and the textual status information along the bottom are standard for all LM Detailed Status Displays.

The display lists all SoftFails supported by the LM, and highlights all which are currently active.

Targets:

LM Status, VERS/REVIS, LM Config, UCN STATS, MAINT SUPPORT, SOFT FAILURE, UCN xxx - see section on LM Blank Status Display.

Refer to table 5-7 for an explanation of the various soft failure codes.

Table 5-7 — Logic Manager Soft Failure Codes

FAILURE CODE	DESCRIPTION	CORRECTIVE ACTION
21	Secondary Not Synched. When the secondary LM comes on line, this softfail is set for Å 30 seconds. If it persists, the secondary is not ready for a failover. The LC is ready for failover, but the LMM database was never transferred.	If communications are good, reissue a STARTUP or IDLE command to the primary LM. This will initiate three more attempts at synchronization. If this is unsuccessful, it is likely that network communication is not the problem. The status of the UCN and RCM communication is available via the Node Info Status on the Maintenance Information Display.
22	LMM Battery Failure	Replace battery
34	UCN Overrun. Another UCN node on the network is not available or is overloaded. This problem is the result of a failed peer-to-peer communication via logic points.	Correct the problem in the other node(s).
35	LMM Processor Overrun. The LM point database is consistently taking longer than 500 milliseconds to process. Usually this is a result of overloading the LM with too many parameter requests.	Cut back on peer-to-peer and/or LCN access.
51	LM Processing Overrun. The LMM access to the LC has been cut off. The LM should go to the FAILED state if communication is lost. This condition should never exist.	Call the Honeywell Technical Assistance Center (TAC).
52	LMM I/O Not On Line. Ladder logic must exist in the LC to monitor the status of serial I/O (must also be inserted for parallel I/O systems). When this softfail exists, the serial I/O system is not being scanned.	Load ladder logic or correct the serial I/O problem.
53	LM Processor I/O Disable. Indicates the Keyswitch is in the DISABLE mode. On a redundant system, setting the LC in DISABLE would cause a failover before the softfail is asserted.	If ready, change the Keyswitch to another mode.

Table 5-7 — Logic Manager Soft Failure Codes (Continued)

FAILURE CODE	DESCRIPTION	CORRECTIVE ACTION
54	Not Scanning Ladder Logic Program. The LM Processor is in the Software Program Mode.	Check the keyswitch on the LM. Make sure it is in the Run/Program position. Use the MSDOS Loader and clear the Software Program Mode in the LM Processor.
57	I/O System Power Fail Condition. A power failure condition has been detected by the LM and the LM has shut down.	For parallel I/O systems, check for correct input and output of power on the I/O rack power supplies. For serial I/O systems, check for link faults and clear them. After clearing the fault(s), cycle the input power on the LM to reset the fault condition in the SLM. (Note: This soft failure will only occur if the dipswitch setting on the SLM is set to shut down all serial links <u>and</u> to send a CLEAR OUTPUT signal to the I/O upon detection of a link fault.)
58	Scan Watchdog Timer Expired Logic Controller Ladder Logic Scan Timer has expired.	The Logic Controller's scan time has exceeded 140ms. Either disable the watchdog timer or reduce the amount of relay ladder logic in the Logic Controller.
59	LM Processor Battery Failure.	Replace battery in LM Processor.
61	SIOM Failed. The LM has detected anSIOM failure and depending on the SLM dip switch settings, may have caused a switchover.	Check wiring and connectors for breaks or loose connections. Also check power on failed card file. If no problems are found, replace suspected faulty SIOM. Restart the Serial I/O by cycling power on the LM or by PUSHING a restart value to the SLM.
63	I/O Card Fault. An output type module has failed or certain analog type cards have gone into overcurrent mode.	For hot insertion modules, place the Freeze/Run switch on the SIOM to Freeze. Replace faulty I/O module or correct overcurrent condition. Returning the Freeze/Run switch to Run will then clear the fault. For SIOMs without the Freeze/Run switch, disconnect power, replace faulty module, and then reconnect power.

DIAGNOSTICS Section 6

This section describes the Logic Manager System's self-diagnostics and error reporting.

The Logic Manager System self-diagnostics helps maintenance personnel to locate system faults at the module level. LED indicators alert you to system faults when they occur and direct you to fault locations. Most system faults can be repaired by replacing a single module.

The self-diagnostic tests are conducted in a series beginning with the Processor Module, which is the core of the system, and proceeds through the processor backplane, other processor modules, the I/O bus, individual I/O card files, interface I/O modules, and individual process I/O modules. The Processor Module thoroughly tests itself before proceeding with the self-diagnostics to assure the validity of the testing. The Processor Module executes functional and comprehensive memory location tests. The functional test assures that hardware paths, control functions, and system registers are operational. The memory location test verifies the ability of the memory to receive and retain stored information.

The diagnostics discussion is broken into three categories: powerup diagnostics, on-line diagnostics, and monitoring diagnostics using the Loader/Terminal.

6.1 SYSTEM POWERUP SELF-TEST

The Processor Module (PM) begins executing the self-diagnostic test program stored in the executive ROM. The red LED labeled **TESTING** illuminates, indicating the initiation of diagnostic testing. This LED remains illuminated until the diagnostic testing procedure is complete. The test procedure is as follows:

1. Test the data flow through the microprocessor and the operation of the microprocessor register.
2. Compute the checksum of the executive ROM.
3. Test the read/write ability of Processor Module's working memory.
4. Test the bus operation.
5. Test the program memory address register in the System Control Module (SCM).
6. Test the Processor Module control and contact data flow in the Register Module.
7. Test the Control Logic Unit (CLU) in the System Control Module (SCM).
8. Test the Address Register in the Memory Module (MM) and the data flow through the MM.

9. Test the register data flow through the Register Module (RM).
10. Test the single bit output transmission of the Input/Output Control Module (IOCM).
11. Test the Sendout Output Status transmission in the IOCM.
12. Test the IOCM and Parallel Link Driver Module (PLDM) for correct system configuration.
13. Test the system mode. The System Mode Keyswitch is examined. The diagnostic tests continue with Test 14 if the Keyswitch is in **PROGRAM**. The diagnostic tests end if the Keyswitch is in the **RUN**, **RUN/PROGRAM**, or **DISABLE** mode. The results of the diagnostic tests are stored in the System Status Table.
14. Test the Memory Module and Register Module. Every location is comprehensively tested in the MM and RM. The processor executes a series of read, write, and complement procedures, comparing the data stored in every memory and register location. The results of the diagnostic tests are stored in the System Status Table.

6.2 PROGRAM MEMORY CHECK

The user program memory integrity is checked through a comparison of checksum calculations. If any on-line changes occur, the initial checksum is recalculated. The checksum calculation and comparison check is as follows:

1. The processor calculates the initial checksum of the user program during the retentive scan.
2. The processor calculates a second checksum for the user program while the program is running by reading six memory words per scan. If any on-line changes occur, flag bits are set to cause the processor to start over by recalculating the initial checksum.
3. If the two checksum calculations match, the process starts over since no error has been detected.
4. If the two checksum calculations do not match, the processor sets the error flag and is forced into the **PROGRAM** mode. The Loader/Terminal will display a PC Diagnostics Fail message. The recovery procedure for a checksum error is to reload program memory. The Loader/Terminal will display Program Checksum Error when the processor is placed in the **PROGRAM** mode.

Table 6-1 describes the function of System Status Table registers involved in the checksum calculation and comparison procedure.

Table 6-1 — Checksum Errors Functions

REGISTER	CONTENT	FUNCTION
2395	Checksum (MSB)	These registers store the calculated user memory program checksum.
2394	Checksum (LSB)	
2393	Checksum	The most significant bit (MSB) is 0 when no error is detected. The MSB is 1 when an error is detected.
2392	0	Not used
2391	Initial Flag	System use only
2390	Initial Pass	The most significant bit (MSB) is 0 when the initial checksum is being calculated. The MSB is 1 when the checksum is complete.

6.3 ON-LINE CHECKS

On-line checks are functional tests performed before contacts are collected at the beginning of every scan. These tests are abbreviated versions of the diagnostics performed at system powerup.

1. Check the processor backplane.
2. Check the System Control Module.
3. Check the main memory.
4. Check the integrity of the Memory Module.
5. Check the Contact Logic Unit in the System Control Module.
6. Check the Parallel Link Driver Module.
7. Check the program checksum.

The processor immediately goes into self-diagnostics if any of these tests are unsuccessful.

6.4 I/O SYSTEM TEST

The Logic Manager Processor also performs a diagnostic function with every output command. Output modules have diagnostic circuitry that receive data over two different routes on the data bus. When an output is solved, not only is the address data and output status sent to the I/O system, but also the complement of this information. These two groups of data are transmitted to appropriate output modules on separate data lines. At the output module, a comparison between the data and its complement is made. If a miscompare occurs, signifying a module or I/O bus fault, the output module sets a fault flag. This flag is read by the processor during Input Status Scan. The address of the faulted module is then posted in the System Status Table. The output module data fault LED illuminates if a miscompare is detected.

6.5 MONITORING DIAGNOSTICS

The model 623-60 MS-DOS Loader allows the user to examine the results of the processor diagnostics. Logic Manager hardware and software status may be accessed and monitored using the PC's CRT. Descriptions of the CRT displays follow.

- **HARDWARE STATUS DISPLAY** provides the user with data associated with various logic controller DIP switch settings and other processor information.
- **SELF-TEST DISPLAY** indicates pass/fail status of the individual hardware elements involved in the self-test routine.
- **I/O MODULE STATUS DISPLAY** provides the total I/O module faults at any given time and lists the most significant addresses of up to eight modules at fault. The displayed faults are cleared and additional faults are displayed if more than eight faults are present.

REMOVAL AND REPLACEMENT Section 7

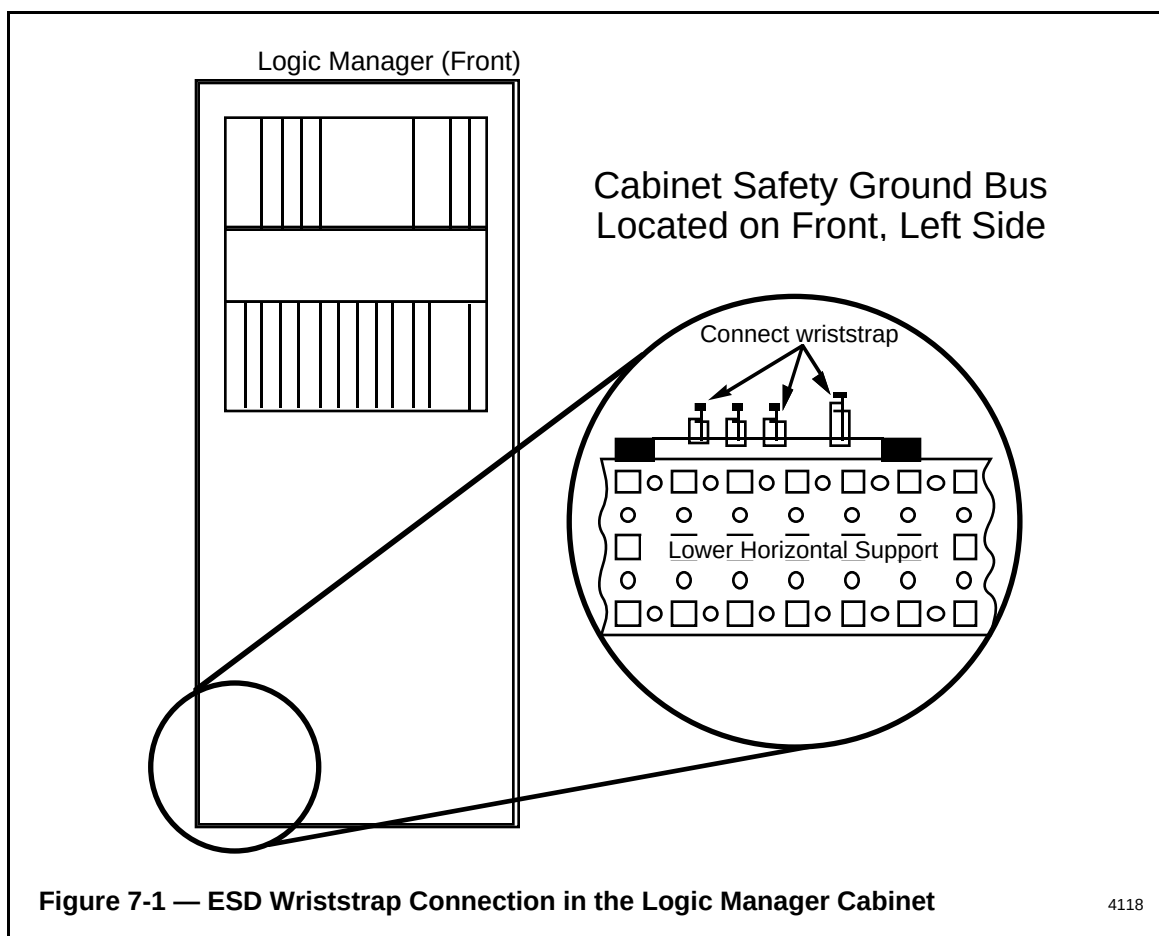
This section describes how to remove and replace components in the Logic Manager System.

7.1 ELECTROSTATIC DISCHARGE PROTECTION

CAUTION

While removing, handling, and installing Logic Manager System components, it is extremely important that you wear an Electrostatic Discharge (ESD) wriststrap that is properly connected to ground. Be sure power to the equipment is off. Slip the strap on your wrist like a wristwatch and connect its clip to the ground bus located inside the front, left side of the cabinet. See Figure 7-1. Wearing an approved ESD wriststrap does not increase the danger of electrical shock. Upon removal, be sure to place modules in a static-safe carrying pouch.

Honeywell can supply a wriststrap manufactured by 3M Company that has a maximum resistance of 10 megaohms and a minimum resistance of 500 k ohms.



4118

7.2 MODULE REMOVAL AND REPLACEMENT

1. Remove power from the Logic Manager processor card file by placing the power breakers in the off position.
2. Before removing a Logic Manager System module, connect your ESD wriststrap to the cabinet ground bar, or if the equipment is panel mounted, connect your ESD wriststrap to the panel ground.
3. Cabling must first be disconnected from the front of the module to be removed. The connection method varies. Power supply module cable disconnection requires the removal of individual wires from its terminal strip. Control modules are connected to cables with friction fit connectors, requiring a small amount of pulling force to release the cable. Figures 7-2 through 7-6 show various typical system cable configurations for both redundant and nonredundant serial or parallel I/O. The I/O point modules in the I/O card file employ terminal block connectors which pivot away from the front of the module and thereby eliminate the need to disconnect each individual terminal block wire.
- 3A. I/O modules may be removed from an I/O card file while power is applied to the card file IF:
 1. the I/O system is a serial I/O configuration.
 2. a 621-9938R SIOM is used in each of the I/O car files.
 3. the I/O module has an "R" suffix in the model number.
4. Modules are released and removed by two different methods. Five modules, those behind a five module wide metal coverplate on the Processor card file, are removed by simultaneously pulling on the upper and lower extractor levers. The metal coverplate is first removed by loosening the four thumbscrews that hold the plate in place. All other modules are removed by loosening their thumbscrews and pulling the module gently forward. Very little force is required.
5. When the module is free from its connector, carefully slide it out on the card rails and into your hands. Immediately place the module in a static safe pouch for transport.
6. To install a module, connect your ESD wriststrap before removing the module from its pouch.
7. Carefully slide the module onto the card file rails, checking to be sure that the module is positioned in both rails.
8. Slide the module in until it has made contact with the backplane connector. Then firmly press forward on the front of the card with the heel of your hand, stopping when the module is fully engaged in the connector. **Do not use excessive force!** If the card does not slip easily into its connector, remove the card and locate the source of the obstruction. Check for bent pins in the backplane connector.

WARNING

Service personnel must not use the power supply module fuse as a means of disconnecting power from the card file. Each card file must have its own individual circuit breaker for this purpose. The fuse holder is not designed for such excessive usage and will deteriorate.

CAUTION

If you are replacing a defective module with a new one, be sure to check whether there are configuration switches or termination resistors on the module being replaced. Configuration switch settings must be duplicated on the replacement module. Some of the modules which may require termination resistors are the SLM, SIOM, SLS, and PBS. Refer to Section 5 of the *Logic Manager Installation* manual in the *Implementation/Logic Manager - 1* binder TDC 2070 for termination resistor guidelines. Similarly, if you are adding serial I/O modules to an existing system, you will likely have to reposition the I/O link termination resistors.

NOTES

1. The serial communications links require either 200 or 300 ohm resistors. When Belden 9729 cable is used (0-4,000 feet), 200 ohm resistors are required. This is a single run of two twisted pair. When Belden 9182 cable is used (0-10,000 feet), 300 ohm resistors required. This is a single twisted pair, and two runs must be used.
2. The last SIOM in the link must have two resistors. The SLM requires one resistor because it is at one end of the link.
3. UCN coaxial cables to the LMM are color coded. Cable A is orange, and cable B is brown.
4. Serial I/O cables less than 50 meters (164 feet) are assigned model numbers ML-KLSxxx depending on the length.

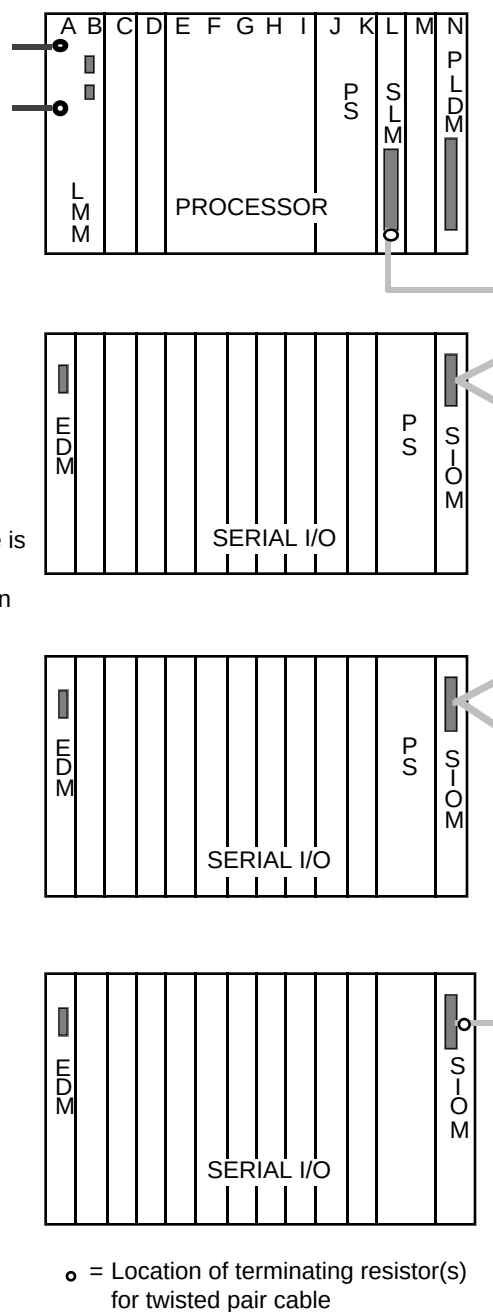
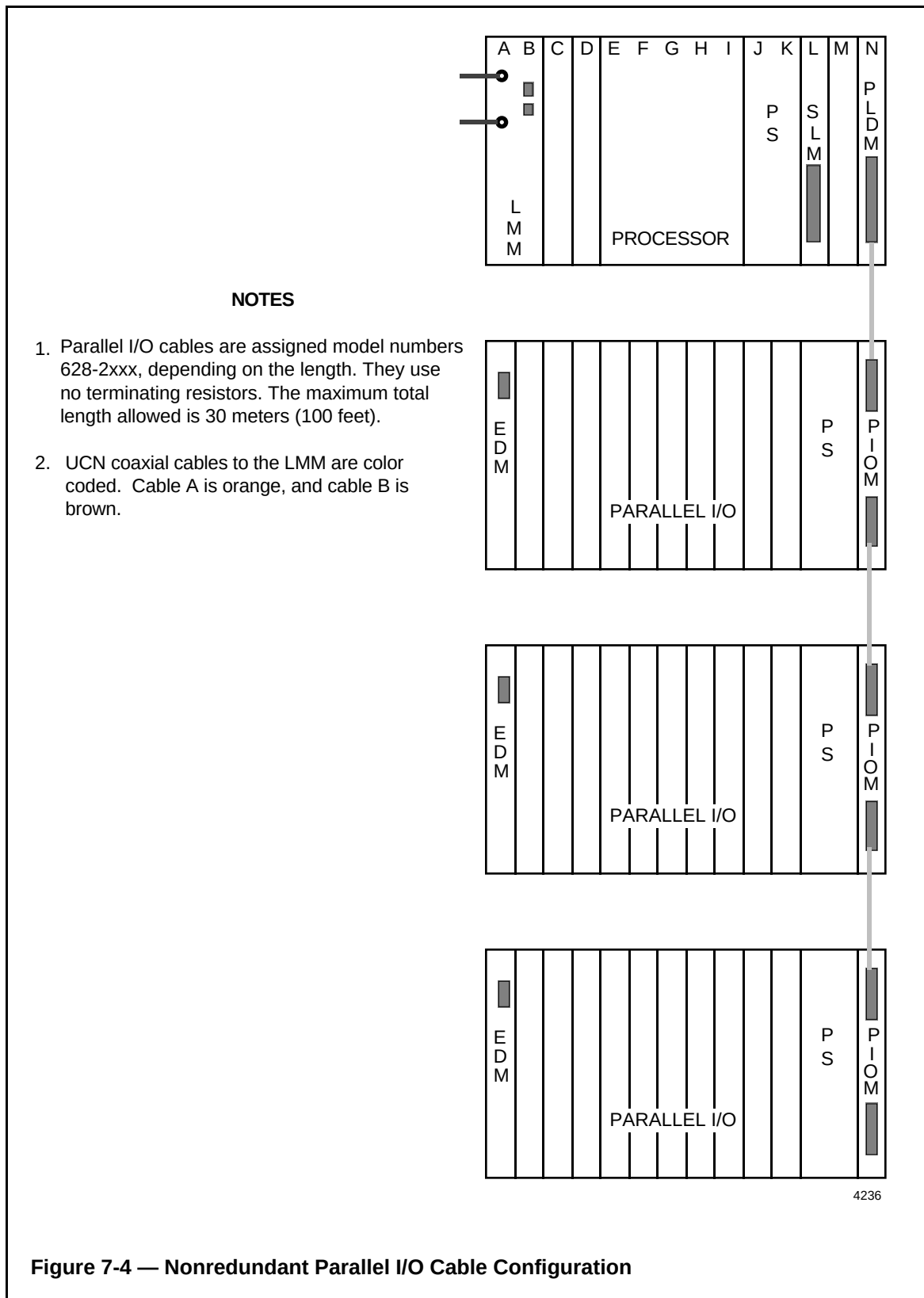
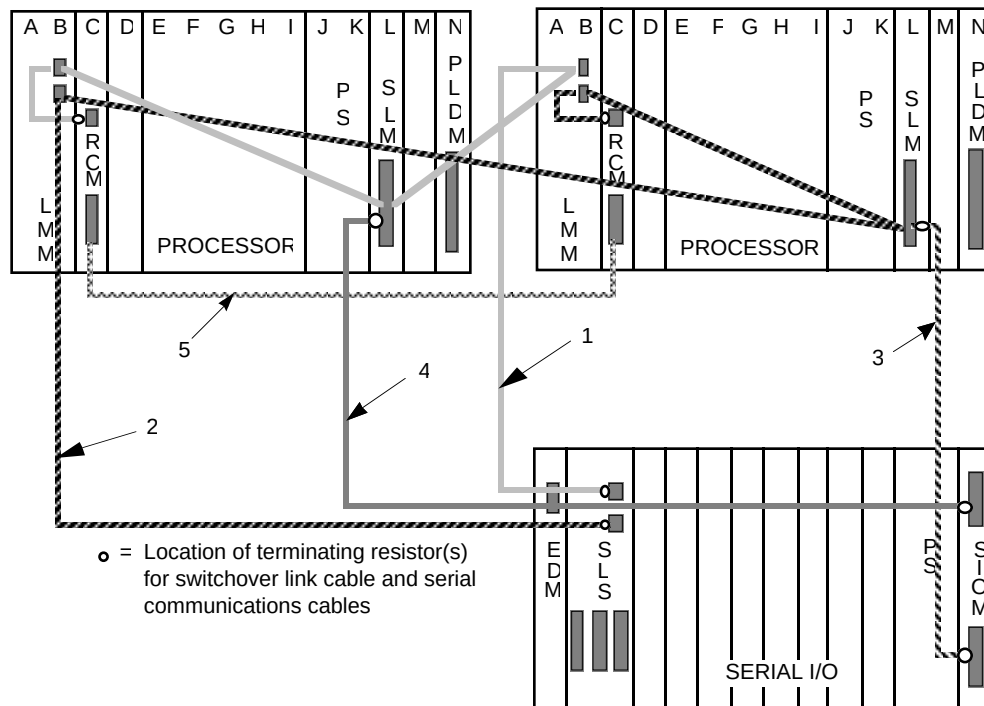


Figure 7-2 — Nonredundant Single Channel Serial I/O Multidrop Cable Configuration

4234





NOTES

1. The RCMs in the Processor card files have built-in terminating resistors.
2. The SLS in the I/O card file has terminating resistors in the module. A DIP switch selects the resistors.
3. The last SIOM in the link must have two terminating resistors per port. The SLM requires one resistor because it is at one end of the serial I/O communications link.
4. Terminating resistor values depend on the type of control link and/or the cable as follows:
 - Control switchover links (Cables 1 and 2) require Belden 8761 cable and 100 ohm resistors. Cables 1 and 2 are limited to 6 meters (20 feet) in length.
 - Serial communications links (Cables 3 and 4) require either 200 or 300 ohm resistors. When Belden 9729 cable is used (0-4,000 feet), 200 ohm terminating resistors are required. This is a single run of two twisted pair. When Belden 9182 cable is used (0-10,000 feet), 300 ohm terminating resistors are required. This is a single twisted pair cable and two runs must be used.
5. Cable 5 is a model 628-5000 RCM Data Cable supplied with the RCMs. The cable is a standard 3 meters (10 feet) in length.
6. The following terminating resistors are supplied with each Logic Manager System.
 - one 100 ohm
 - eight 200 ohm
 - eight 300 ohm
7. Serial I/O cables of less than 50 meters (164 feet) are assigned model numbers ML-KLSxxx depending on the length.

Figure 7-5 — Redundant Serial I/O Cable Configuration

4237

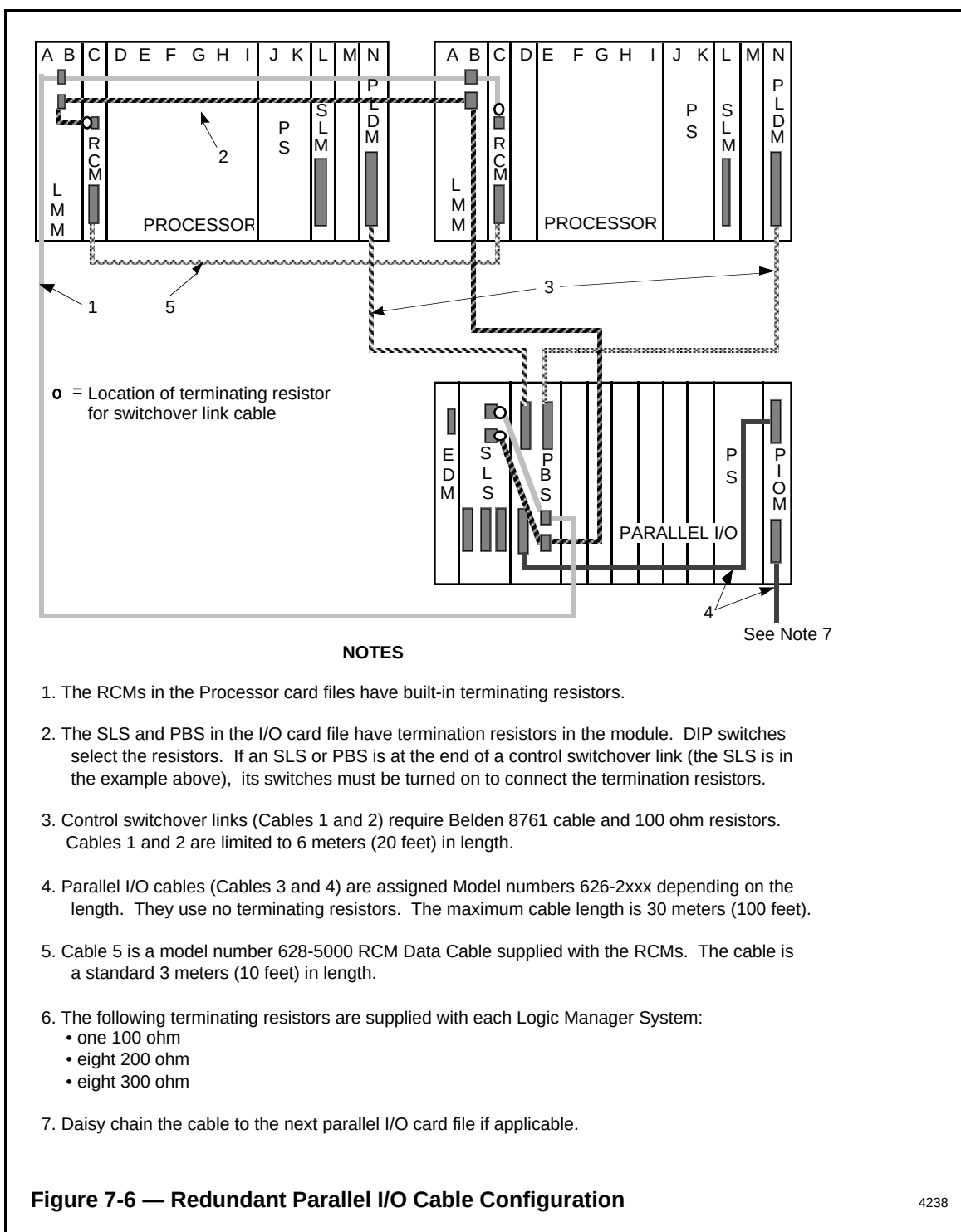


Figure 7-6 — Redundant Parallel I/O Cable Configuration

4238

7.3 BATTERY REMOVAL AND REPLACEMENT

There is a size D lithium battery in the Processor Card File Power Supply Module that provides backup power for the Logic Manager System's memory and registers, and a size 2/3 A lithium battery in the Logic Manager Module (LMM) that provides backup power for the Logic Manager Module's memory and registers. These batteries have a shelf life of several years as long as they are not required to supply backup power. However, when power is off from the LM card file, the batteries discharge more rapidly as they supply power to the system memories. The Power Supply battery may require replacement after 1.5 years of cumulative power off time, while the LMM battery may require replacement after six months of cumulative power off time. For this reason, batteries should be replaced periodically.

The LMM has a green LED labeled **BAT** and the CPM has a green LED labeled **BATTERY PASS**. These LEDs are illuminated when power is on and the battery condition is adequate to provide backup in the event of power failure. In addition, the LM Node Detail Status Display has soft failure indications for Power Supply battery failure. See Section 3 for information.

To replace the batteries, use the following procedure:

1. Release and remove the battery coverplates by loosening their thumbscrews.
2. Remove the old batteries from the battery holders and replace with fresh batteries.
3. It is not necessary to remove equipment power while the batteries are being replaced. Observe the polarity marked on the case of each battery and the holder. After the battery is in place, rotate the battery slightly with your finger to assure that good contact is made with the holder.

WARNING

Lithium batteries are prohibited on passenger aircraft. The following conditions are potentially hazardous and must be avoided:

Recharge
Incineration
Short circuit
Forced discharge (reversal)
Use or storage outside of the specified temperature range
Puncturing, crushing, or disassembly



7.4 CARD FILE REMOVAL AND REPLACEMENT

1. Remove power from the equipment.
2. Disconnect all cables attached to the card file modules.
3. Loosen, but do not remove the four screws holding the card file to its support.
4. Grasp the card file firmly and lift it about 1 cm (1/2 in.) to release the file into your hands.
5. Install the card file in the reverse order, making sure it is engaged by all four mounting screws. Tighten the screws when the card file is in place. Reconnect the cables disconnected in Step 1.

SPARE PARTS

Section 8

8.1 INTRODUCTION

The following tables list the spare parts for the Logic Manager System. The parts are grouped in tables of parts that are associative.

Table 8-1 — Card File Assemblies

ASSEMBLY DESCRIPTION	HONEYWELL PART NUMBER
Nonredundant Processor Card File (115/230 Vac), see Note 1	ML-LMSA35
Nonredundant Processor Card File (24 Vdc), see Note 1	ML-LMSD35
Redundant Processor Card File (115/230 Vac), see Notes 1 and 2	ML-LMRA35
Redundant Processor Card File (24 Vdc), see Notes 1 and 2	ML-LMRD35
Processor Card File (without modules)	620-3591
Serial I/O Full Size Card File, see Note 1	ML-LIOS01
Parallel I/O Full Size Card File, see Note 1	ML-LIOP01
I/O Full Size Card File (without modules)	621-9990
I/O Half Size Card File (without modules)	621-9991

Notes:

1. The Processor, Serial I/O, and Parallel I/O card files include the standard card modules as defined in Section 2 of this manual.
2. The model 628-5000 RCM Data Control Cable is included.

Table 8-2 — Processor Card File Modules

MODULE DESCRIPTION	HONEYWELL PART NUMBER
Memory Module (24 K words)	620-0024
Memory Module (8 K words)	620-0027
Power Supply Module (115/230 Vac 15 A)	620 0036
Control Network Module	620-0038
ModBus Computer Interface Module	620-0043
Power Supply Module (24 Vdc 15 A)	620-0047
ABC Data Collection Module	620-0048
ABC Data Collection Module (Dual port)	620-0052
System Control Module	620-0054
Register Module (4 K x 4 K)	620-0056
Redundancy Control Module	620-0059
Logic Manager Module	620-0071
Processor Module	620-0080
I/O Control Module (2 K I/O)	620-0085
Parallel Link Driver Module	620-0088
Serial Link Module	621-9939

Table 8-3 — Input/Output Card File Interface and Power Supply Modules

MODULE DESCRIPTION	HONEYWELL PART NUMBER
Enhanced Diagnostics Module	621-0021R
Serial Link Selector Module	621-9928R
Parallel Bus Selector Module	621-9929
Power Supply Module (24 Vdc 8 A)	621-9932
Power Supply Module (115/230 Vdc 15 A)	621-9933
Power Supply Module (115/230 Vdc 8 A)	621-9934
Parallel I/O Module	621-9937
Serial I/O Module (Dual port)	621-9938R
Serial I/O Module (Single port)	621-9940

Table 8-4 — Input/Output Card File Process I/O Modules

HONEYWELL PART NUMBER	MODULE DESCRIPTION	NUMBER OF POINTS
621-0010-AR	4-20 mA Analog Output	4
621-0010-VR	0-10 V Analog Output	4
621-1101R	115 Vac/Vdc Isolated Discrete Input	6
621-1201R	230 Vac/Vdc Isolated Discrete Input	6
621-0007R	Reed Relay Output	6
621-2101R	115 Vac Isolated Discrete Output	6
621-2201R	230 Vac Isolated Discrete Output	6
621-0022-AR	0-20 mA Isolated Analog Input	8
621-0022-VR	0-10 V Isolated Analog Input	8
621-1100R	115 Vac/Vdc Discrete Input	8
621-1200R	230 Vac/Vdc Discrete Input	8
621-1500R	24 Vac/Vdc Discrete Input	8
621-3300R	5 Vdc Sink Discrete Input	8
621-3500R	12-24 Vdc Sink Discrete Input	8
621-3502R	12-24 Vdc Sink Fast Response Discrete Input	8
621-3600R	48 Vdc Sink Discrete Input	8
621-4300R	5 Vdc Source Discrete Input	8
621-4500R	12-24 Vdc Source Discrete Input	8
621-4502R	12-24 Vdc Source Fast Response Discrete Input	8
621-2100R	115 Vac Discrete Output	8
621-2102R	115 Vac Self-Protected Discrete Output	8
621-2200R	230 Vac Discrete Output	8
621-2500R	24 Vac Discrete Output	8
621-6300R	5 Vdc Source Discrete Output	8
621-6500R	12-24 Vdc Source Discrete Output	8
621-6503R	12-24 Vdc Source Self Protected Discrete Output	8
621-6600R	48 Vdc Source Discrete Output	8
621-6700R	120 Vdc Source 0.5 A Discrete Output	8
621-6701R	120 Vdc Source 2 A Discrete Output	8

(Continued)

Table 8-4 — Input/Output Card File Process I/O Modules (Continued)

HONEYWELL PART NUMBER	MODULE DESCRIPTION	NUMBER OF POINTS
621-0020R	Universal Analog Input	16
621-1160R	115 Vac Discrete Input	16
621-1250R	240 Vac Discrete Input	16
621-1251R	240 Vac Isolated Discrete Input	16
621-1550R	24 Vac/Vdc Discrete Input	16
621-3450R	12 Vdc Sink Discrete Input	16
621-3552R	24 Vdc Sink Fast Response Discrete Input	16
621-3560R	24 Vdc Sink Discrete Input	16
621-3660R	48 Vdc Sink Discrete Input	16
621-4350R	5 Vdc TTL Discrete Input	16
621-4550R	24 Vdc Source Discrete Input	16
621-2150R	115 Vac Discrete Output	16
621-2550R	24 Vac Discrete Output	16
621-6350R	5 Vdc TTL Source Discrete Output	16
621-6450R	12 Vdc Source Discrete Output	16
621-6550R	24 Vdc Source Discrete Output	16
621-6551R	24 Vdc Source Low Power Discrete Output	16
621-6650R	48 Vdc Source Discrete Output	16
621-1180R	115 Vac Discrete Input	32
621-3580R	24 Vdc Sink Discrete Input	32
621-2175R	115 Vac Discrete Output	32
621-6575R	24 Vdc Source Discrete Output	32

Table 8-5 — Special Function Modules

HONEYWELL PART NUMBER	MODULE DESCRIPTION
621-0006R	BCD Conversion Module
621-0009R	Input Simulator Module
621-0012R	ASCII Communication Module
621-0018R	Absolute Encoder Module
621-0019R	Pulse/Quadrature Input Module
621-0024R	Pulse Input Module
621-0025R	Resistance Temperature Detector Module
621-0307R	High-Speed Counter Module
621-0576R	24 Vdc 24-Sink Input, 8-Source Output Module
627-1002R	3-Port MiniCOP Module
627-1003R	4-Port MiniCOP Module

Table 8-6 — Paper I/O Address Labels

ADDRESS LABEL DESCRIPTION	HONEYWELL PART NUMBER
8-Point Address 0-511	629-9980
8-Point Address 0-2047	629-9981
16-Point Address 0-511	629-9982
16-Point Address 0-2047	629-9983
6-Point Address 0-511	629-9988
32-Point Address 0-2047	629-9997

Table 8-8 — Point I/O Terminal Block Sets

DESCRIPTION	HONEYWELL PART NUMBER
8- Point I/O Terminal Block Set	621-9949
16-Point I/O Terminal Block Set	621-9950
32-Point I/O Terminal Block Set With LEDs (24 Vdc)	621-9976
32-Point I/O Terminal Block Set	621-9977
Terminal block for 621-0018R, 621-0020R	621-9953
Terminal block for 621-0022VR, 621-0022AR	621-9954

Table 8-9 — Miscellaneous Card File Hardware

DESCRIPTION	HONEYWELL PART NUMBER
I/O Extender Board	621-9962
Blank Plastic Coverplate	621-9900
Redundancy Data Control Cable (3 m/10 ft.)	628-5000
Blank Metal Coverplate	629-4004

Table 8-10 — Cabinet and Mounting Accessories

DESCRIPTION	HONEYWELL PART NUMBER
NEMA 1 Dual Access Cabinet (Markhon UCN)	ML-CBDM01
NEMA 1 Single Access Cabinet (Markhon UCN)	ML-CBSM01
Forklift Dual Access Cabinet Base (Markhon UCN)	ML-CFDM01
Forklift Single Access Cabinet Base (Markhon UCN)	ML-CFSM01
Dual Cabinet Complexing Kit (Markhon UCN)	51109532-100
Single Cabinet Complexing Kit (Markhon UCN)	51109532-200
NEMA 1 Dual Access Cabinet (Rittal UCN)	ML-CBDX01
NEMA 1 Single Access Cabinet (Rittal UCN)	ML-CBSX01
Cabinet Complexing Kit (Rittal UCN)	51109524-100
Plastic Wireway for 1 Card File	MP-PCWD11
Plastic Wireway for 2 Card Files	MP-PCWD12
Plastic Wireway for 3 Card Files	MP-PCWD13
Plastic Wireway for 4 Card Files	MP-PCWD14
Cabinet Fan Assembly (240 Vac 50 Hz)	ML-FAN501
Cabinet Fan Assembly (120 Vac 60 Hz)	ML-FAN601
AC Circuit Breaker Box ((4 individual breakers)	MP-PCPE04

Table 8-11 — Module Fuse Replacements

MODULE MODEL NUMBER	FUSE TYPE	HONEYWELL PART NUMBER*
620-0036 P/S (230 Vac)	1 A slow blow – 5 x 20 mm (250 V)	629-1002
620-0036 P/S (115 Vac) 621-9933 P/S (230 Vac)	2 A slow blow – 5 x 20 mm (250 V)	629-1003
621-2100R AC Output † 621-2102R AC Output 621-2150R AC Output	3 A slow blow – 5 x 20 mm (250 V)	629-1022
621-6500 DC Output 621-6600 DC Output 621-6701 DC Output 621-9932 P/S (24 Vdc)	8 A fast blow – 5 x 20 mm (250 V)	629-1007
621-6575 DC Output	10 A fast blow – subminiature	629-1008
621-6300R DC Output 621-9934 P/S (230 Vac)	0.8 A slow blow – 5 x 20 mm	629-1009
621-2200R AC Output † 621-2500R AC Output † 621-2550R AC Output 621-6503R DC Output †† 621-6551R DC Output 621-6700R DC Output † 621-9929 DC Output 621-2101 DC Output	3 A fast blow – subminiature 3A slow blow	629-1012 629-1023

(Continued)

* Honeywell part number is for a package of 5 fuses.

† Fuse type depends on the product release date. Current models use fuse 629-1012.

†† More than one fuse is installed.

Table 8-11 — Module Fuse Replacements (Continued)

MODULE MODEL NUMBER	FUSE TYPE	HONEYWELL PART NUMBER*
621-0018R Ab. Encoder 621-6450R DC Output 621-6503R DC Output 621-6550R DC Output 621-6650R DC Output	7 A fast blow – subminiature	629-1016
621-9933 P/S (115 Vac)	4 A slow blow – 5 x 20 mm	629-1017
621-0034 P/S (115 Vac)	1.6 A slow blow – 5 x 20 mm	629-1018
621-0047 P/S (24 Vdc)	6.3 A slow blow – 5 x 20 mm	629-1019
621-0018R Ab. Encoder 621-2175R AC Output 620-0071	5 A fast blow – subminiature	629-1020

* Honeywell part number is for a package of 5 fuses.

Table 8-12 — Module Battery Replacements

MODULE MODEL NUMBER	BATTERY TYPE	HONEYWELL PART NUMBER
620-0036 P/S (115/230 Vac) 620-0047 P/S (24 Vdc)	Size D lithium	629-3001
621-0012 ASCII Communication 621-1002R 621-1003R	Size 1/2 AA lithium	629-3003
620-0071 Logic Manager	Size 2/3 A lithium	629-3004

Table 8-13 — Serial I/O Bulk Cables

CABLE TYPE	CABLE LENGTH	HONEYWELL PART NUMBER
Belden 9505	152 m (500 ft.)	ML-KLS061
	305 m (1000 ft.)	ML-KLS062
	915 m (3000 ft.)	ML-KLS063
Belden 9182 (Single twisted pair)	152 m (500 ft.)	ML-KLS064
	305 m (1000 ft.)	ML-KLS065
	915 m (3000 ft.)	ML-KLS066
Belden 9729 (Two twisted pair)	152 m (500 ft.)	ML-KLS067
	305 m (1000 ft.)	ML-KLS068
	915 m (3000 ft.)	ML-KLS069
Belden 9730 (Three twisted pair)	152 m (500 ft.)	ML-KLS070
	305 m (1000 ft.)	ML-KLS071
	915 m (3000 ft.)	ML-KLS072
Belden 9731 (Six twisted pair)	152 m (500 ft.)	ML-KLS073
	305 m (1000 ft.)	ML-KLS074
	915 m (3000 ft.)	ML-KLS075

NOTE

Belden 9729 cable is a four conductor (two twisted pair) cable, and the amount of cable needed is equal to the total cable distance between card files. The Belden 9182 cable is a two conductor cable (single twisted pair), and twice as much cable must be ordered since the requirement is for two twisted pair. Belden cable is the preferred cable because of its exact electrical characteristics. If a substitute cable is considered, check with the Honeywell Technical Assistance Center (TAC) for approval before the change is made.

Table 8-14— Parallel I/O Cables

HONEYWELL PART NUMBER	CABLE LENGTH	HONEYWELL PART NUMBER	CABLE LENGTH
628-2000	20 cm (9 in.)	628-2017	12 m (39 ft.)
628-2001	35 cm (14 in.)	628-2018	14 m (46 ft.)
628-2002	120 cm (4 ft.)	628-2019	16 m (52 ft.)
628-2003	180 cm (6 ft.)	628-2021	20 m (65 ft.)
628-2004	3 m (10 ft.)	628-2022	22 m (72 ft.)
628-2010	4 m (13 ft.)	628-2023	24 m (78 ft.)
628-2011	5 m (16 ft.)	628-2024	26 m (85 ft.)
628-2012	4 m (13 ft.)	628-2026	30 m (98 ft.)
628-2016	10 m (32 ft.)		

Table 8-15 — UCN Cables, Taps, and Kits

DESCRIPTION	HONEYWELL PART NUMBER
UCN 2 Drop Tap Assembly (Pair)	MU-NTAP02
UCN 4 Drop Tap Assembly (Pair)	MU-NTAP04
UCN 8 Drop Tap Assembly (Pair)	MU-NTAP08
UCN Drop Cable Pair (2 m)	MU-NKD002
UCN Drop Cable Pair (5 m)	MU-NKD005
UCN Drop Cable Pair (10 m)	MU-NKD010
UCN Drop Cable Pair (20 m)	MU-NKD020
UCN Drop Cable Pair (30 m)	MU-NKD030
UCN Drop Cable Pair (40 m)	MU-NKD040
UCN Drop Cable Pair (50 m)	MU-NKD050
UCN Trunk Cable Pair (2.6 m)	MU-NKT002
UCN Trunk Cable Pair (5 m)	MU-NKT005
UCN Trunk Cable Pair (10 m)	MU-NKT010
UCN Trunk Cable Pair (20 m)	MU-NKT020
UCN Trunk Cable Pair (30 m)	MU-NKT030
UCN Trunk Cable Pair (50 m)	MU-NKT050
UCN Trunk Cable Pair (100 m)	MU-NKT100
UCN Trunk Cable Pair (200 m)	MU-NKT200
UCN Trunk Cable Pair 400 m)	MU-NKT400
UCN Trunk Cable Pair (600 m)	MU-NKT600
UCN Trunk Cable Raw Material Reel (152.4 m/500 ft.)	51190899-152
UCN Trunk Cable Raw Material Reel (304.9 m/1000 ft.)	51190899-305
UCN Trunk Cable Raw Material Reel (914.4 m/3000 ft.)	51190899-914
UCN RG-11 Trunk Connector Kit (24 Connectors)	MU-NKTK01
RG-11 Trunk Cable Crimp Tool Kit*	MU-NKTT01
UCN Trunk Cable Splice Kit (10 Splices)	MU-NCSK01
UCN RG-6 Drop Connector Kit (24 Connectors)	MU-NKDK01
RG-6 Drop Cable Crimp Tool Kit*	MU-NKDT01

*Contact the Honeywell Technical Assistance Center (TAC) for a procedure guide.

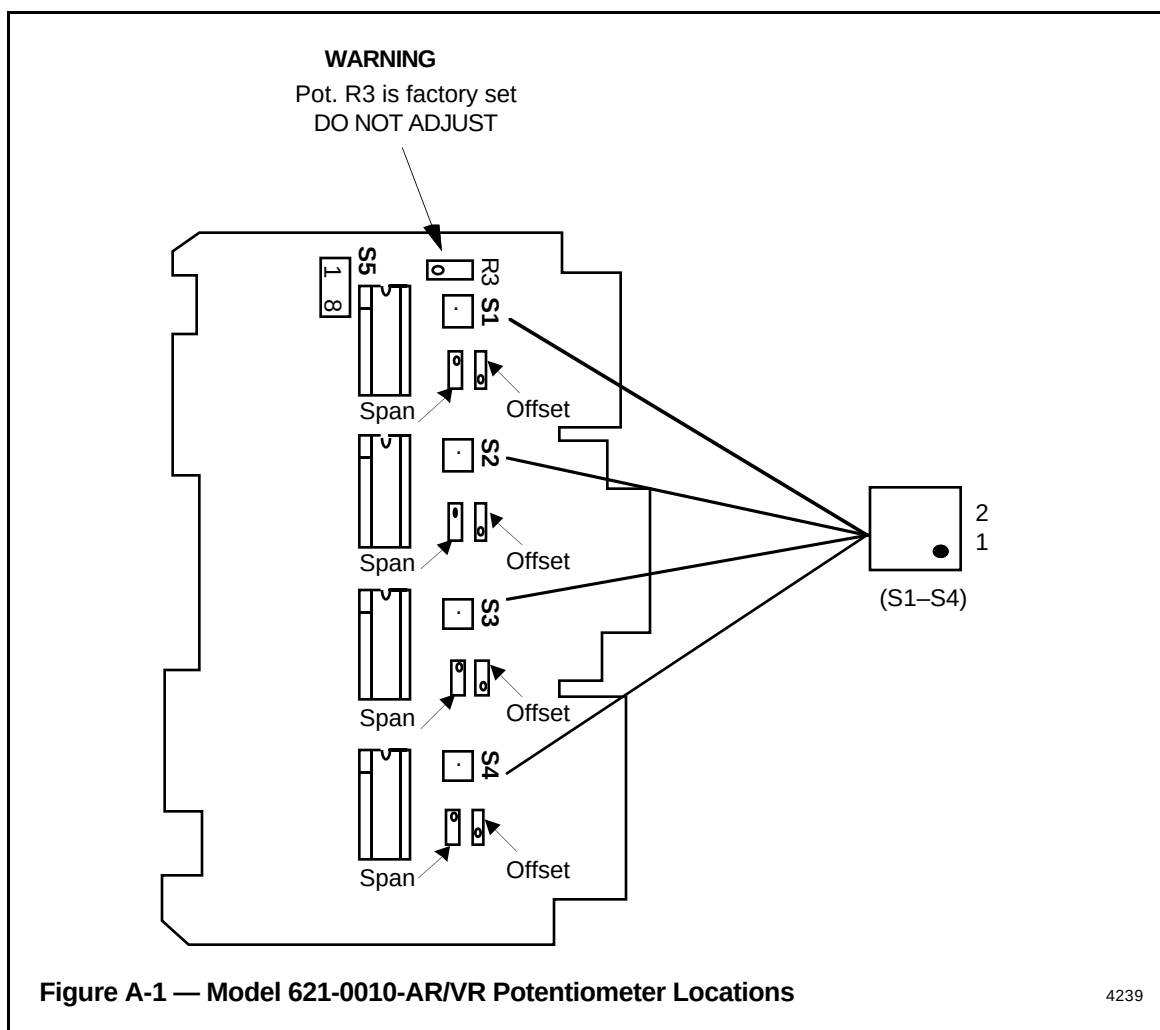
CALIBRATION PROCEDURES

Appendix A

This appendix describes the calibration procedure used to calibrate the analog output modules

A.1 MODEL 621-0010-AR/VR ANALOG OUTPUT MODULES

Each model 621-0010-AR analog output channel has been factory calibrated for 4 to 20 mA operation. Each model 621-0010-VR analog output channel has been factory calibrated for 0 to 10 Vdc operation. Output operation at other ranges may require that the output be recalibrated. Each output channel contains a separate **Offset** and **Span** adjustment potentiometer for this purpose. Refer to Figure A-1 for the adjustment potentiometer locations.



A.1.1 Equipment Required

Calibration of the module channels requires the following equipment:

- Model 623-60 MS DOS Loader
- Model 621-9962 I/O Extender Board
- Small flat blade screwdriver

A.1.2 Calibration Adjustment Procedure

You must remove the module's component side cover to access the **Offset** and **Span** potentiometers.

Install the model 621-0010 in the I/O card file using the model 621-9962 I/O Extender Board. The Logic Manager Processor must be in the **RUN** mode before you calibrate the model 621-0010. Also, be sure to adjust the **Offset** before setting the **Span**.

Offset Adjustment

Adjust each channel to produce the following output, depending on the output range selection, when a zero data value is PUSHed to the channel.

<u>Range Selection</u>	<u>Output</u>
0–10 Vdc	0.0000 Vdc
4–20 mA	4.0000 mA
±10 Vdc	-10.0000 Vdc
±5 Vdc	-5.00000 Vdc

Span Adjustment

Adjust each channel to produce the following output, depending on output range selection, when a 4095 data value is PUSHed to the channel.

<u>Range Selection</u>	<u>Output</u>
0–10 Vdc	9.9976 Vdc
4–20 mA	19.9976 mA
±10 Vdc	9.9951 Vdc
±5 Vdc	4.9987 Vdc

Index

Topic	Section Heading
4 K x 4 K Register Module	
Description	2.3.5
Memory Map	2.3.5
8 K Memory Module	
Configuration Switches	2.3.3
Description	2.3.3
24 K Memory Module	
Configuration Switches	2.3.4
Description	2.3.4
Adhesive Backed I/O Address Labels	8.1
Allowable Leakage Current	2.4.7.1
Analog Input Modules	
Description	2.4.7.2
Analog Output Modules	
Description	2.4.7.4
Battery	
Installation	2.3.1.3, 2.3.9
Removal and Replacement	7.3
Cabinet and Mounting Accessories Parts	8.1
Cable Removal and Replacement	7.2
Calibration Procedures	A
Card File	
Assemblies Parts List	8.1
Removal and Replacement	7.4
Checksum Error Functions	6.2
Connect Test Detail Display	4.7
Connect Test Summary Display-Page 1	4.5.4
Connect Test Summary Display-Page 2	4.5.4
Diagnostics Description	6
Discrete Input Modules	2.4.7.1
Allowable Leakage Current	2.4.7.1
Description	2.4.7.1
Discrete Output Modules	2.4.7.3
Description	2.4.7.3
Display	
Connect Test Detail Display	4.7
Connect Test Summary Display-Page 1	4.5.4
Connect Test Summary Display-Page 2	4.5.4
I/O System Status Display	5.4
UCN Stats Display	5.8
Logic Manager Configuration Display	5.7
Logic Manager Detailed Status Display	5.2
Logic Manager Hardware Status Display	5.5
Logic Manager Self Test	5.3
Maintenance Support Displays	5.9
Revisions and Personality Display	5.6
Soft Failure Display	5.10
UCN Exerciser Overview	4.5.2
UCN Exerciser Help Display	4.5.3
UCN Status Display	5.1

Index

Topic	Section Heading
Electrostatic Discharge Protection	7.1
Enhanced Diagnostic Module	
Configuration Switches	2.4.1.2
Description	2.4.1
Front Panel	2.4.1.1
Failure Types	3.1
Fault Isolation	3
Four Channel Serial I/O Multidrop Network	2.1.3
Half Size Nonredundant Parallel I/O Card File	
Description	2.2.2.3
Module Slot Assignments	2.2.2.3
Half Size Nonredundant Serial I/O Card File	
Description	2.2.2.2
Module Slot Assignments	2.2.2.2
I/O 24 Vdc 8 A Power Supply Module	
Description	2.4.4.1
Fuse	2.4.4.1
I/O 115/230 Vac 8 A Power Supply Module	
Description	2.4.4.3
Fuses	2.4.4.3
I/O 115/230 Vac 15 A Power Supply Module	
Description	2.4.4.2
Fuses	2.4.4.2
I/O Card File	
Modules	2.4
Power Supply Modules	2.4.4
I/O Card File Modules	
115/230 Vac 15 A Power Supply	2.4.4.1
115/230 Vac 8 A Power Supply	2.4.4.1
24 Vdc 8 A Power Supply	2.4.4.1
Enhanced Diagnostic	2.4.1
Parallel Bus Selector	2.4.3
Parallel I/O	2.4.6
Serial I/O	2.4.5
Serial Link Selector	2.4.2
I/O Card Files	2.2.2
I/O Control Module	
Description	2.3.8
I/O System Test	6.4
I/O System Status Display	5.4
I/Output Card File Interface and Power Supply Modules Parts List	8.1
Input/Output Card File Process I/O Modules Parts List	8.1
Logic Manager Configuration Display	5.7
Logic Manager Hardware Status Display	5.5
Logic Manager I/O System Description	2.1.2
Logic Manager Self Test Display	5.3

Index

Topic	Section Heading
Logic Manager Module	
Battery Installation	2.3.1.3
Circuit Board Layout	2.3.1.2
Configuration Switches	2.3.1.3
Description	2.3.1
Front Panel	2.3.1.1
Logic Manager Processor	
Description	2.1.1
Specifications	2.1.1
Logic Manager Self-Test Display	5.3
Miscellaneous Card File Hardware Parts List	8.1
Model 621-0010-A/V Modules	A.1
Calibration Adjustment Procedure	A.1.2
Equipment Required	A.1.1
Offset Adjustment	A.1.2
Span Adjustment	A.1.2
Module	
Battery Replacement Parts List	8.1
Fuse Replacement Parts List	8.1
Removal and Replacement	7.2
Monitoring Diagnostics	6.5
Nonredundant	
Four Channel Serial I/O Multidrop Cable Configuration	7.2
Parallel I/O Cable Configuration	7.2
Single Channel Serial I/O Multidrop Cable Configuration	7.2
Nonredundant Parallel I/O Card Files	
Description	2.2.2.3
Front View	2.2.2.3
Full Size Nonredundant Parallel I/O Card File	2.2.2.3
Half Size Nonredundant Parallel I/O Card File	2.2.2.3
Slot Assignments	2.2.2.3
Nonredundant Processor Card File	
Description	2.2.1.1
Front View	2.2.1.1
Slot Assignments	2.2.1.1
Nonredundant Serial I/O Card Files	
Description	2.2.2.1
Front View	2.2.2.1
Full Size Nonredundant Serial I/O Card File	2.2.2.1
Half Size Nonredundant Serial I/O Card File	2.2.2.1
Slot Assignments	2.2.2.1
On Line Checks	6.3
Paper I/O Address Labels	8.1
Parallel Bus Selector Module	
Configuration Switches	2.4.3.2
Description	2.4.3
Front Panel	2.4.3.1

Index

Topic	Section Heading
Parallel I/O	
Cables	8.1
System Description	2.1.4
Parallel I/O Module	
Configuration Switches	2.4.6.2
Description	2.4.6
Front Panel	2.4.6.1
Parallel Link Driver Module	
Configuration Switches	2.3.11.2
Description	2.3.11
Front Panel	2.3.11.1
Parts List	
Adhesive Backed I/O Address Labels	8.1
Cabinet and Mounting Accessories	8.1
Card File Assemblies	8.1
Input/Output Card File Interface and Power Supply Modules	8.1
Input/Output Card File Process I/O Modules	8.1
Miscellaneous Card File Hardware	8.1
Module Battery Replacements	8.1
Module Fuse Replacements	8.1
Paper I/O Address Labels	8.1
Parallel I/O Cables	8.1
Point I/O Terminal Block Sets	8.1
Processor Card File Modules	8.1
Serial I/O Bulk Cables	8.1
UCN,Cables, Taps, and Kits	8.1
Point I/O Terminal Block Sets	8.1
Process I/O Modules	2.4.7
Processor Card File 115/230 Vac 15 A Power Supply Module	
Battery Installation	2.3.9
Description	2.3.9.1
Fuses	2.3.9.1
Processor Card File 24 Vdc 15 A Power Supply Module	
Battery Installation	2.3.9
Description	2.3.9.2
Fuse	2.3.9.2
Processor Card File	
Modules	2.3
Modules Parts List	8.1
Option Modules Available	2.2.1.3
Power Supply Modules	2.3.9

Index

Topic	Section Heading
Processor Card File Module	
115/230 Vac 15 A Power Supply	2.3.9.1
24 K Memory	2.3.4
24 Vdc 15 A Power Supply	2.3.9.2
4 K X 4 K Register	2.3.5
8 K Memory	2.3.3
I/O Control	2.3.8
Logic Manager	2.3.1
Parallel Link Driver	2.3.11
Processor	2.3.7
Redundancy Control	2.3.2
Serial Link	2.3.10
System Control	2.3.6
Processor Card Files	2.2.1
Processor Module	
Description	2.3.7
Program Memory Check	6.2
Purpose of this Manual	1.1
Redundancy Control Module	
Configuration Switches	2.3.2.2
Description	2.3.2
Fault Display	2.3.2.2
Front Panel	2.3.2.1
Redundant	
Control System Description	2.1.5
Parallel I/O Cable Configuration	7.2
Serial I/O Cable Configuration	7.2
Redundant Parallel I/O Card File	
Description	2.2.2.4
Front View	2.2.2.4
Module Slot Assignments	2.2.2.4
Redundant Processor Card File	
Description	2.2.1.2
Front View	2.2.1.2
Module Slot Assignments	2.2.1.2
Redundant Serial I/O Card File	
Description	2.2.2.2
Front View	2.2.2.2
Module Slot Assignments	2.2.2.2
Related Publications	1.2
Removal and Replacement	7
Revisions and Personality Data Display	5.6
Serial I/O Bulk Cables Parts List	8.1
Serial I/O Modules	
Configuration Switches	2.4.5.2
Description	2.4.5
Front Panel	2.4.5.1
Serial I/O System	
Description	2.1.3
Specifications	2.1.3

Index

Topic	Section Heading
Serial Link Module	
Configuration Switches	2.3.10.2
Description	2.3.10
Front Panel	2.3.10.1
Terminators	2.3.10
Serial Link Selector Module	
Configuration Switches	2.4.2.2
Description	2.4.2
Front Panel	2.4.2.1
Single Channel Serial I/O Multidrop Network	2.1.3
Soft Failures	3.1
Spare Parts	8
Special Function Modules	
Description	2.4.7.5
System Control Module	
Description	2.3.6
System Overview	2.1
System Powerup Self-Test	6.1
Terminal Removal and Replacement	7.2
Terminating Resistors	2.4.5.2
Test	
Request Data Response (RDR)	4.3
System Powerup	6.1
Test Operation	4.5
Triangulation	4.4
Typical UCN System	2.1
UCN Cables, Taps, and Kits Parts List	8.1
UCN Exerciser	4
Analyzing Results	4.7
Applications	4.2
Request Data Response (RDR) Test	4.3
Test Operation	4.5
Triangulation Test	4.4
UCN Exerciser Overview Display	4.5.2
UCN Status Display	5.1

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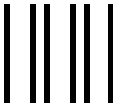
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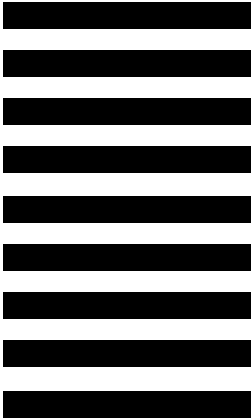
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